US 20250283150A1

(19) **United States**

(12) **Patent Application Publication**

(10) **Pub. No.: US 2025/0283150 A1**

(43) **Pub. Date: Sep. 11, 2025**

KUZUYA et al.

(54) **NUCLEIC ACID STRUCTURE**

(71) Applicants: **CRANEBIO CO., LTD.**, Tokyo (JP); **THE SCHOOL CORPORATION KANSAI UNIVERSITY**

(72) Inventors: **Akinori KUZUYA**, Suita-shi, Osaka (JP); **Nozomu TAKAHASHI**, Suita-shi, Osaka (JP); **Yuta OKAMOTO**, Suita-shi, Osaka (JP); **Sachiko KONO**, Ibaraki-shi, Osaka (JP); **Chihiro KATO**, Osaka-shi, Osaka (JP); **Keita SAITO**, Tokyo (JP)

(21) Appl. No.: **18/019,375**

(22) PCT Filed: **Jul. 30, 2021**

(86) PCT No.: **PCT/JP2021/028281**

§ 371 (c)(1),
(2) Date: **Feb. 2, 2023**

(30) **Foreign Application Priority Data**

Aug. 5, 2020 (JP) 2020-132829

Publication Classification

(51) **Int. Cl.**
C12Q 1/6816 (2018.01)
C12N 15/115 (2010.01)

(52) **U.S. Cl.**
CPC *C12Q 1/6816* (2013.01); *C12N 15/115* (2013.01); *C12N 2310/16* (2013.01)

(57) **ABSTRACT**

An object is to provide a technique for detecting a test substance more easily using a nucleic acid structure. The object is achieved by a nucleic acid structure comprising a partial nucleic acid structure 1 and a partial nucleic acid structure 2, the partial nucleic acid structure 1 and the partial nucleic acid structure 2 being linked by a Holliday junction, the partial nucleic acid structure 1 comprising a protruding structure 1a containing a nucleic acid aptamer sequence, and a protruding structure 1b containing a split domain 1 of an enzyme, and the partial nucleic acid structure 2 comprising a protruding structure 2a containing a sequence complementary to part or all of the nucleic acid aptamer sequence, and a protruding structure 2b containing a split domain 2 paired with the split domain 1.

Specification includes a Sequence Listing.

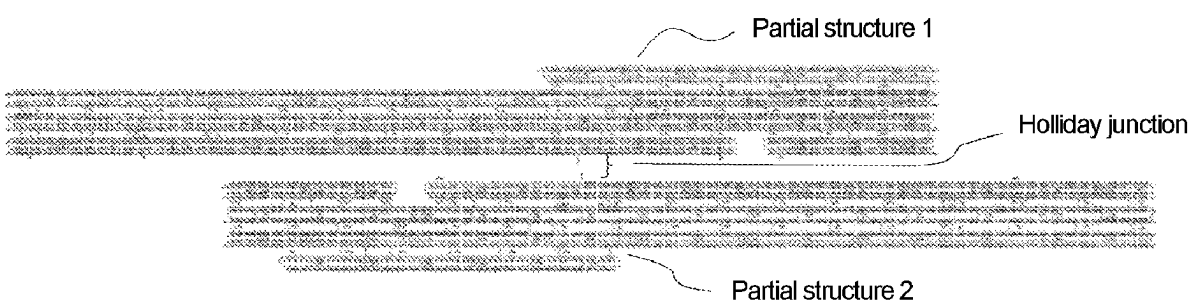


Fig. 1

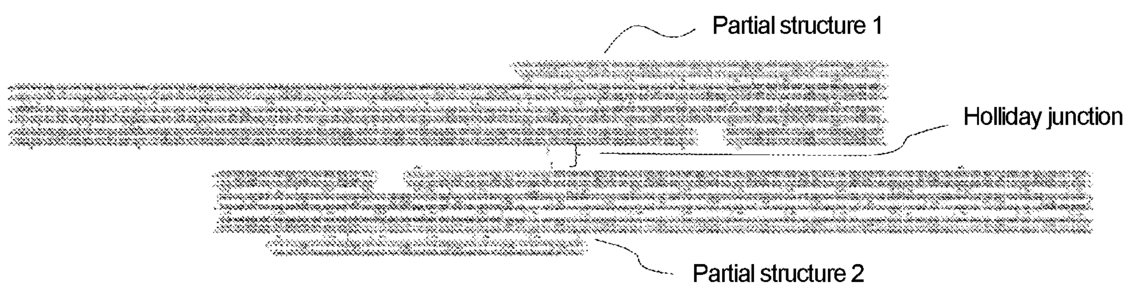

Partial structure 1
Holliday junction
Partial structure 2

Fig. 2

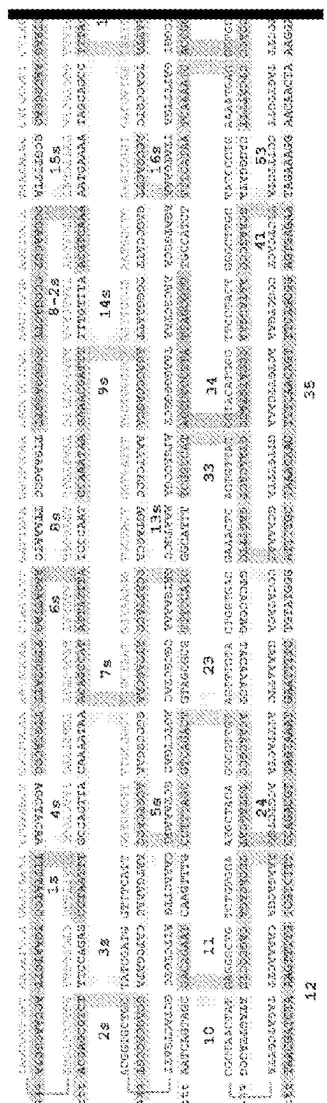

1s
2s
3s
4s
5s
6s
7s
8s
8-2s
9s
10
11
12
13s
14s
15s
16s
23
24
33
34
35
41
53

Fig. 3

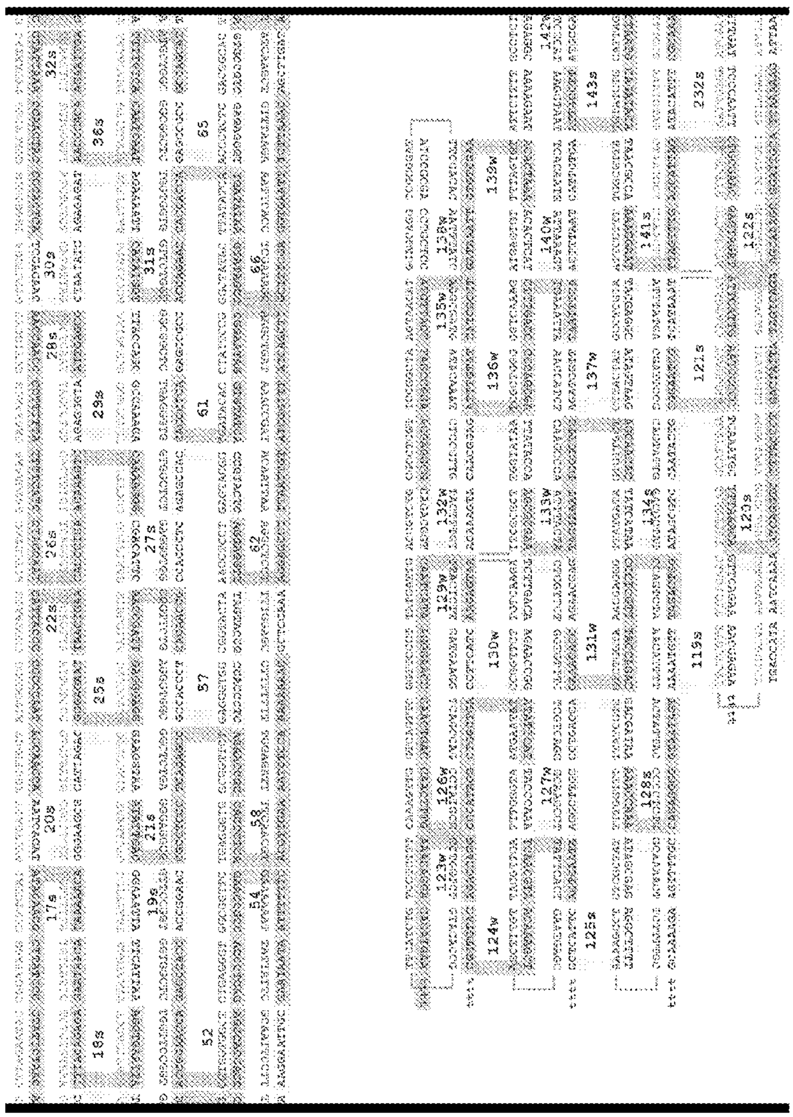


Fig. 4

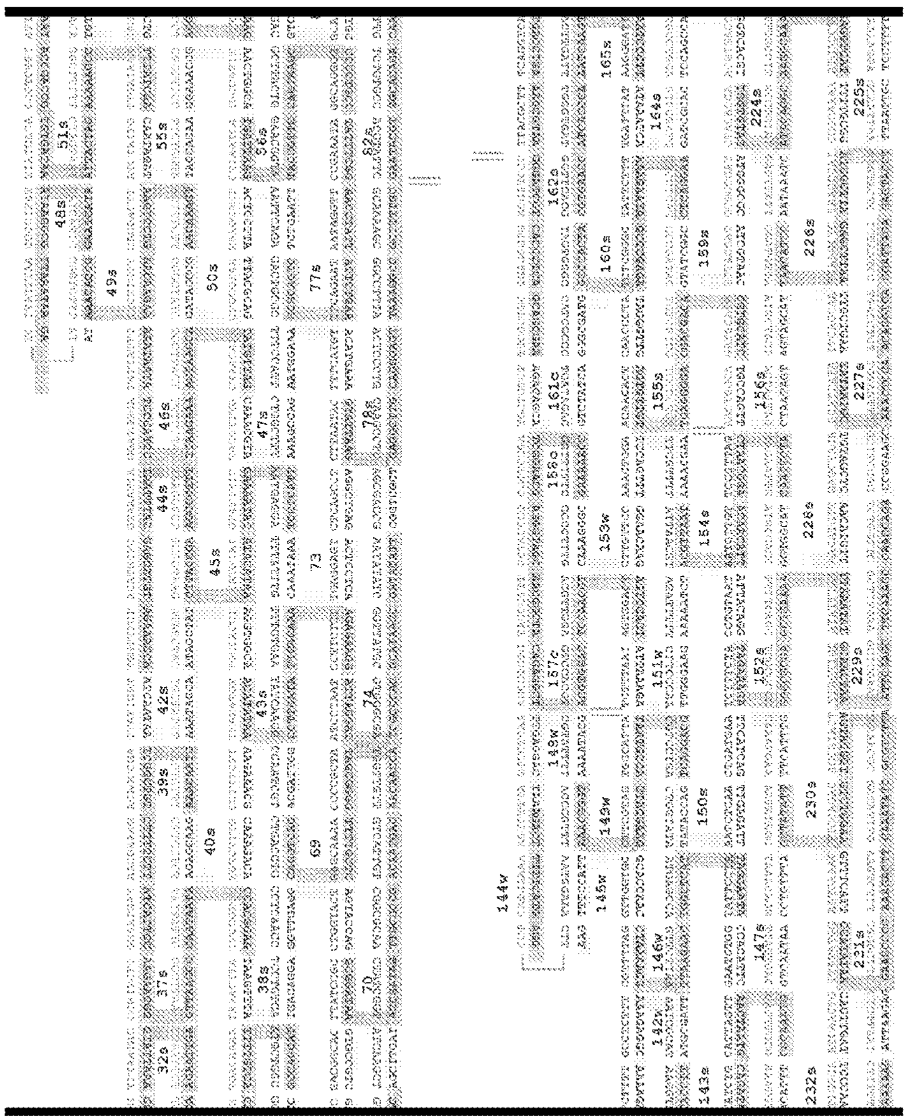


Fig. 5

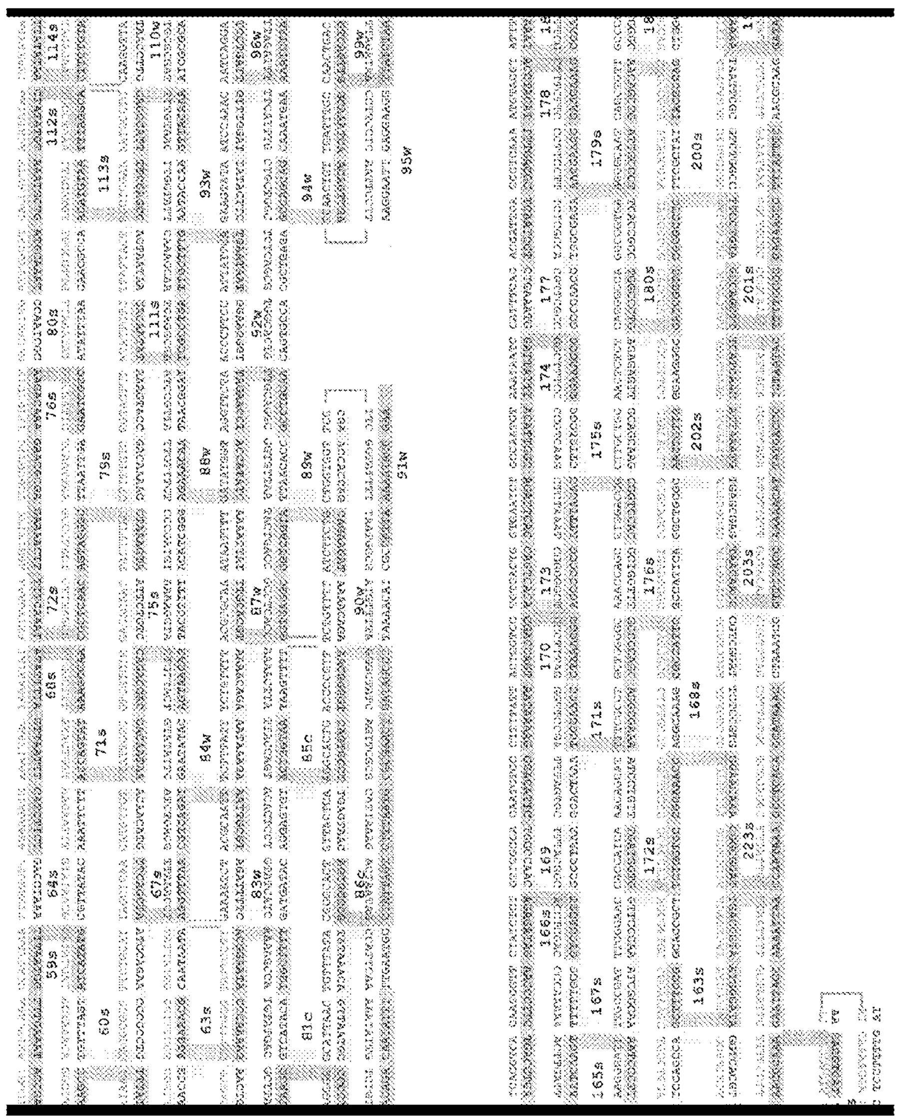


Fig. 6

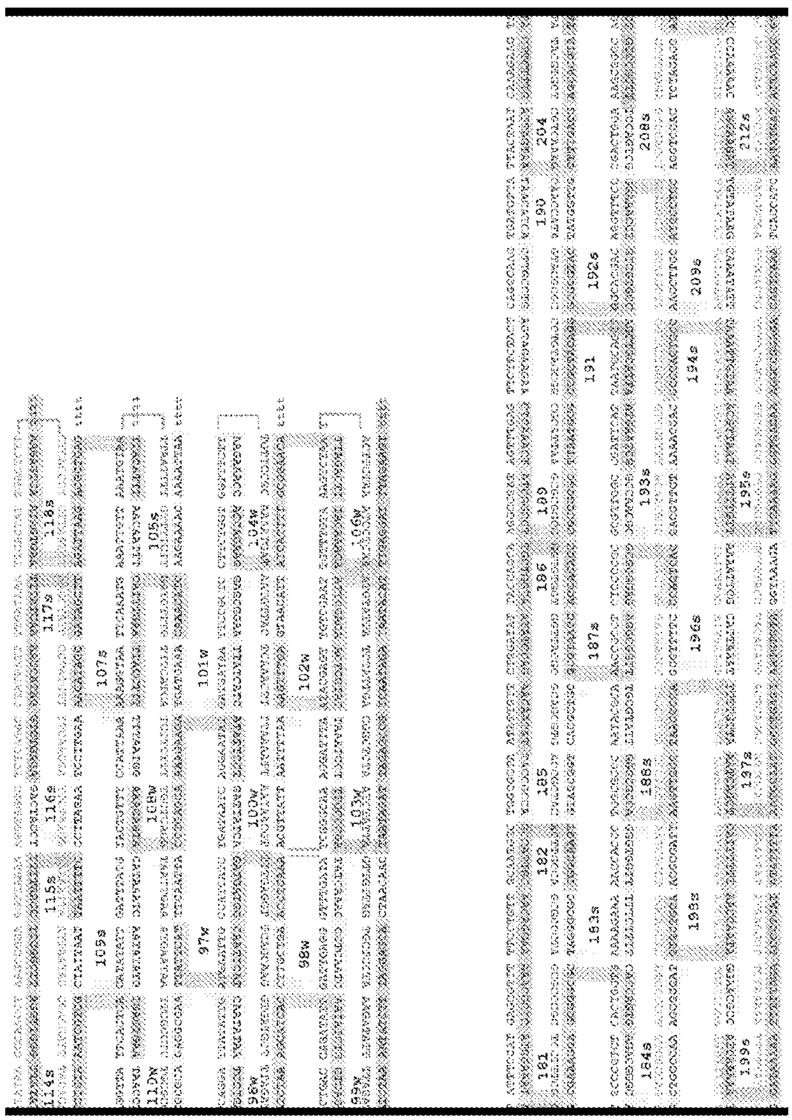


Fig. 7

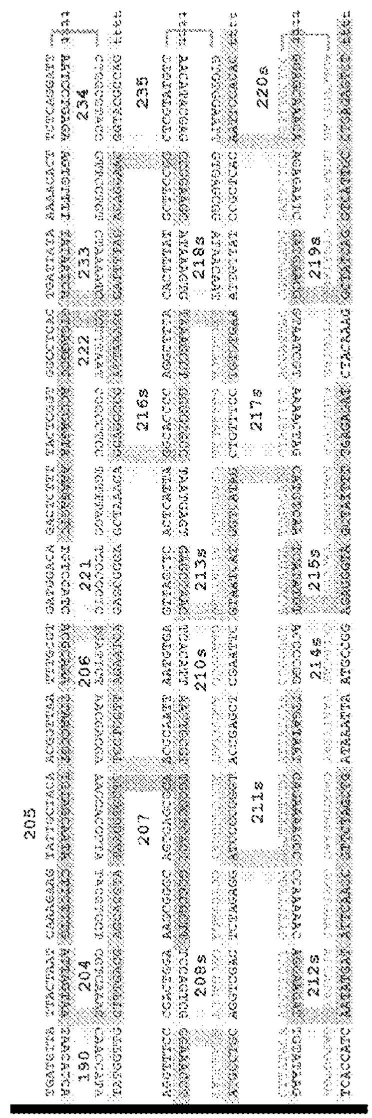

205
190
204
206
207
221
222
233
234
235
208s
210s
211s
213s
216s
217s
218s
219s
220s
212s
214s
215s

Fig. 8

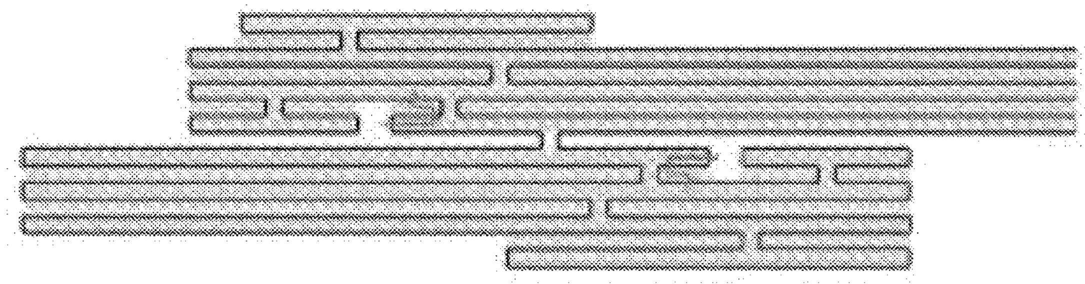


Fig. 9

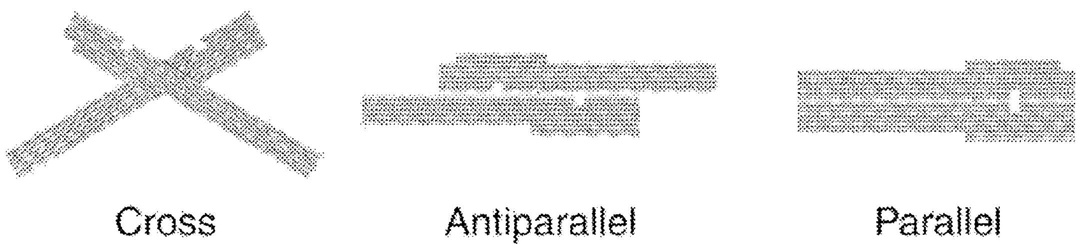


Fig. 10

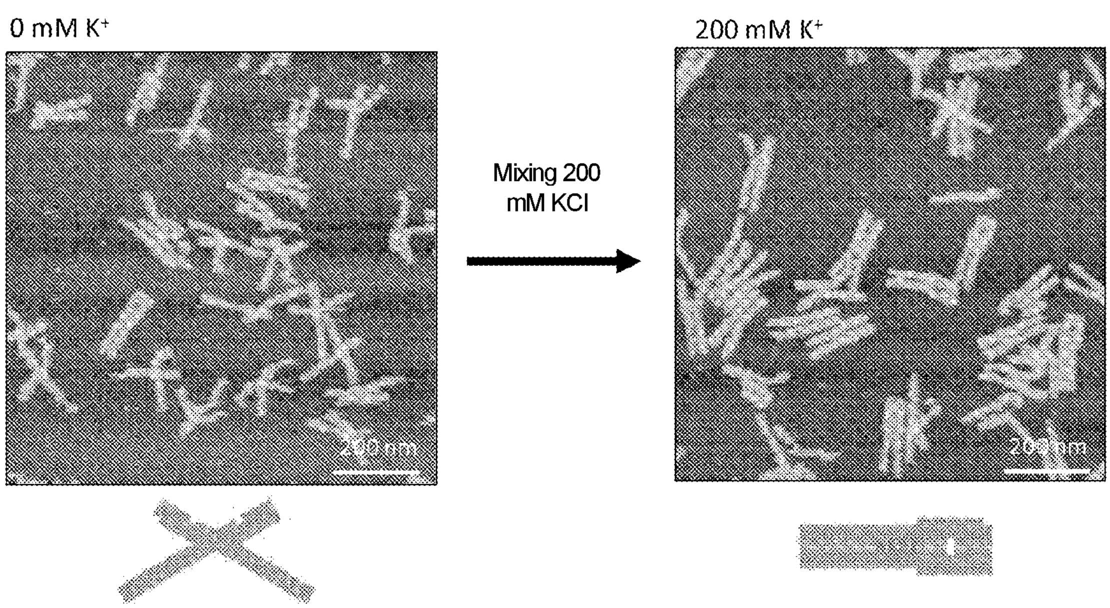


Fig. 11

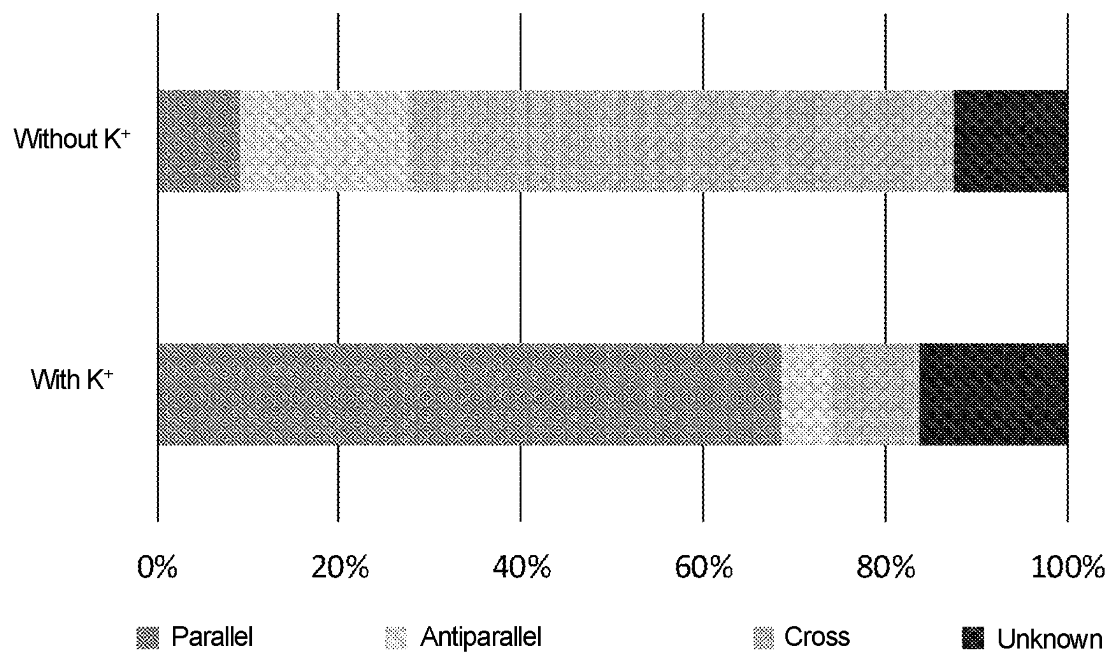

Without K⁺
With K⁺
0%
20%
40%
60%
80%
100%
Parallel
Antiparallel
Cross
Unknown

Fig. 12

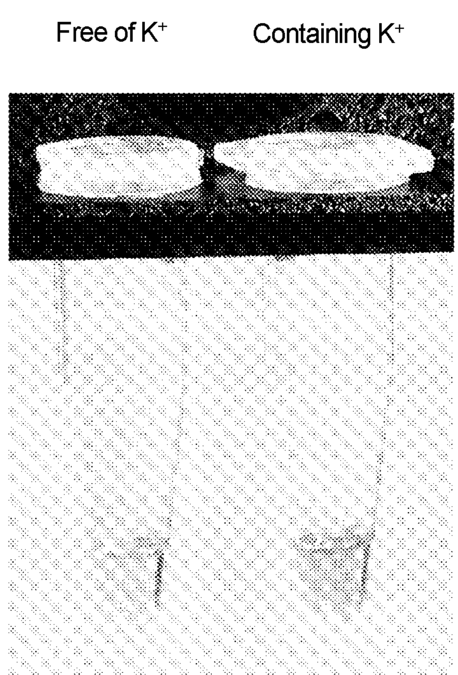

Free of K⁺
Containing K⁺

Fig. 13

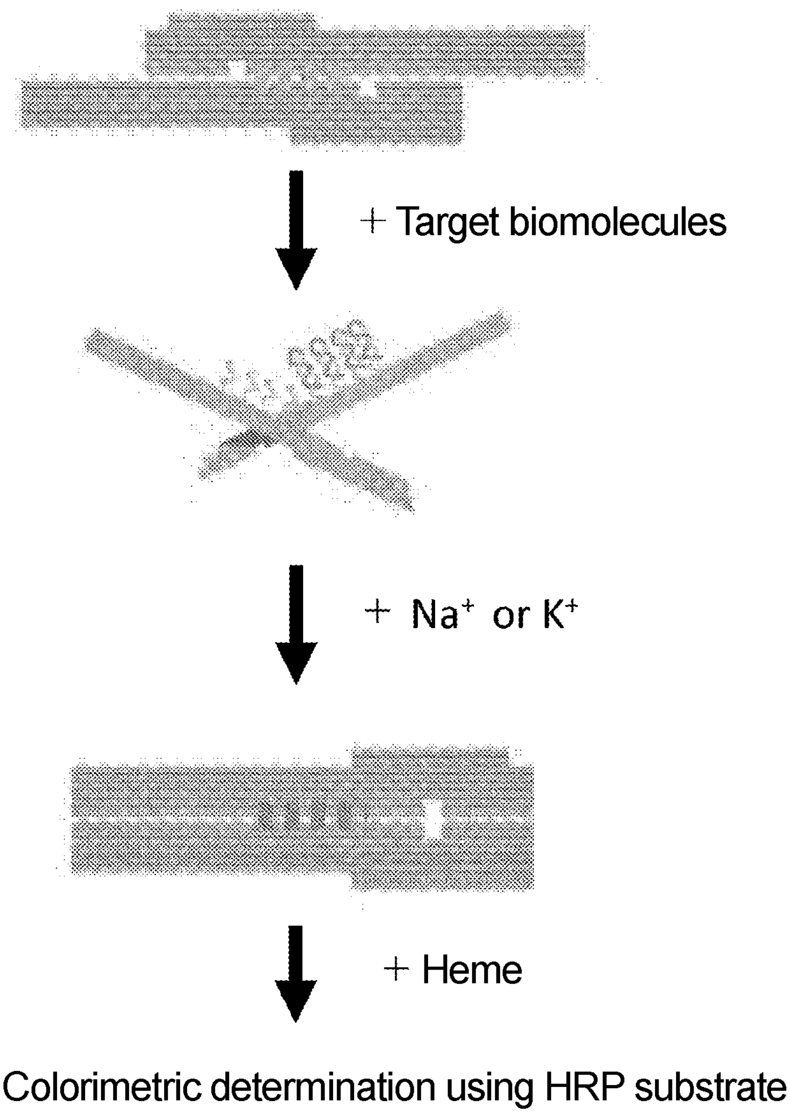

+ Target biomolecules
+ Na⁺ or K⁺
+ Heme
Colorimetric determination using HRP substrate

Fig. 14

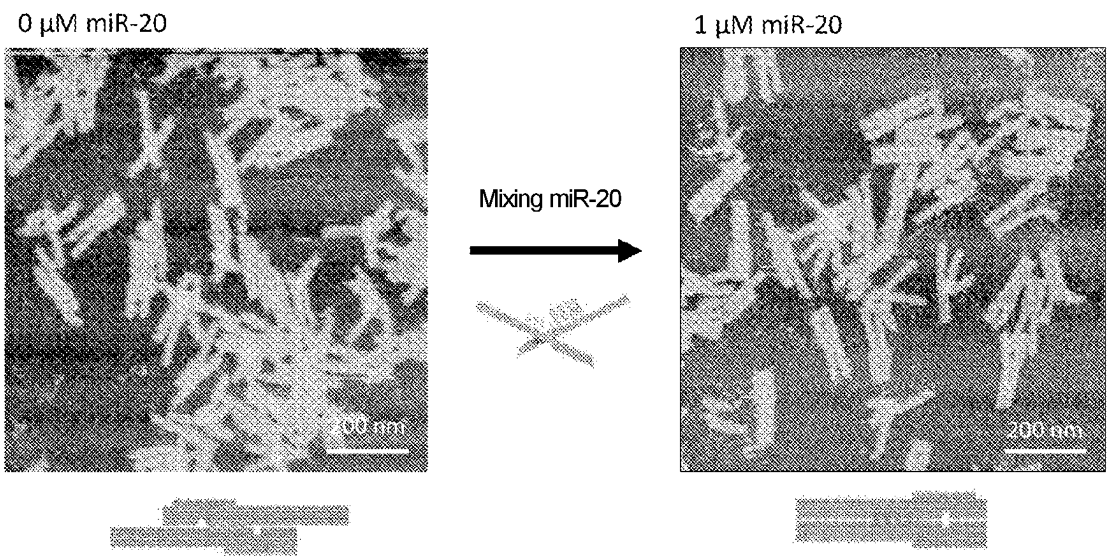

0 μM miR-20
1 μM miR-20
Mixing miR-20
200 nm
200 nm

Fig. 15

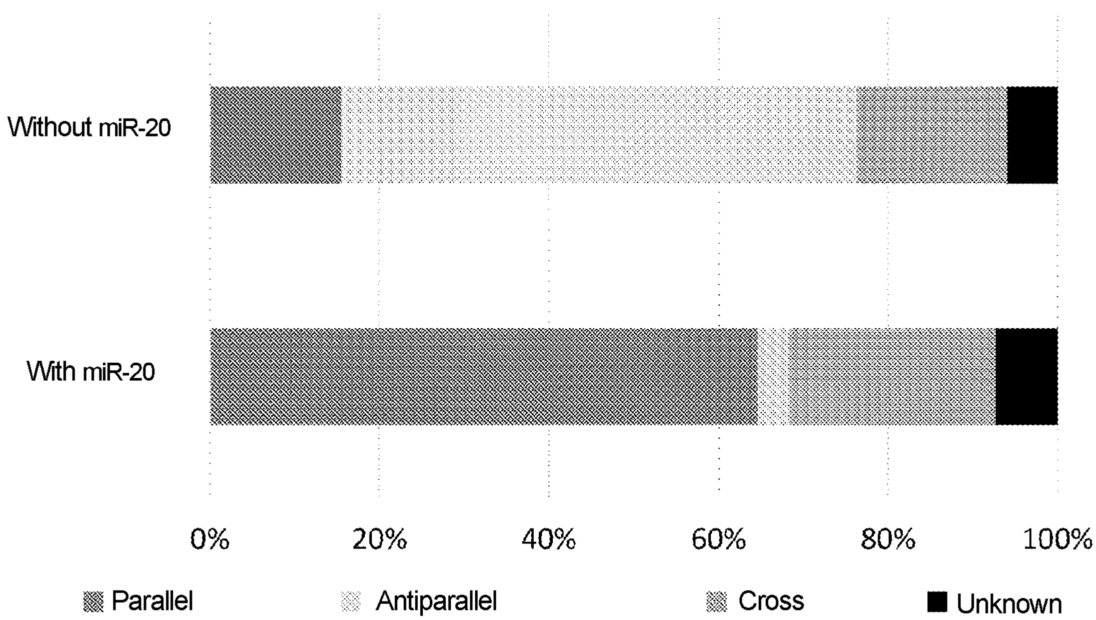

Without miR-20
With miR-20
0%
20%
40%
60%
80%
100%
Parallel
Antiparallel
Cross
Unknown

Fig. 16

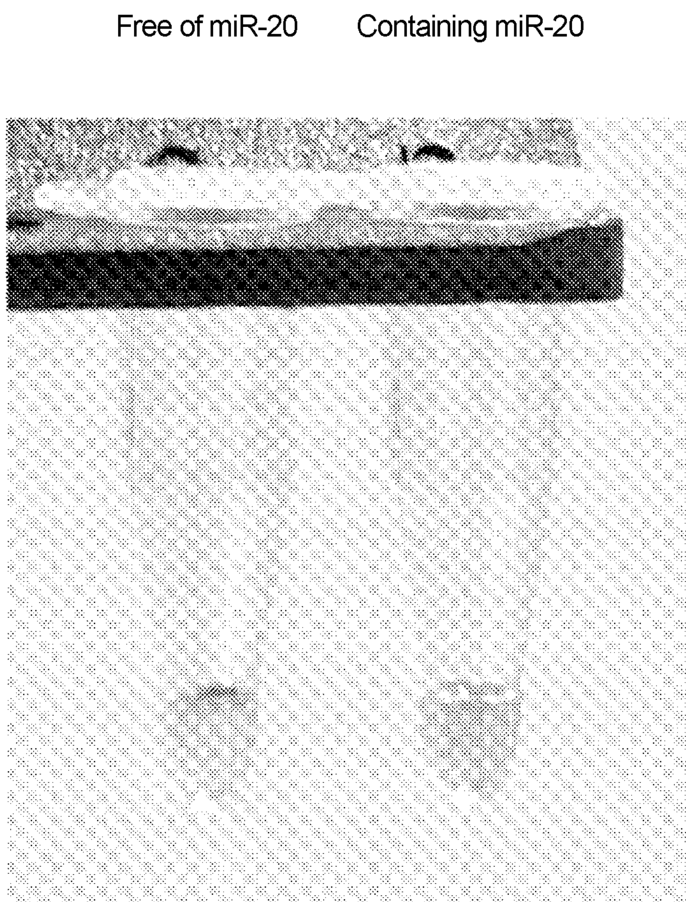


Fig. 17

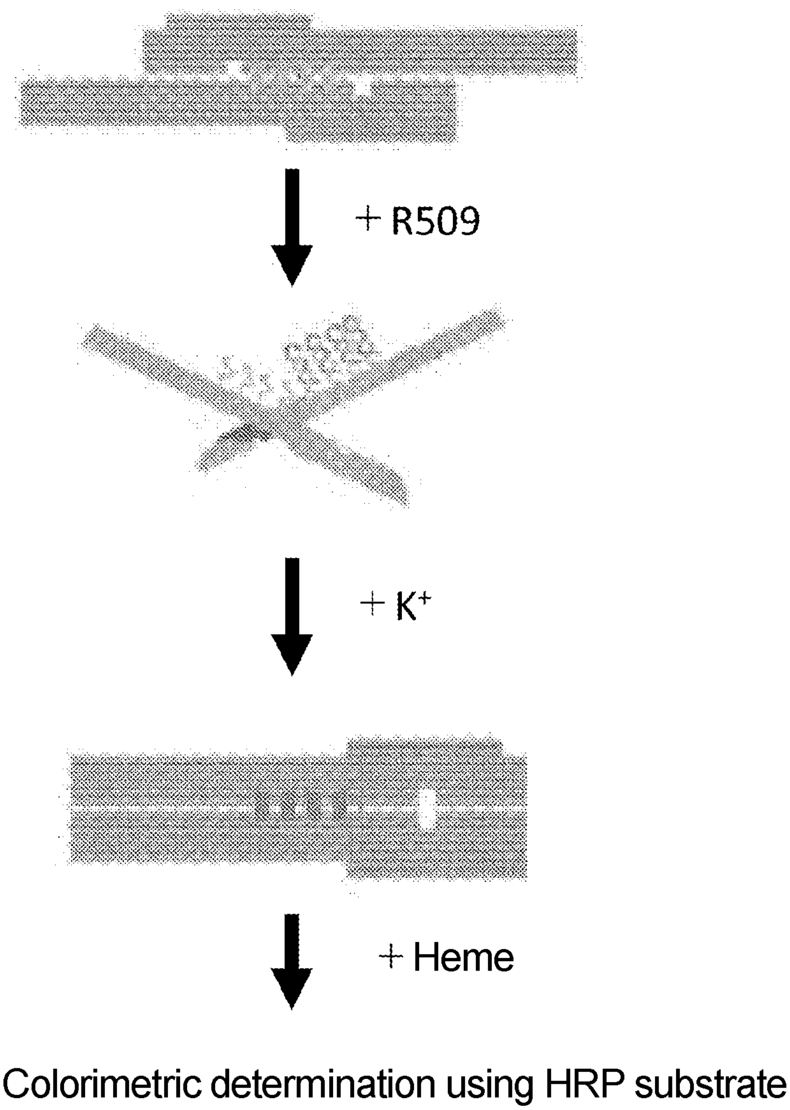

+ R509
+ K+
+ Heme
Colorimetric determination using HRP substrate

Fig. 18

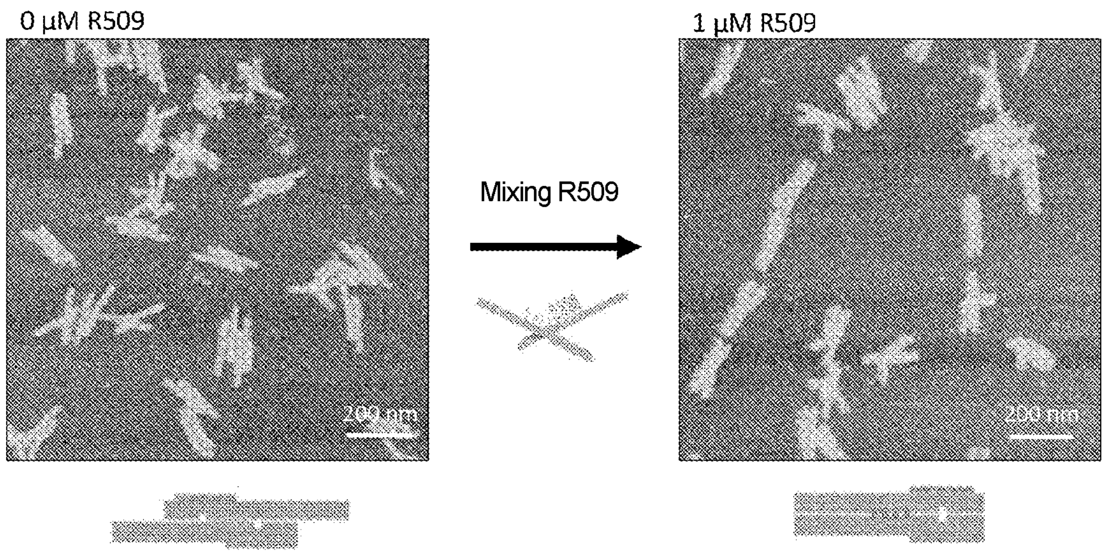

0 μM R509
1 μM R509
Mixing R509
200 nm
200 nm

Fig. 19

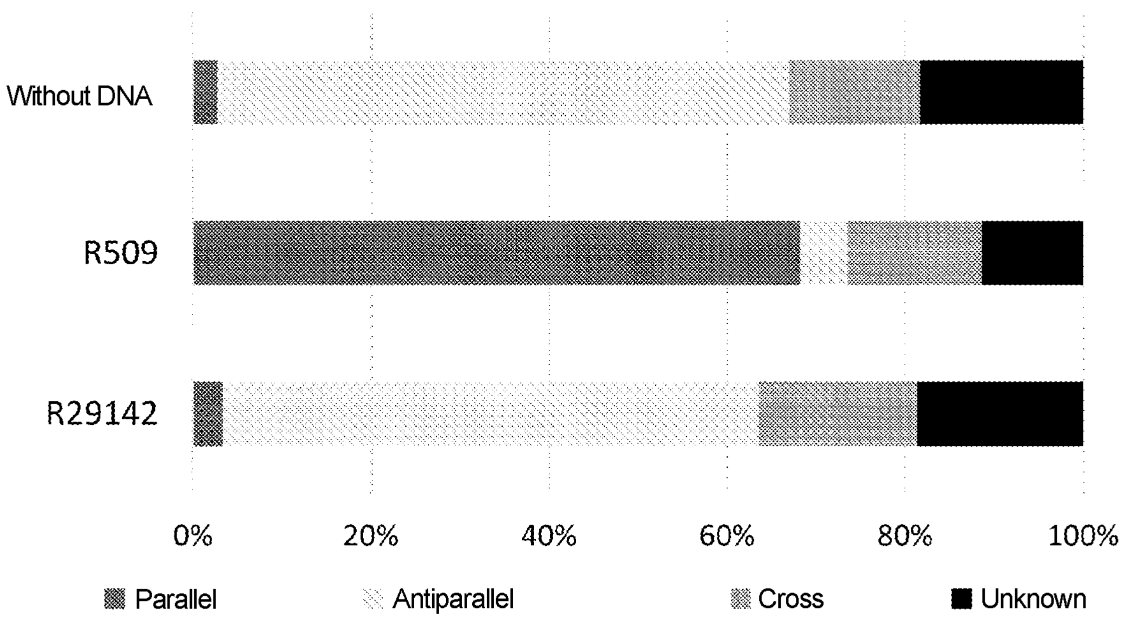

Without DNA
R509
R29142
0%
20%
40%
60%
80%
100%
Parallel
Antiparallel
Cross
Unknown

Fig. 20

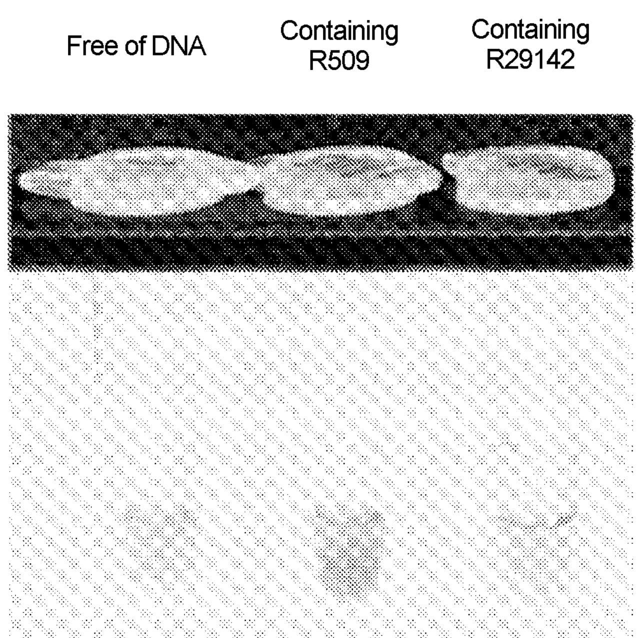


Fig. 21

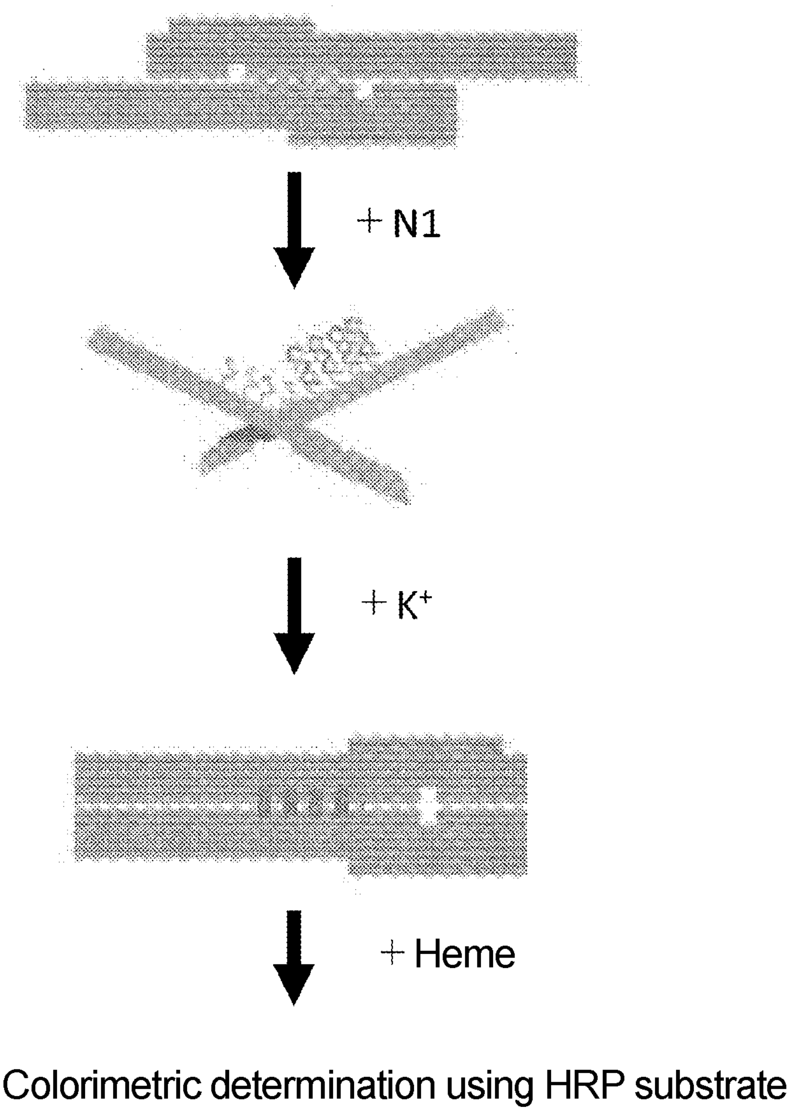


Fig. 22

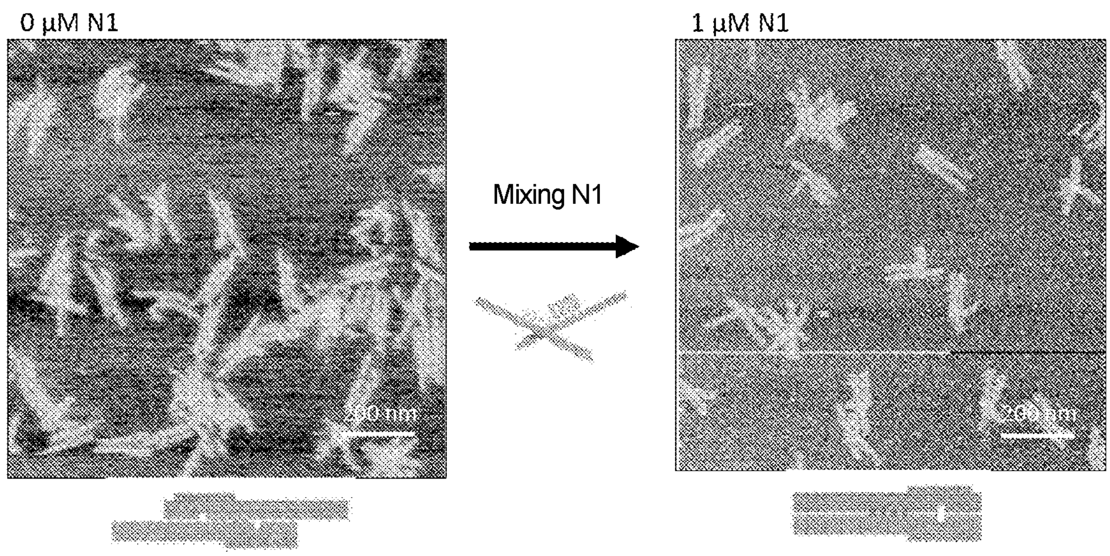

0 μM N1
1 μM N1
Mixing N1
200 nm
200 nm

Fig. 23

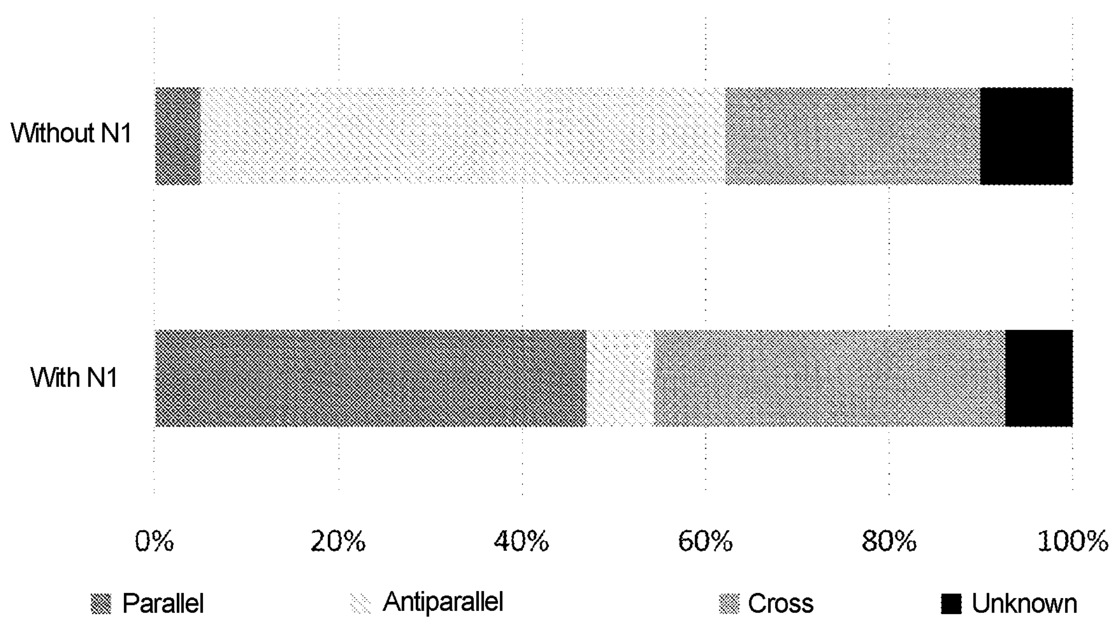

Without N1
With N1
0%
20%
40%
60%
80%
100%
Parallel
Antiparallel
Cross
Unknown

Fig. 24

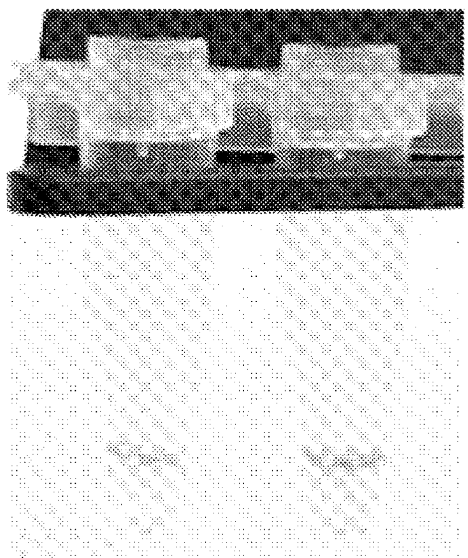


Fig. 25

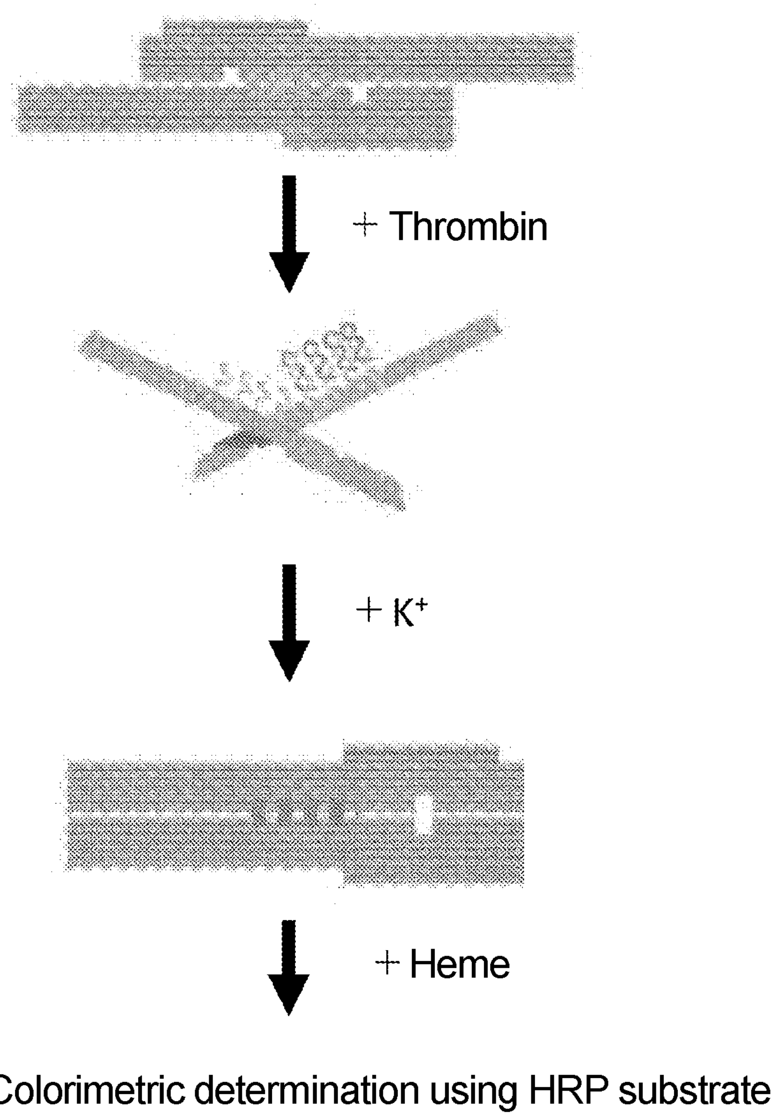

+ Thrombin
+ K+
+ Heme
Colorimetric determination using HRP substrate

Fig. 26

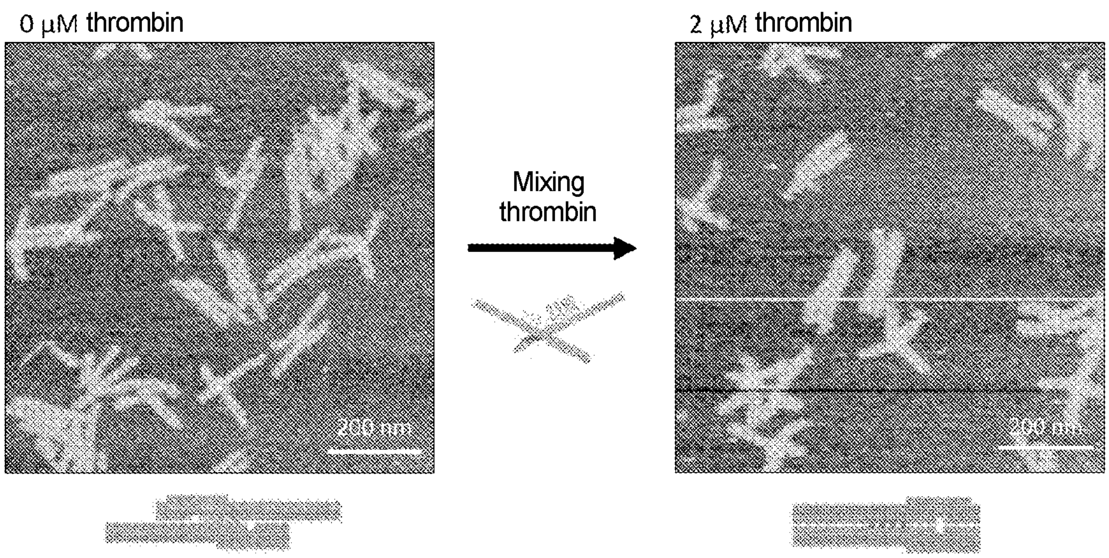

0 μM thrombin
2 μM thrombin
Mixing
thrombin
200 nm
200 nm

Fig. 27

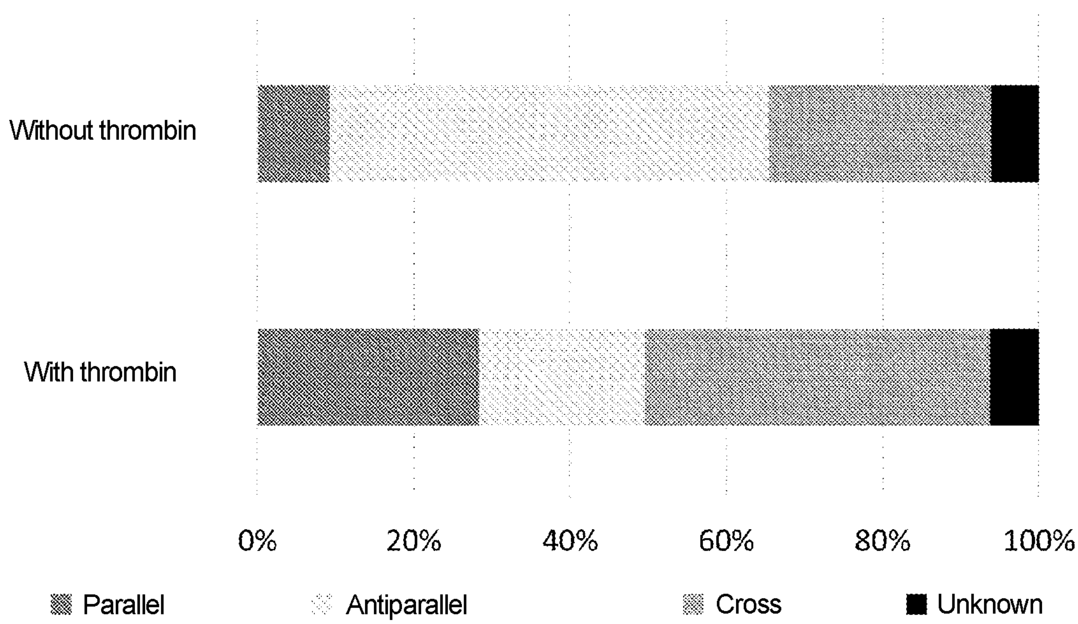

Without thrombin
With thrombin
0%
20%
40%
60%
80%
100%
Parallel
Antiparallel
Cross
Unknown

Fig. 28

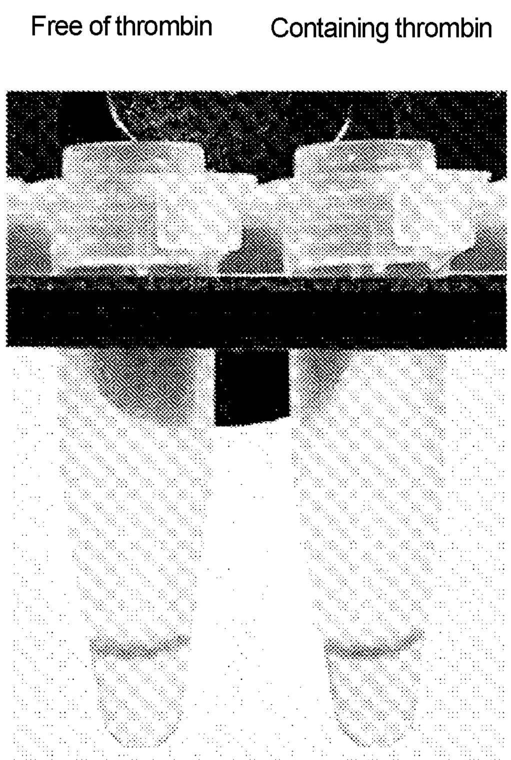


Fig. 29

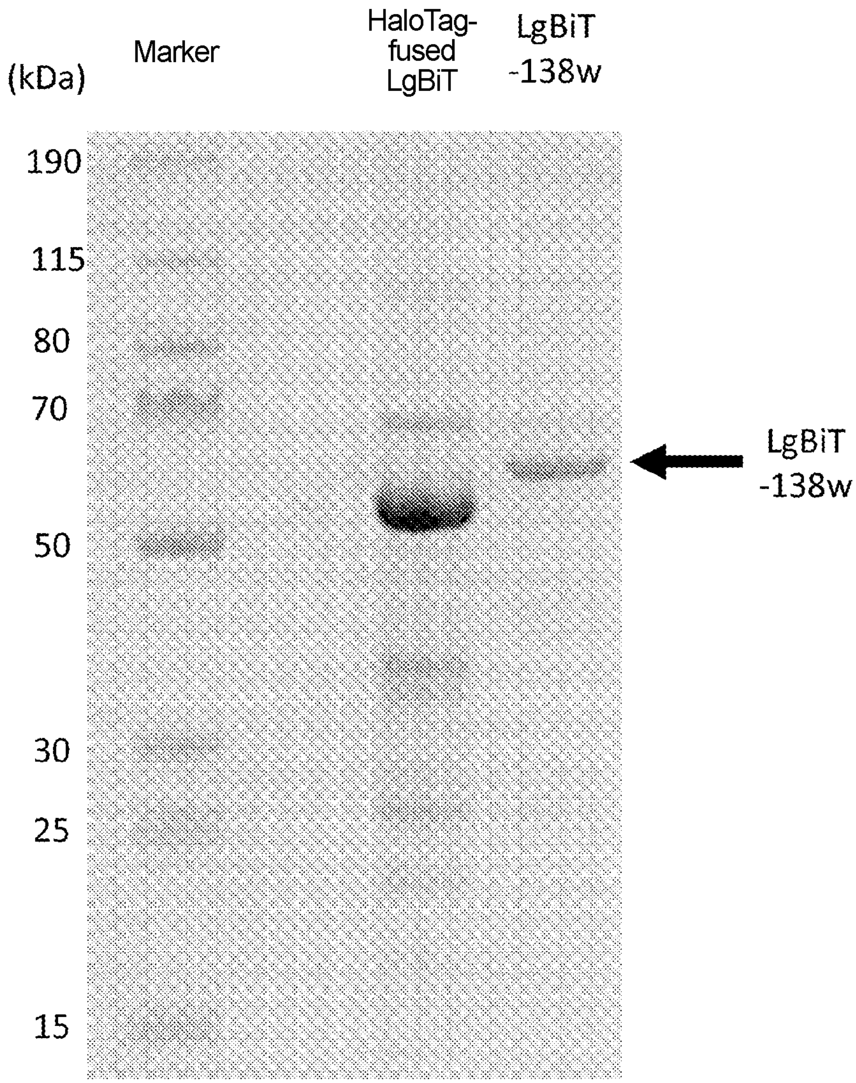

(kDa)
Marker
HaloTag-fused LgBiT
LgBiT -138w
190
115
80
70
50
30
25
15
LgBiT -138w

Fig. 30

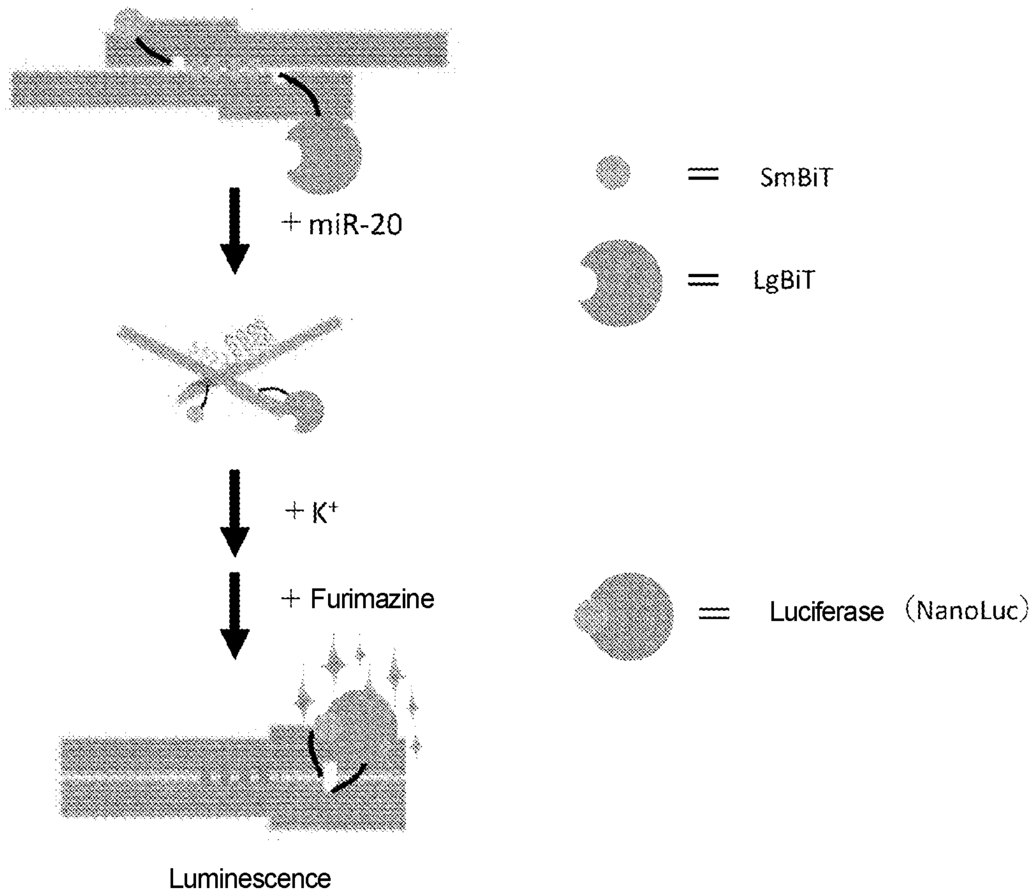

+ miR-20
+ K+
+ Furimazine
Luminescence
= SmBiT
= LgBiT
= Luciferase (NanoLuc)

Fig. 31

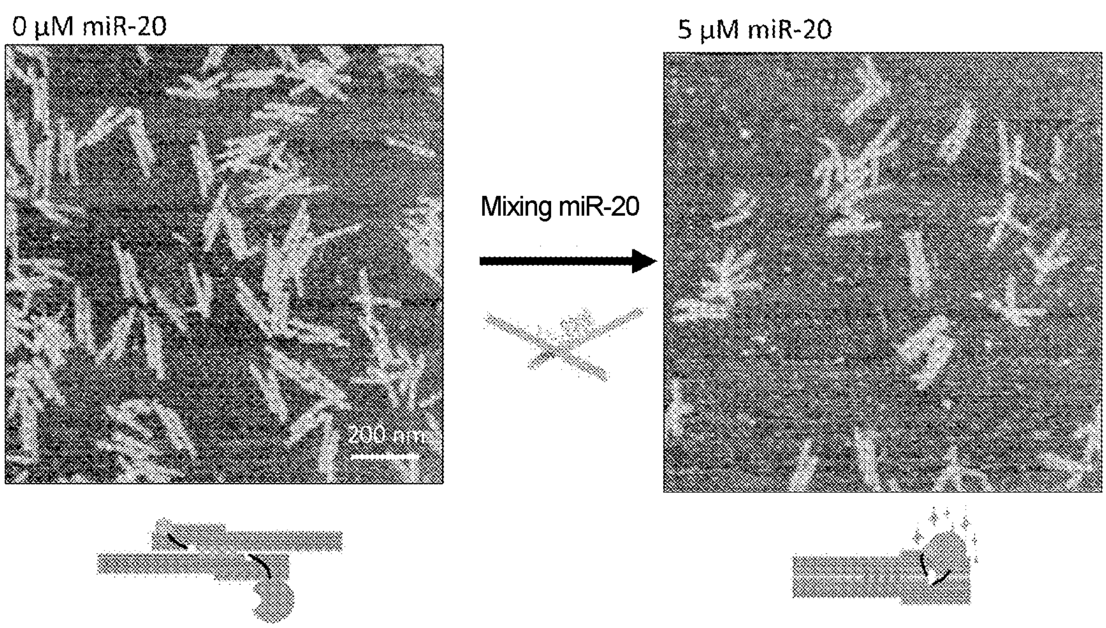

0 μM miR-20
5 μM miR-20
Mixing miR-20
200 nm

Fig. 32

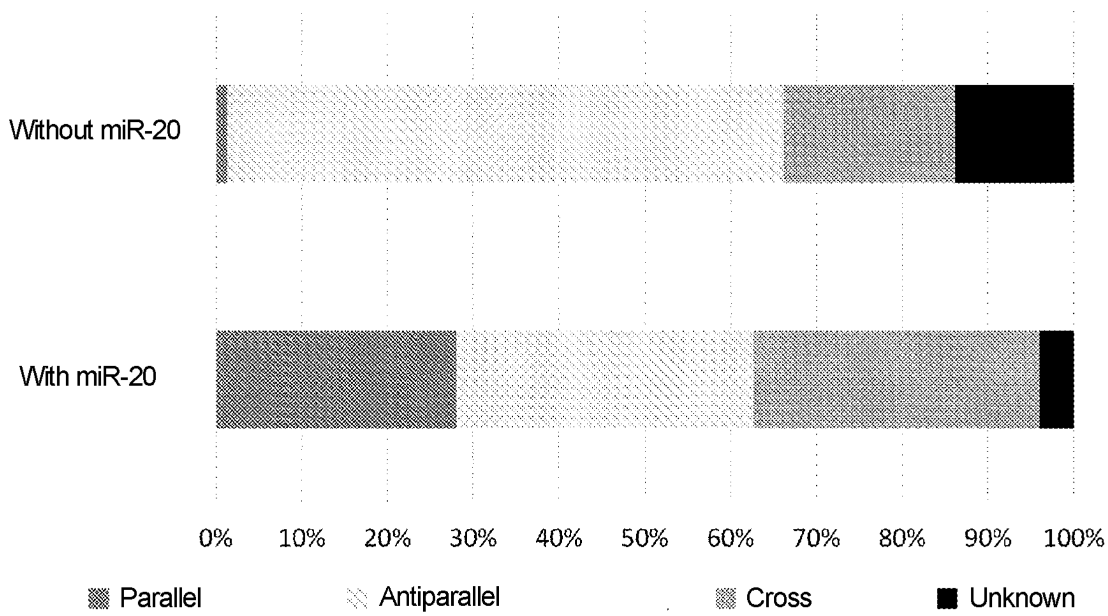
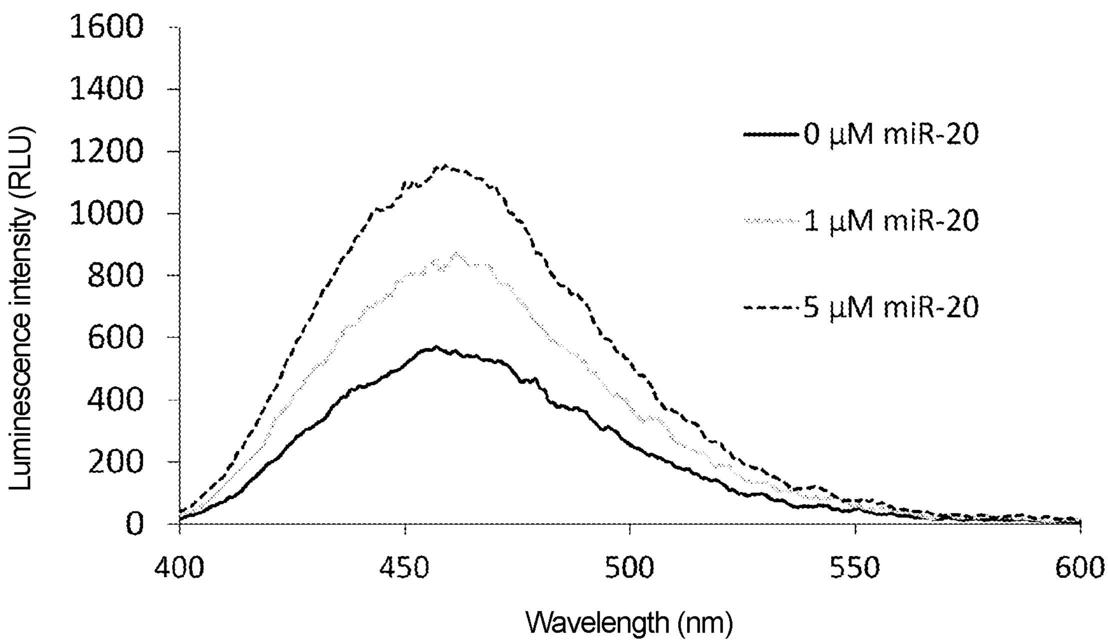

Without miR-20
With miR-20
0%
10%
20%
30%
40%
50%
60%
70%
80%
90%
100%
Parallel
Antiparallel
Cross
Unknown

Fig. 33


Luminescence intensity (RLU)
1600
1400
1200
1000
800
600
400
200
0
0 μM miR-20
1 μM miR-20
5 μM miR-20
400
450
500
550
600
Wavelength (nm)

NUCLEIC ACID STRUCTURE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a U.S. National Stage application of PCT/JP2021/028281 filed 30 Jul. 2021, which claims priority to Japanese application No. 2020-132829 filed 5 Aug. 2020, the entire disclosures of which are hereby incorporated by reference in their entireties.

SEQUENCE LISTING

[0002] The present application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on 11 Jun. 2023, is named 2023-06-11SEQUENCELISTING.txt and is 83 kilobytes in size.

TECHNICAL FIELD

[0003] The present invention relates to a nucleic acid structure and the like.

BACKGROUND ART

[0004] Techniques for detecting nucleic acids, proteins, peptides, low-molecular-weight compounds, metal ions, and the like are used in various fields, such as medical and environmental fields. For example, disease tests may be performed using the presence and amount of specific nucleic acids and other biomolecules as indicators.

[0005] On the other hand, DNA origami is a nucleic acid structure formed by folding single-stranded circular DNA into many short DNA strands (staple DNAs) that can complementarily bind to the DNA. Various structures can be created at will, depending on the design of the sequence of staple DNA. NPL 1 has reported DNA origami (DNA origami pliers (or DNA pliers)) having a structure in which two levers are joined at a fulcrum (Holliday junction). DNA origami pliers include three possible structures (FIG. **9**: antiparallel, cross, and parallel types) based on the conformation of the Holliday junction. So far, there have been reports on techniques for detecting Na+, Ag+, streptavidin, IgG, ATP, and miRNA using structural changes of DNA origami pliers. However, it is necessary for these techniques to fluorescently detect small fluorescence intensity changes based on structural changes, or to detect structural changes by an atomic force microscope.

CITATION LIST

Non-Patent Literature

[0006] NPL 1: Kuzuya, A., (2011) Nat. Commun. doi: 10.1038/ncomms1452

SUMMARY OF INVENTION

Technical Problem

[0007] An object of the present invention is to provide a technique for detecting a test substance more easily using a nucleic acid structure.

Solution to Problem

[0008] As a result of intensive research in view of the above object, the present inventors found that the object can be achieved by a nucleic acid structure comprising a partial nucleic acid structure 1 and a partial nucleic acid structure 2, the partial nucleic acid structure 1 and the partial nucleic acid structure 2 being linked by a Holliday junction, the partial nucleic acid structure 1 comprising a protruding structure 1a containing a nucleic acid aptamer sequence, and a protruding structure 1b containing a split domain 1 of an enzyme, and

[0009] the partial nucleic acid structure 2 comprising a protruding structure 2a containing a sequence complementary to part or all of the nucleic acid aptamer sequence, and a protruding structure 2b containing a split domain 2 paired with the split domain 1.

Upon further research based on this finding, the present inventors have completed the present invention. More specifically, the present invention includes the following embodiments.

[0010] Item 1. A nucleic acid structure comprising a partial nucleic acid structure 1 and a partial nucleic acid structure 2,

[0011] the partial nucleic acid structure 1 and the partial nucleic acid structure 2 being linked by a Holliday junction,

[0012] the partial nucleic acid structure 1 comprising a protruding structure 1a containing a nucleic acid aptamer sequence, and a protruding structure 1b containing a split domain 1 of an enzyme, and

[0013] the partial nucleic acid structure 2 comprising a protruding structure 2a containing a sequence complementary to part or all of the nucleic acid aptamer sequence, and a protruding structure 2b containing a split domain 2 paired with the split domain 1.

[0014] Item 2. The nucleic acid structure according to Item 1, wherein the partial structure 1 and the partial structure 2 each have an elongated shape.

[0015] Item 3. The nucleic acid structure according to Item 1 or 2, which is made of a material containing a single-stranded circular nucleic acid and a staple nucleic acid containing a sequence complementary to the single-stranded circular nucleic acid.

[0016] Item 4. The nucleic acid structure according to any one of Items 1 to 3, wherein the protruding structures are arranged so that, among three structures (structural type 1, structural type 2, and structural type 3: provided that the structural type 2 is an intermediate structure that mediates structural changes between the structural type 1 and the structural type 3) based on the conformation of the Holliday junction, when taking the structural type 1, the protruding structure 1a and the protruding structure 2a face each other, and when taking the structural type 3, the protruding structure 1b and the protruding structure 2b face each other.

[0017] Item 5. The nucleic acid structure according to Item 4, wherein the protruding structures are arranged so that the protruding structure 1a and the protruding structure 2a face each other and bind together based on complementary base pairing in the absence of a recognition substance for the nucleic acid aptamer sequence.

[0018] Item 6. The nucleic acid structure according to Item 4 or 5, wherein the protruding structures are arranged so that the protruding structure 1b and the protruding structure 2b face each other to form the enzyme in the presence of a recognition substance for the nucleic acid aptamer sequence and in the presence of a monovalent cation.

[0019] Item 7. The nucleic acid structure according to any one of Items 1 to 6, wherein the enzyme is a nucleic acid enzyme containing two or more G-quartets.

[0020] Item 8. The nucleic acid structure according to any one of Items 1 to 7, wherein the recognition substance for the nucleic acid aptamer sequence is a biological material.

[0021] Item 9. The nucleic acid structure according to any one of Items 1 to 8, wherein the recognition substance for the nucleic acid aptamer sequence is at least one member selected from the group consisting of nucleic acids, proteins, peptides, low-molecular-weight compounds, physiologically active substances, and metal ions.

[0022] Item 10. The nucleic acid structure according to any one of Items 1 to 9, which is a nano-structure.

[0023] Item 11. A test substance detection composition comprising the nucleic acid structure according to any one of Items 1 to 10.

[0024] Item 12. The test substance detection composition according to Item 11, which is a reagent or a test agent.

[0025] Item 13. A method for detecting a test substance, comprising bringing the nucleic acid structure according to any one of Items 1 to 10 into contact with a sample that may contain the test substance.

Advantageous Effects of Invention

[0026] According to the present invention, a technique for detecting a test substance more easily using a nucleic acid structure can be provided.

BRIEF DESCRIPTION OF DRAWINGS

[0027] FIG. **1** shows the overall view of the nucleic acid structure of Reference Example 1.

[0028] FIG. **2** shows the leftmost diagram among the six diagrams obtained by dividing FIG. **1** into six parts. The sequences of single-stranded circular DNA and staple DNAs, and staple numbers are shown. The marker parts indicate staple DNAs. The black bar indicates the boundary with the adjacent divided diagram.

[0029] FIG. **3** shows the second diagram from the left among the six diagrams obtained by dividing FIG. **1** into six parts. The sequences of single-stranded circular DNA and staple DNAs, and staple numbers are shown. The marker parts indicate staple DNAs. The black bar indicates the boundary with the adjacent divided diagram.

[0030] FIG. **4** shows the third diagram from the left among the six diagrams obtained by dividing FIG. **1** into six parts. The sequences of single-stranded circular DNA and staple DNAs, and staple numbers are shown. The marker parts indicate staple DNAs. The black bar indicates the boundary with the adjacent divided diagram.

[0031] FIG. **5** shows the fourth diagram from the left among the six diagrams obtained by dividing FIG. **1** into six parts. The sequences of single-stranded circular DNA and staple DNAs, and staple numbers are shown. The marker parts indicate staple DNAs. The black bar indicates the boundary with the adjacent divided diagram.

[0032] FIG. **6** shows the fifth diagram from the left among the six diagrams obtained by dividing FIG. **1** into six parts. The sequences of single-stranded circular DNA and staple DNAs, and staple numbers are shown. The marker parts indicate staple DNAs. The black bar indicates the boundary with the adjacent divided diagram.

[0033] FIG. **7** shows the sixth (rightmost) diagram from the left among the six diagrams obtained by dividing FIG. **1** into six parts. The sequences of single-stranded circular DNA and staple DNAs, and staple numbers are shown. The marker parts indicate staple DNAs. The black bar indicates the boundary with the adjacent divided diagram.

[0034] FIG. **8** shows a schematic diagram showing only single-stranded circular DNA in a linear form in the nucleic acid structure of Reference Example 1.

[0035] FIG. **9** shows three structures based on the conformation of the Holliday junction in the nucleic acid structure of Reference Example 2.

[0036] FIG. **10** shows AFM observation images showing structural changes due to the addition of K^+ in the nucleic acid structure of Reference Example 2.

[0037] FIG. **11** shows changes in the proportion of each structure due to the addition of K^+ in the nucleic acid structure of Reference Example 2.

[0038] FIG. **12** shows a photographic image of the detection results in Reference Example 3.

[0039] FIG. **13** shows a schematic diagram of structural changes in the nucleic acid structure of Example 1.

[0040] FIG. **14** shows AFM observation images showing structural changes due to the addition of miR-20 in the nucleic acid structure of Example 1.

[0041] FIG. **15** shows changes in the proportion of each structure due to the addition of miR-20 in the nucleic acid structure of Example 1.

[0042] FIG. **16** shows a photographic image of the detection results in Example 2.

[0043] FIG. **17** shows a schematic diagram of structural changes in the nucleic acid structure of Example 3.

[0044] FIG. **18** shows AFM observation images showing structural changes due to the addition of R509 in the nucleic acid structure of Example 3.

[0045] FIG. **19** shows changes in the proportion of each structure due to the addition of R509 in the nucleic acid structure of Example 3.

[0046] FIG. **20** shows a photographic image of the detection results in Example 4.

[0047] FIG. **21** shows a schematic diagram of structural changes in the nucleic acid structure of Example 5.

[0048] FIG. **22** shows AFM observation images showing structural changes due to the addition of N1 in the nucleic acid structure of Example 5.

[0049] FIG. **23** shows changes in the proportion of each structure due to the addition of N1 in the nucleic acid structure of Example 5.

[0050] FIG. **24** shows a photographic image of the detection results in Example 6.

[0051] FIG. **25** shows a schematic diagram of structural changes in the nucleic acid structure of Example 7.

[0052] FIG. **26** shows AFM observation images showing structural changes due to the addition of thrombin in the nucleic acid structure of Example 7.

[0053] FIG. **27** shows changes in the proportion of each structure due to the addition of thrombin in the nucleic acid structure of Example 7.

[0054] FIG. **28** shows a photographic image of the detection results in Example 8.

[0055] FIG. **29** shows an electrophoretic image of LgBiT and LgBiT-138w of Example 9.

[0056] FIG. **30** shows a schematic diagram of structural changes in the nucleic acid structure of Example 9.

[0057] FIG. 31 shows AFM observation images showing structural changes due to the addition of miR-20 in the nucleic acid structure of Example 9.

[0058] FIG. 32 shows changes in the proportion of each structure due to the addition of miR-20 in the nucleic acid structure of Example 9.

[0059] FIG. 33 shows the detection results in Example 10.

DESCRIPTION OF EMBODIMENTS

1. Definition

[0060] In the present specification, the terms "comprise" and "contain" include the concepts of "comprising," "containing," "consisting essentially of," and "consisting of."

[0061] In the present specification, the "identity" of base sequences refers to the degree of matching between two or more comparable base sequences. Therefore, the higher the match between two base sequences, the higher the identity or similarity of those sequences. The level of identity between base sequences is determined, for example, using a sequence analysis tool, FASTA, with default parameters. Alternatively, the identity level can be determined using the algorithm BLAST by Karlin and Altschul (Karlin S, Altschul SF. "Methods for assessing the statistical significance of molecular sequence features by using general scoring schemes" Proc Natl Acad Sci USA. 87:2264-2268 (1990), Karlin S, Altschul SF. "Applications and statistics for multiple high-scoring segments in molecular sequences." Proc Natl Acad Sci USA. 90:5873-7 (1993)). A program called BLASTX has been developed based on the BLAST algorithm. Specific methods of these analysis methods are known, and reference may be made to the website of the National Center of Biotechnology Information (NCBI) (http://www.ncbi.nlm.nih.gov/).

[0062] In the present specification, nucleic acids, such as DNA and RNA, may be subjected to known chemical modification, as shown below. In order to prevent degradation by hydrolases, such as nucleases, the phosphate residue (phosphate) of each nucleotide can be replaced by a chemically modified phosphate residue, such as phosphorothioate (PS), methylphosphonate, or phosphorodithionate. The hydroxyl group at position 2 of the sugar (ribose) of each ribonucleotide may be replaced by —OR (R is, for example, CH3(2'-O-Me), CH2CH2OCH3(2'-O-MOE), CH2CH2NHC(NH)NH2, CH2CONHCH3, or CH2CH2CN). Further, a base moiety (pyrimidine, purine) may be chemically modified, and examples include introduction of a methyl group or a cationic functional group into position 5 of the pyrimidine base, substitution of the carbonyl group at position 2 with thiocarbonyl, and the like. Other examples include, but are not limited to, those in which the phosphate moiety or hydroxyl moiety is modified with biotin, an amino group, a lower alkylamine group, an acetyl group, or the like. In addition, BNA (LNA), in which the conformation of the sugar moiety is fixed at the N-type by bridging the 2' oxygen and 4' carbon of the sugar moiety of the nucleotide, can also be used.

2. Nucleic Acid Structure

[0063] The present invention relates to a nucleic acid structure comprising a partial nucleic acid structure 1 and a partial nucleic acid structure 2,

[0064] the partial nucleic acid structure 1 and the partial nucleic acid structure 2 being linked by a Holliday junction,

[0065] the partial nucleic acid structure 1 comprising a protruding structure 1a containing a nucleic acid aptamer sequence, and a protruding structure 1b containing a split domain 1 of an enzyme, and

[0066] the partial nucleic acid structure 2 comprising a protruding structure 2a containing a sequence complementary to part or all of the nucleic acid aptamer sequence, and a protruding structure 2b containing a split domain 2 paired with the split domain 1 (also referred to as "the nucleic acid structure of the present invention" in the present specification). This is described below.

[0067] The nucleic acid structure is not particularly limited as long as it mainly contains a nucleic acid. The content of the nucleic acid in the nucleic acid structure is, for example, 70 mass % or more, preferably 80 mass % or more, more preferably 90 mass % or more, even more preferably 95 mass % or more, still more preferably 98 mass % or more, still even more preferably 99 mass % or more, and particularly preferably 100 mass %. Substances other than the nucleic acid in the nucleic acid structure are not particularly limited, and examples include proteins, peptides, sugars, labeling substances, metal ions, and the like.

[0068] The nucleic acid structure may be a three-dimensional structure, a two-dimensional structure (planer structure), or a combination thereof.

[0069] The nucleic acid structure is preferably formed by folding a nucleic acid. In terms of being suitable for the formation of the nucleic acid structure of the present invention, the nucleic acid structure is preferably a structure (DNA origami) made of a material containing a single-stranded circular nucleic acid and a staple nucleic acid containing a sequence complementary to the single-stranded circular nucleic acid.

[0070] The single-stranded circular nucleic acid that can form the nucleic acid structure is not particularly limited as long as it can be folded to form DNA origami. A single-stranded circular nucleic acid that may be composed of any base sequence can form DNA origami by designing the sequence of staple DNA according thereto. The number of bases in the single-stranded circular nucleic acid is not particularly limited, and is, for example, 500 to 50000, preferably 1000 to 25000, more preferably 2000 to 15000, even more preferably 3000 to 12000, and still more preferably 4000 to 10000. The base sequence of the single-stranded circular nucleic acid is not particularly limited, and is, for example, a base sequence containing a M13 bacteriophage DNA sequence or a sequence derived from the DNA sequence. Specific examples of the base sequence include a base sequence containing a base sequence represented by SEQ ID NO: 273, and a base sequence containing a base sequence having 70% or more (preferably 80% or more, more preferably 90% or more, even more preferably 95% or more, still more preferably 98% or more, and still even more preferably 99% or more) identity to the base sequence represented by SEQ ID NO: 273.

[0071] The staple nucleic acid that can form the nucleic acid structure is not particularly limited as long as it contains a sequence complementary to the single-stranded circular nucleic acid, and can fold the single-stranded circular nucleic acid by complementary base pairing with the single-stranded circular nucleic acid to form DNA origami. The number of bases in the staple nucleic acid is not particularly limited, and is, for example, 10 to 500, preferably 20 to 250, and more preferably 30 to 100. In the staple nucleic acid, the number of bases in the sequence complementary to the single-stranded circular nucleic acid is not particularly limited, and is, for example, 10 to 500, preferably 20 to 250, and more preferably 30 to 100. Like the staple DNAs indicated by markers in FIGS. **2** to **7**, one staple nucleic acid generally contains sequences complementary to a plurality of (e.g., 2 to 5, preferably 2 to 4, and more preferably 2 or 3) adjacent strands that are present when a single-stranded circular nucleic acid is folded. A staple nucleic acid having such sequence characteristics can further stabilize the single-stranded circular nucleic acid in a folded state through complementary base pairing with the single-stranded circular nucleic acid. However, in the nucleic acid structure of the present invention, two staple nucleic acids involved in Holliday junction formation (82s and 162s in Reference Example 1, provided later) exceptionally, rather than forming a complementary base pair across a plurality of adjacent strands that are present when the single-stranded circular nucleic acid is folded, preferably form a complementary base pair with only one of these strands. Various DNA origami structures have been reported so far, and the sequence of staple nucleic acid for forming a nucleic acid structure of the desired shape and configuration can be designed according to known methods and information.

[0072] The size of the nucleic acid structure is not particularly limited, and can be suitably adjusted depending on the shape, size, etc. of the test substance. The nucleic acid structure is preferably nano-sized (nano-structure). For example, the longest diameter (major diameter) of the nucleic acid structure is 1000 nm or less. In an embodiment of the present invention, the major diameter of the nucleic acid structure is, for example, 5 to 1000 nm, preferably 20 to 800 nm, more preferably 50 to 600 nm, and even more preferably 100 to 400 nm.

[0073] The nucleic acid structure contains a partial nucleic acid structure 1 and a partial nucleic acid structure 2. The partial nucleic acid structure 1 and the partial nucleic acid structure 2 are each part of the above nucleic acid structure, and linked to each other by a Holliday junction (e.g., FIG. **1**). That is, the partial nucleic acid structure 1 has two of four arms that form the Holliday junction, and the partial nucleic acid structure 2 has the other two arms. In Reference Example 1, provided later, in the partial nucleic acid structure 1, double-stranded DNA formed by staple DNA with staple number 82s and a single-stranded circular nucleic acid forms two arms, and in the partial nucleic acid structure 2, double-stranded DNA formed by staple DNA with staple number 162s and a single-stranded circular nucleic acid forms the other two arms.

[0074] The nucleic acid structure can take three structures (structural type 1, structural type 2, and structural type 3) based on the conformation of the Holliday junction. The structural type 2 is an intermediate structure that mediates structural changes between the structural type 1 and the structural type 3. That is, the structural type 1 can be structurally changed to the structural type 3 via the structural type 2, and the structural type 3 can be structurally changed to the structural type 1 via the structural type 2 (e.g., FIG. **13**). As in the Reference Examples and Examples provided later, when the structural type 1 and the structural type 3 differ in the shape of the nucleic acid structure, as an embodiment, the structural type 1 can be expressed as antiparallel type (or parallel type), and the structural type 3 can be expressed as parallel type (or antiparallel type). The structural type 2 can be expressed as cross type because the partial nucleic acid structure 1 and the partial nucleic acid structure 2 are generally fixed in a crossed state.

[0075] The shape of the partial structure 1 and partial structure 2 is not particularly limited. Examples of the shape include rectangle, triangle, polygon, ellipse, fan, polyhedron, cone, column, ellipse, etc., or a combination of these shapes. In an embodiment of the present invention, the shape is preferably elongated (e.g., the longest diameter is, for example, 1.5 times or more, 2 times or more, or 4 times or more the shortest diameter (excluding the thickness)).

[0076] The size of the partial structure 1 and partial structure 2 is not particularly limited, and can be suitably adjusted depending on the shape, size, etc. of the test substance. The partial structure 1 and partial structure 2 are each preferably nano-sized (nano-structure). For example, the longest diameter (major diameter) of each of the partial structure 1 and partial structure 2 is 1000 nm or less. In an embodiment of the present invention, the major diameter of each of the partial structure 1 and partial structure 2 is, for example, 5 to 1000 nm, preferably 20 to 800 nm, more preferably 50 to 600 nm, and even more preferably 100 to 400 nm.

[0077] The partial nucleic acid structure 1 contains a protruding structure 1a containing a nucleic acid aptamer sequence, and the partial nucleic acid structure 2 contains a protruding structure 2a containing a sequence complementary to part or all of the nucleic acid aptamer sequence.

[0078] The nucleic acid aptamer sequence is not particularly limited as long as it is a nucleic acid sequence that can bind (preferably specifically bind) to a substance (recognition substance). The recognition substance is not particularly limited as long as it can bind to the nucleic acid aptamer, and examples include nucleic acids, proteins, peptides, low-molecular-weight compounds, physiologically active substances, metal ions, and the like. In an embodiment of the present invention, the recognition substance is preferably a biological material. When the recognition substance is a nucleic acid, the nucleic acid aptamer sequence contains a sequence complementary to the nucleic acid, which is a recognition substance. The nucleic acid aptamer sequence can be a nucleic acid aptamer sequence known for a substance, or a sequence obtained by screening using a known method (e.g., SELEX method). The number of bases in the nucleic acid aptamer sequence is not particularly limited, and is, for example, 5 to 500, preferably 10 to 250, and more preferably 20 to 100.

[0079] The sequence complementary to part or all of the nucleic acid aptamer sequence is not particularly limited, but is preferably a sequence complementary to part of the nucleic acid aptamer sequence (e.g., 70% or less, 60% or less, or 50% or less of the nucleic acid aptamer sequence).

[0080] The protruding structure 1a and the protruding structure 2a are not particularly limited as long as they contain a nucleic acid aptamer sequence or a sequence complementary to part or all of the nucleic acid aptamer sequence, and protrude in the partial nucleic acid structure 1 and partial nucleic acid structure 2 (e.g., not incorporated into the structure due to complementary pairing etc.). For example, the protruding structure 1a and protruding structure 2a are part of the staple nucleic acid. In this example, the staple nucleic acid contains a sequence complementary to the single-stranded circular nucleic acid, and a nucleic acid aptamer sequence or a sequence complementary to part or all of the nucleic acid aptamer sequence, and the latter forms a protruding structure consisting of a single-stranded nucleic acid without complementarily binding to the single-stranded circular nucleic acid.

[0081] The number of protruding structures 1a is not particularly limited, and is, for example, 1 to 30, 2 to 20, 3 to 10, or 2 to 7. The number of protruding structures 2a is not particularly limited, and is, for example, 1 to 30, 2 to 20, 3 to 10, or 2 to 7.

[0082] The partial nucleic acid structure 1 contains a protruding structure 1b containing a split domain 1 of an enzyme, and the partial nucleic acid structure 2 contains a protruding structure 2b containing a split domain 2 paired with the split domain 1.

[0083] The enzyme used in the present invention is not particularly limited as long as it is a nucleic acid or protein that itself has enzymatic activity or that can exhibit enzymatic activity by binding to other substances, such as cofactors, and as long as it can be split into two (i.e., even if split, it can reassociate and exhibit enzymatic activity). The enzyme is preferably a nucleic acid enzyme containing two or more G-quartets, a photoprotein, or the like. The G-quartet is a planar structure formed by four guanine bases through Hoogsteen hydrogen bonding. Two or more of these are stacked to form a higher-order structure known as a guanine quadruplex. The higher-order structure can exhibit peroxidase activity in the presence of heme. Examples of photoproteins include NanoLuc (Promega). NanoLuc is a luciferase with furimazine as a substrate, and its luminescence intensity is very strong, about 100 times the luminescence intensity of firefly luciferase. LgBiT (Promega) and SmBiT (Promega) are subunits of NanoLuc, and they are reconstituted as a luciferase by interacting with each other. LgBiT is a large subunit of approximately 18 kDa, and SmBIT is a small subunit of 11 amino acid residues. LgBiT and SmBiT have a dissociation constant of 190 μM and extremely low affinity, and they do not express luciferase activity when simply mixed (Andrew, D., (2016) ACS Chem. Biol. 11, 400-408).

[0084] The split domain 1 and the split domain 2 are obtained by dividing an enzyme into two. The split position is not particularly limited as long as these domains can reassociate and exhibit enzymatic activity. When the enzyme is a nucleic acid enzyme containing two or more G-quartets, the split domain 1 (split nucleic acid sequence 1) and split domain 2 (split nucleic acid sequence 2) are not particularly limited as long as they are nucleic acid sequences that can associate to form G-quartets in the presence of a monovalent cation (e.g., potassium ion or sodium ion). Examples of such nucleic acid sequences include nucleic acid sequences in which a structural unit consisting of two or more (e.g., 2 to 4, or 2 or 3) consecutive guanine bases and a structural unit consisting of one or more (e.g., 1 to 5, or 2 or 3) consecutive non-guanine bases are alternately arranged. Typical examples of such nucleic acid sequences include telomere element sequences. When the enzyme is a photoprotein such as NanoLuc, the split domain 1 (split peptide sequence 1) and split domain 2 (split peptide sequence 2) are not particularly limited as long as they are peptide sequences that can express luciferase activity under conditions that allow both domains to remain stable in close proximity. The split peptide sequences can be determined pursuant to or according to known information, depending on the type of photoprotein, the desired dissociation constant, etc.

[0085] The protruding structure 1b and the protruding structure 2b are not particularly limited as long as they contain a split domain 1 or a split domain 2, and protrude in the partial nucleic acid structure 1 and the partial nucleic acid structure 2 (e.g., not incorporated into the structure due to complementary pairing etc.). For example, the protruding structure 1b and the protruding structure 2b are part of the staple nucleic acid. In this example, the staple nucleic acid contains a sequence complementary to the single-stranded circular nucleic acid, and the split domain 1 or split domain 2, and the latter forms a protruding structure consisting of a single-stranded nucleic acid without complementarily binding to the single-stranded circular nucleic acid.

[0086] The number of protruding structures 1b is not particularly limited, and is, for example, 1 to 30, 2 to 20, 3 to 10, or 2 to 7. The number of protruding structures 2b is not particularly limited, and is, for example, 1 to 30, 2 to 20, 3 to 10, or 2 to 7.

[0087] The arrangement of the protruding structure 1a and protruding structure 2a is not particularly limited as long as the protruding structure 1a and protruding structure 2a are arranged to face each other and bind together based on complementary base pairing in the absence of a recognition substance for the nucleic acid aptamer sequence. For example, in the case of taking the structural type 1, the protruding structure 1a and the protruding structure 2a are arranged to face each other. When the protruding structure 1a and the protruding structure 2a face each other, it is preferable that the protruding structure 1b and the protruding structure 2b do not face each other.

[0088] The arrangement of the protruding structure 1b and protruding structure 2b is not particularly limited as long as the protruding structure 1b and the protruding structure 2b are arranged to face each other to form a nucleic acid enzyme in the presence of a recognition substance for the nucleic acid aptamer sequence (when the nucleic acid enzyme contains two or more G-quartets, further in the presence of a monovalent cation). For example, in the case of taking the structural type 3, the protruding structure 1b and the protruding structure 2b are arranged to face each other. When the protruding structure 1b and the protruding structure 2b face each other, it is preferable that the protruding structure 1a and the protruding structure 2a do not face each other.

[0089] The protruding structure 1a, protruding structure 2a, protruding structure 1b, and protruding structure 2b are preferably arranged near the Holliday junction, which is the fulcrum. For example, these protruding structures are preferably arranged, for example, within 300 bases, within 200 bases, or within 150 bases, from the boundary between the two arms that form the Holliday junction in the partial nucleic acid structure 1 or partial nucleic acid structure 2. This makes it possible to keep the target structure more stable in the absence or presence of a test substance, and thus further increase the detection sensitivity of the test substance.

[0090] The nucleic acid structure of the present invention can be produced by or according to a known method. For example, the nucleic acid structure of the present invention can be produced by heating a solution containing a single-stranded circular nucleic acid and a staple nucleic acid containing a sequence complementary to the single-stranded circular nucleic acid to a high temperature (e.g., 80° C. or higher, 85° C. or higher, or 90° C. or higher) where base pairing hardly occurs, followed by gradual cooling (so that the temperature drops by 1° C., for example, for 15 seconds to 5 minutes, 30 seconds to 3 minutes, or 45 seconds to 2 minutes).

3. Use

[0091] As an example is shown in FIG. **13**, in the nucleic acid structure of the present invention, in the absence of a test substance, the protruding structure 1a and the protruding structure 2a face each other and bind together based on complementary base pairing, and the structural type 1 (or structural type 3) becomes dominant. On the other hand, in the presence of a test substance, the bond between the protruding structure 1a and the protruding structure 2a is dissociated by the binding of the protruding structure 1a to the test substance. Along with this, the protruding structure 1b and the protruding structure 2b face each other to form a nucleic acid enzyme, and the structural type 3 (or structural type 1) becomes dominant. Accordingly, the test substance can be detected by detecting the activity of the nucleic acid enzyme. Therefore, the nucleic acid structure of the present invention can be used for detecting a test substance (e.g., as a constituent of a test substance detection composition).

[0092] From this point of view, the present invention, in an embodiment, relates to a test substance detection composition (the composition of the present invention) comprising the nucleic acid structure of the present invention, and a method for detecting a test substance, comprising bringing the nucleic acid structure of the present invention into contact with a sample that may contain the test substance (the detection method of the present invention). These are described below.

[0093] The test substance is not particularly limited as long as it can bind to a nucleic acid aptamer. Examples include nucleic acids, proteins, peptides, low-molecular-weight compounds, physiologically active substances, metal ions, and the like. In an embodiment of the present invention, the test substance is preferably a biological material.

[0094] When the test substance is a biological material, in an embodiment thereof, organisms containing the biological material can be detected. Examples of such organisms include viruses. Examples of viruses include enveloped viruses (viruses with envelopes), such as influenza viruses (e.g., type A and type B), rubella virus, Ebola virus, coronaviruses, measles virus, varicella-zoster virus, mumps virus, arbovirus, RS virus, SARS virus, hepatitis viruses (e.g., hepatitis B virus and hepatitis C virus), yellow fever virus, AIDS virus, rabies virus, hantavirus, dengue virus, Nipah virus, and Lyssa virus; non-enveloped viruses (viruses without envelopes), such as adenovirus, norovirus, rotavirus, human papillomavirus, poliovirus, enterovirus, coxsackievirus, human parvovirus, encephalomyocarditis virus, poliovirus, and rhinovirus; and the like. Particularly preferred among these are coronaviruses.

[0095] Coronaviruses belong to the subfamily Orthocoronavirinae. Examples of coronaviruses include alphacoronaviruses, betacoronaviruses, gammacoronaviruses, deltacoronaviruses, and the like. Preferred among these are betacoronaviruses. Examples of betacoronaviruses include SARS-associated coronavirus (SARSr-CoV), human coronavirus HKU1, MERS coronavirus, and the like. Preferred among these is SARSr-CoV. Examples of SARSr-CoV include SARS-CoV-2, SARS-CoV-1, and the like. Preferred among these is SARS-CoV-2. The agent of the present invention can be preferably used particularly against SARS-CoV-2.

[0096] The composition of the present invention may contain other components, if necessary. Examples of other components include bases, carriers, solvents, dispersants, emulsifiers, buffers, stabilizers, excipients, binders, disintegrants, lubricants, thickeners, humectants, colorants, fragrances, chelating agents, and the like.

[0097] The composition of the present invention may constitute a kit. The kit may include instruments, reagents, etc. for use in the implementation of the test method of the present invention. Examples of instruments include test tubes, microtiter plates, agarose particles, purification columns, epoxy-coated glass slides, gold colloid-coated glass slides, microtubes, and the like. Examples of reagents include cofactors of the nucleic acid enzyme, monovalent cation-containing buffers, standard samples (positive control and negative control), and the like.

[0098] The sample that may contain a test substance used in the detection method of the present invention is not particularly limited. Examples include biological samples. Specific examples include body fluids, such as whole blood, serum, plasma, follicular fluid, menstrual blood, saliva, cerebrospinal fluid, synovial fluid, urine, interstitial fluid, sweat, tears, and saliva; samples derived from these body fluids (e.g., purified products), tissue pieces, samples derived from tissue pieces (e.g., crushed products and purified products), and the like.

[0099] Before, during, or after contact of the nucleic acid structure of the present invention with the sample, a monovalent cation can be added, if necessary. For example, when the nucleic acid enzyme contains two or more G-quartets, G-quartets are formed by the presence of a monovalent cation.

[0100] Before, during, or after contact of the nucleic acid structure of the present invention with the sample, a cofactor of the nucleic acid enzyme can be added, if necessary. For example, when the nucleic acid enzyme contains two or more G-quartets, peroxidase activity is exerted by the presence of heme, such as hemin.

[0101] In an embodiment of the present invention, contact is performed in a solution, but is not limited thereto. The concentration of the nucleic acid structure of the present invention in the solution is not particularly limited, and is, for example, 0.1 to 100 nM, preferably 0.5 to 20 nM, and more preferably 1.5 to 10 nM. When the solution contains a monovalent cation, the concentration thereof is, for example, 10 mM to 1 M, preferably 50 to 600 mM, and more preferably 100 to 300 mM.

[0102] The pH of the solution in which contact is performed is preferably 2 to 10, more preferably 3.5 to 7.5, and even more preferably 4.5 to 5.5, in terms of allowing detection in a shorter time.

[0103] After contact, the enzymatic activity of the nucleic acid enzyme can be detected to detect the test substance. The enzymatic activity can be detected by or according to a known method. In a preferred embodiment of the present invention, it is preferable to add a substrate that becomes colored due to the enzymatic activity of the nucleic acid enzyme. As a result, the presence or absence of color change (i.e., the presence or absence and amount of test substance) can be detected or measured by visual observation or simple absorbance measurement. When the enzymatic activity is peroxidase activity, examples of such substrates include TMB (3,3',5,5'-tetramethylbenzidine), OPD (o-phenylenediamine dihydrochloride), ABTS (2,2'-azinobis [3-ethylbenzothiazoline-6-sulfonic acid]-diammonium salt), and the like.

EXAMPLES

[0104] The present invention is described in detail below based on Examples; however, the present invention is not limited by these Examples.

Reference Example 1: Preparation 1 of Nucleic Acid Structure

[0105] As single-stranded circular DNA, M13mp18 (SEQ ID NO: 273) was purchased and prepared from tilibit nanosystems. Further, staple DNAs (SEQ ID NOs: 1 to 236) constituting a nucleic acid structure by complementary binding to the single-stranded circular DNA were purchased and prepared from IDT. M13mp18 and each staple DNA were mixed in a solution containing 40 mM of Tris, 20 mM of acetic acid, 2 mM of EDTA, and 12.5 mM of magnesium acetate (1×TAE/Mg^{2+} solution) so that the final concentration of M13mp18 was 4 nM and the final concentration of each staple DNA was 20 nM. The mixed solution was heated to 90° C. by a PCR thermal cycler and then cooled to 25° C. at a rate of—1.0° C. per minute. Excess staple was removed using a centrifugal ultrafiltration filter (Amicon Ultra 0.5 mL-100 K, Millipore).

[0106] Atomic force microscope (AFM) observation was performed, and it was confirmed that a pincher- or plier-like nucleic acid structure was formed. The major diameter was around 200 nm.

[0107] Tables 1 to 6 show the staple numbers, base sequences, and SEQ ID Nos of the staple DNAs used. FIG. **1** shows the overall view of the nucleic acid structure, and FIGS. **2** to **7** show diagrams obtained by dividing the overall view into six parts. Further, FIG. **8** shows a schematic diagram showing only single-stranded circular DNA in a linear form in the nucleic acid structure.

TABLE 1

staple No.	Sequence	SEQ ID No
1s	CCTAATTTTTTTATCCTGAATCTTACCAACGCTATTIT	1
2s	TTTTACGAGCGTCTGCAGCACCGTTTTT	2
3s	CAAGTTTGAATGAAACCATCGATATTCCAGAG	3
4s	AGCTACAAGCCAGTTACAAAATAAGGCCGGAA	4
5s	ACGTCACCCCTTTAGCGTCAGACTACAACGCC	5
6s	ATTATTTAAAGATTAGTTGCTATTTTGCACCC	6
7s	TTTTCATCCCATTACCATTAGCAAACAGCCAT	7
8s	TTGAAGCCTTAAATCTCCCAAT	8
9s	CCAAATAAGAAACGATTTAGAGCCAGCA	9
10	TITTAATCAGTAGCATAGTTAGCGTTTT	10
11	TCGTCTTTTCCACAGACAGCCCTCGACAGAAT	11
12	TTTTTAACGATCTAAAGTTTTG	12
13s	AAATCACCAGTAGCAGGCATTT	13
14s	TGCCATCTGAGCCATTTGGGAATTTTTGTTTA	14
15s	GCGTTTTAAATGAAAATAGCAGCCTCACCGTC	15
16s	ACCGACTTTTTCATAATCAAAATCCTCATTTT	16
17s	TAAAAACAATAGAAGGCTTATCCGGTATTCTAAGAACGCGAG	17
18s	AGGTGAATTATTTACAGAGAGAATAACA	18
19s	ACCGGAACGGAAATTATTCATTAA	19
20s	AATCAGATGGGAAGCGCATTAGACGAAGGTAA	20
21s	ATATTGACCGCCTCCCTCAGAGCCAGAACCGC	21
22s	TAACTGAACATTACCGCGCCCAATAGCAAGCA	22
23	TGTATGGGGTCACCAGTACAAACTGTAGCGCG	23

TABLE 1-continued

staple No.	Sequence	SEQ ID No
24	TGTAGCATCCAGACGTTAGTAAATGAATTTTC	24
25s	CAGAACCGAACCGATTGAGGGAGGGGGAGAAT	25
26s	GTAGGAATCACCCTGAACAAAGTCCAAAAGGG	26
27s	CGACATTCCCACCCTCAGAGCCACCCGTACTC	27
28s	ATTGAGCGAAGCAAGCCGTTTTTATTTTCATC	28
29s	GAGCCGCCTTACCAGCGCCAAAGAAGAGGGTA	29
30s	TCGAGAACCTAATATCAGAGAGATAGAAAATT	30
31s	CATATGGTACCAGAACCACCACCATGATATAA	31
32s	AGAATTGAGTATTAAACCAAGTACCGCACTCA	32
33	TCGGTCATGTAACACTGAGTTTCATTTTGC	33
34	CCCATGTACCAGCCCCCTTATTAGCGTT	34
35	TAAACAACTTTCAACAGTTTCAGCGG	35
36s	GCCAGCATTTTTGTCACAATCAATAACCCACA	36
37s	AAGAACGGGTTAAGCCCAATAATACCACGGAA	37
38s	TAAGTTTATGACAGGAGGTTGAGGAGTACCAG	38
39s	AAACAATGTCGGCTGTCTTTCCTTATCATTCC	39
40s	ACGATTGGAAGAAACGCAAAGACAAGAGCAAG	40

TABLE 2

staple No.	Sequence	SEQ ID No
41	AGTGAGAAGCAAGCCCAATAGGAA	41
42s	ATCAATAAAAATAGCAATAGCTATAGGTGGCA	42
43s	ACATATAACCTTGATATTCACAAAAGAGAAGG	43
44s	AGCCCTTTTAATTTACGAGCATGTAGAAACCA	44
45s	TCCTCATTGAAAATACATACATAACTTACCGA	45
46s	AAATAATATCCCATCCTTAAGAAAAGTAAGCATATGTTAG	46
47s	CAAACGTAAAAGCCAGAATGGAAAACATGAAA	47
48s	GAATCATAATAAGGCGTTAAATAAGA	48
49s	AACAAAGTAAGTCCTGAACAAGAAAAACACCG	49
50s	TCTGAATTACTCCTTATTACGCAGGATAGCCG	50
51s	TGTGATAAATTACTAGAAAAAGCCTTTATCAA	51
52	AGCCACCACCACCGGAACCAGAGCCACC	52
53	CAGGGATATAGAAAGGAACAACTAAAGGAATTGC	53
54	GAATAATAATTTTTTCGAACCGCCACCCTCAG	54
55s	CAATAGATTACCAGAAGGAAACCGAACTGGCA	55
56s	TGATTAAGTACCGTTCCAGTAAGCCCCCTGCC	56

TABLE 2-continued

staple No.	Sequence	SEQ ID No
57	GCTCCAAATAGTACCGCCACCCTCGCCACCCT	57
58	CACCCTCAACGTTGAAAATCTCCAAAAAAAAG	58
59s	ATCATATGTTTAATGGTTTGAAATACCGACCG	59
60s	CAATAATAATGCAGAACGCGCCTGTGTTTAGT	60
61	ATCAGCTTCGGAATAGGTGTATCACACCCTCA	61
62	AGGAGGTTAGGAGCCTTTAATTGTATCGGTTT	62
63s	TGGCTTTTACGGAATACCCAAAAGAGGAAACG	63
64s	GACCTAAACGTTATACAAATTCTTACAACATG	64
65	AGCTTGATGTGCCGTCGAGAGGGTGAGCCGCC	65
66	GTATAGCCGCTTTCGAGGTGAATTTCTTAAAC	66
67s	TTCAGCTAAGGTTTAACGTCAGATAATTGCGT	67
68s	AAAGCCAAATATTTTAGTTAATTTCATCTTCT	68
69	AACAACCATAGCGGGGTTTTGCTCCAGGTCAG	69
70	GCGGATAAACCGATAGTTGCGCCGACAATGAC	70
71s	AGTAACAGCAGACGACGACAATAAACCAGTAT	71
72s	TTTCAAATCGCTCAACAGTAGGGCGTAAAGTA	72
73	CGGTCGCTAGGCTGAGACTCCTCACAAATAAA	73
74	ATTAGGATTCGCCCACGCATAACCGATATATT	74
75s	ATTCTGTCTACCTTTTACATCGGGAAAATTAT	75
76s	GAATCGCCAAGACAAAGAACGCGAGAAAACTT	76
77$	TATTTCGGAACCTATTATTCTGAAGCGCAGTC	77
78s	GTATTAAGGAGGCTTGCAGGGAGT	78
79s	TAACGGATAAAGTACCGACAAAAGTTAATTGA	79
80s	CCAATCGCATATTTAACAACGCCAAGTAATAA	80

TABLE 3

staple No.	Sequence	SEQ ID No
81c	TTGAATGGTATAAACAGTTAATGCGTCATACA	81
82s	TAAAGGCCGCTTTTGCGAATACGTGGCACAGACAATATTT	82
83w	AGATTTTCGATGATACAGGAGTGTTGAGTAAC	83
84w	TAAGTTTTAAAACAGAAATAAAGAGAATATAC	84
85c	GATAGCCCAACGGGGTCAGTGCCTACTGGTAA	85
86c	AGTGCCCGCTATTAGTCTTTAATGCGCGAACT	86
87w	TTGCACGTGGTGAGGCGGTCAGTACAGAAGAT	87
88w	GCCTGCAATAGAACCTACCATATCAGAAACAA	88
89w	CGAACCACCAGTTAACACC	89

TABLE 3-continued

staple No.	Sequence	SEQ ID No
90w	AAAACAGATAAAACATCGC	90
91w	CATTAAAAATACCGAA	91
92w	GGAAGGGTCAGTGCCACGCTGAGA	92
93w	CAAATGAAGTTTGGATTATACTTCAATACCAA	93
94w	GCAAATCAACAGTTGAGCCAGCAG	94
95w	AAGGAATTGAGGAAGG	95
96w	TCCTGATTAAATCTAAAGCATCACAATATCTG	96
97w	ACCTCAAAGATGATGGCAATTCATTTATTCAT	97
98w	CTAACAACTATCAAACCCTCAATCCTTGCTGA	98
99w	GTCAGTTGTTATCTAAAATATCTTTAGGAGCA	99
100w	GATTATCAACGTTATTAATTTTAATAAATCCT	100
101w	GTAACATTGAGCGGAATTATCATCTGATGAAA	101
102w	TAATACATATTCGACAACTCGTATAAGTTTGA	102
103w	TTGCCCGATAATAGATTAGAGCCGTCAATAGA	103
104w	ACCAGAAGATCATTTTGCGGAACATTTTTTAGACTT	104
105s	AACAATTTAAGAAAACAAAATTAATTTTAAGAAACC	105
106w	TACAAACATTGAGGATTTAGAAGTTTTT	106
107s	CAAACATCCATTTGAATTACCTTTAACATAGC	107
108w	AAACAGTACCTGAGCAAAAGAAGAATATTCCT	108
109s	TTCAATTACATAAATCAATATATGCTATTAAT	109
110w	TAACCTTGATCGCGCAGAGGCGAACAATATAA	110
111s	GAGAATATTCGCCTGATTGCTTTGTGAATAAT	111
112s	TTTAGGCACTATATGTAAATGCTGATGCAAAT	112
113s	GTTACAAAGAGGCATTTTCGAGCCACATGTAA	113
114s	TTATATAACTTCTGTAAATCGTCGTGAGTGAA	114
115s	TAATTTTCTTTAACCTCCGGCTTAGGTTGGG	115
116s	GACTACCTCCTTAGAATCCTTGAATTTAATGG	116
117s	GATAGCTTTTTATCAAAATCATAGGTCTGAGA	117
118s	GAGTCAATAGTGAAAGATTAAGACGCTGAGTTTTTTACATTT	118
119s	ACCATAAATCAAAAGTTCAGAAAACGAGAATTTTAAAATGTT	119
120s	TTTAAACAATCAGGTCTTTACCCTGACTATTA	120

TABLE 4

staple No.	Sequence	SEQ ID No
121s	TAGTCAGAATTCATTGAATCCCCCCGGAATCG	121
122s	GAGTAGATAGCAAAGCGGATTGCATCAAAAAG	122

TABLE 4-continued

staple No.	Sequence	SEQ ID No
123w	AGACCAGGAAAGAGGACAGATGAATTTT	123
124w	AGTGAATATCAACGTAACAAAGCTTTTTCGGTGTAC	124
125s	AAAAGAAGTTTTGCATAGCGAGAGGCTTTTTTTTGCTCATTC	125
126w	CAACTTTGCGCATAGGCTGGCTGAATATTCAT	126
127w	TACCCAAAAGGCTTGCCCTGACGAGACGATAA	127
128s	AAACCAAACAGAGGGGGTAATAGT	128
129w	AAGAGTAACAATCATAAGGGAACCGAACTGAC	129
130w	AGAACGAGTCTTGACAAGAACCGGCCTTCATC	130
131w	TAGACTGGCCCTCGTTTACCAGACGAAACACC	131
132w	CAGACGGTACAAAGTACAACGGAGTTATACCA	132
133w	AGCGCGAATAGTAAATTGGGCTTGAGCAACAC	133
134s	TATCATAAATAGCGTCCAATACTGTCAAATGC	134
135w	CATCGCCTATGTTACTTAGCCGGAACGAGGCG	135
136w	TAATTTCACTTTGACCCCCAGCGAATTTGTAT	136
137w	TCATAAATTACGAGGCATAGTAAGAGATGGTT	137
138w	ATCCGCGACCTGCTCC	138
139w	GATAAATTGTGTCGAAACACTAAA	139
140w	ACACTCATACTTTAATCATTGTGATAACGCCA	140
141s	AAAGGAATTTAGTTTGACCATTAGCTGCGAAC	141
142w	ATGCGATTAAAGAGGCAAAAGAAT	142
143s	CGCAAATGAACTAATGCAGATACAATTACCTT	143
144w	AGACTTTTTCATGAGG	144
145w	AAGTTTCCATTGCACCAAC	145
146w	CTAAAACGTTAAGAACTGGCTCATTAGGAATA	146
147s	CCACATTCGTCAATAACCTGTTTAGTTTCATT	147
148w	AAAATACGTTGAGGACTAA	148
149w	TCAGGACGTAATGCCACTACGAAGAAACGGGT	149
150s	TTCATTTGTTCATCAGTTGAGATTTATACCAG	150
151w	ATTAAAGATTGGGAAGAAAAATCTATTACAGG	151
152s	TAGAAAGAGGGCGCGAGCTGAAAATTAAATAT	152
153w	AAAACGAATCCAGTTTGGAACAAGCAAAGGGC	153
154s	CAATTCTACTAACGGAACAACATTACGTTAAT	154
155s	AGTGTTGTTGAGGGGACGACGACACGTGCATC	155
156s	TGCCAGTTCTAATAGTAGTAGCATTTGCTGAA	156
157c	CAGAGGCTACGTGGACTCCAACGTAGTCCACT	157
158c	GAAAAACCTCGGAACGAGGGTAGCAACGGCTA	158

TABLE 4-continued

staple No.	Sequence	SEQ ID No
159s	AATAAATCATGGGCGCATCGTAACGTATCGGC	159
160s	CTCAGGAAAAAGAATAGCCCGAGAGCCCACTA	160

TABLE 5

staple No.	Sequence	SEQ ID No
161c	AGACAGCAGTCTATCAGGGCGATGTAGGGTTG	161
162s	TGACCTGAAAGCGTAAGGGATCGTCACCCTCAGCAGCGAA	162
163s	GAATTAGCAAAATTAATGACCGTAATGGGATAGCTTTCCG	163
164s	ATAAATCAGATCGCACTCCAGCCAGGTCACGT	164
165s	CGTGAACCATCACCCAAATCAAGTAATCCCTT	165
166s	GTCGAGGTAGAGATAGAACCCTTC	166
167s	GCACCGCTGTTCCGAAATCGGCAATTTTTGGG	167
168s	CTAAATCGCGTCGGATTCTCCGTGAGGCAAAG	168
169	TGGCCAACGCCGTAAAGCACTAAAATCCTGTT	169
170	CTAAAGGGCGACCAGTAATAAAAGGGACATTC	170
171s	CGCCATTCGCCCCAGCAGGCGAAATCGGAACC	171
172s	TGATGGTGTCTGGTGCCGGAAACCGGAACAAA	172
173	CAGTCACAAGCCCCCGATTTAGAGCGGTCCAC	173
174	GGAAAGCCGATTATTTACATTGGCAGATTCAC	174
175s	GGAAGGGCAGAGAGTTGCAGCAAGCTTGACGG	175
176s	GCTGGTTTGCCATTCAGGCTGCGCTGAGCGAG	176
177	CTGAAATGGGCGAACGTGGCGAGATCACCGCC	177
178	GAAGAAAGACCTACATTTTGACGCTCAATCGT	178
179s	TACGCCAGAACAGCTGATTGCCCTAAGGAAGG	179
180s	TGGCCCTGGATCGGTGCGGGCCTCTTCCTGTA	180
181	ATGGAAATCGAAAGGAGCGGGCGCCACCAGTG	181
182	TGGCAAGTGCCATTGCAACAGGAAAAACGCTC	182
183s	AGGCGATTGGGTGGTTTTTTTTTTAGGGCGC	183
184s	AGACGGGCCTGGCGAAAGGGGGATGGAACGCC	184
185	TACCGCCAGTAGCGGTCACGCTGCTGCGTATT	185
186	ACCACACCTGCTGGTAATATCCAGAACAATAT	186
187s	CCAGTCACGGGGGGAGAGGCGGTTGCGTAACC	187
188s	GGGCGCCAAAGTTGGGTAACGCCATTTTGTTA	188
189	CTCAAACTATCGGCCTCGCCGCGCTTAATGCG	189
190	TATGGTTGTAACATCACTTGCCTGAGTAGAAGAA	190
191	CCGCTACAGGAGCTGCATTAATGAATCG	191

TABLE 5-continued

staple No.	Sequence	SEQ ID No
192s	ATGCCTGCGGAAACCTGTCGTGCCGCGCGTAC	192
193s	GCCAACGCGACGTTGTAAAACGAC	193
194s	GGCCAGTGCCTAAATTGTAAACGTTAAT	194
195s	ATTTTGTTTTCAAAAGGGTGAGAAAGGCCGGAGACAGTCAAA	195
196s	GGTAAAGAAAAATTCGCATTAAATGGGTTTTC	196
197s	AATCAGCTAATGCAATGCCTGAGTAATGTGTA	197
198s	ATATTTTACATTTTTTAACCAATAGTGCTGCA	198
199s	ATCAAAAAGATAAAAAATTTTTAGAACCCTCAT	199
200s	AACGCAAGTAATTCGCGTCTGGCCTTCGCTAT	200

TABLE 6

staple No.	Sequence	SEQ ID No
201s	GCCAGCTTTTTTGCGGGAGAAGCCTTTATTTC	201
202s	TGTAATACTCATCAACATTAAATGAACTGTTG	202
203s	TAACAACCGTTGTACCAAAAACATTATGACCC	203
204	ATTAGTAACTTTGACGAGCACGTA	204
205	TTAACCGTTGTAGCAATACTTCTTTG	205
206	AATTGCGTTCCTCGTTAGAATCAACGCAAA	206
207	TAACGTGCTTTGCGCTCACTGCCCGCTT	207
208s	TCCAGTCGAGGTCGACTCTAGAGGCCAAAAAC	208
209s	TCACCATCTGTATAAGCAAATATTAAGCTTGC	209
210s	ACCGAGCTCGAATTCTCACATT	210
211s	TTGATAATCAGAAAAGCCATCCCCGGGT	211
212s	AGGAAGATAATATGATATTCAACCGTTCTAGCTG	212
213s	GAGCTAACGTAATCATGGTCATAGCATGTCAA	213
214s	ATAAATTAATGCCGGACCCCGG	214
215s	TCATATGTAGAGGGTAGCTATTTTTGAGAGAT	215
216s	TGTGTGAATAAAGCCTGGGGTGCCGGAGGCCG	216
217s	CTACAAAGGTAATCGTAAAACTAGCTGTTTCC	217
218s	ATAAAGTGATTGTTATCCGCTCACAGAGAATC	218
219s	GATGAACGGCTATCAGGTCATTGCCTGAGAGTCTTTTT	219
220s	TTTTGGAGCAAACAAATTCCACACTTTT	220
221	TGTCCATCGAGCGGGAGCTAAACATAATGAGT	221
222	ATTAAAGGGTGAGGCCACCGAGTAAAAGAGTC	222
223s	CGGCGGATGCAATAAAGCCTCAGAGCATAAAG	223
224s	TGGTGTAGATACAGGCAAGGCAAAGAGGTCAT	224

TABLE 6-continued

staple No.	Sequence	SEQ ID No
225s	TTTTGCGGTTAATTGCTCCTTTTG	225
226s	GAGTACCTATGGCTTAGAGCTTAATAACATCC	226
227s	TATAATGCAAACTCCAACAGGTCAGGATTAGA	227
228s	CCGGAAGCTGTAGCTCAACATGTTGGTGGCAT	228
229s	GCAACTAAATTCGAGCTTCAAAGCGAACCAGA	229
230s	GCGTTTTAAGTACGGTGTCTGGAAGCTATATT	230
231s	CCATATAAGAAGCCCGAAAGACTTCAAATATC	231
232s	ATTAAGAGCAGTTGATTCCCAATTATACATTT	232
233	TATAATCAGATTITAGACAGGAACCCGGAAGC	233
234	TITTAATCCTGAGAAGTGTTTT	234
235	TITTAACATACGAGGGTACGCCAGTTTT	235
8s-2	ACGTCAAAGCGAACCTCCCGACTTGCGGGAGGTT	236

Reference Example 2: Preparation 2 of Nucleic Acid Structure

[0108] A nucleic acid structure was prepared in the same manner as in Reference Example 1, except that some staple DNAs were changed to other staple DNAs. Table 7 shows the corresponding staple numbers of Reference Example 1, and the staple numbers, base sequences, and SEQ ID Nos of the staple DNAs used. The configuration of the obtained nucleic acid structure was the same as that of the nucleic acid structure of Reference Example 1, except that the partial structure 1 and the partial structure 2 each contained a protruding structure consisting of single-stranded DNA of the following telomere element sequence.

TABLE 7

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
65	65long	TCTTAAACAGCTTGATGTGCCGTCGAGAGGGTGAGCCGCC	237
66	66shortTelo	GTATAGCCGCTTTCGAGGTGAATTTTTTTAGGGTTAGGGT	238
69	69long	ACAATGACAACAACCATAGCGGGGTTTTGCTCCAGGTCAG	239
70	70shortTelo	GCGGATAAACCGATAGTTGCGCCGTTTTTAGGGTTAGGGT	240
73	73long	GATATATTCGGTCGCTAGGCTGAGACTCCTCACAAATAAA	241
74	74shortTelo	ATTAGGATTCGCCCACGCATAACCTTTTTAGGGTTAGGGT	242
78s	78s-telo	GTATTAAGGAGGCTTGCAGGGAGTTTTTTAGGGTTAGGGT	243
166s	166s-telo	GTCGAGGTAGAGATAGAACCCTTCTTTTTAGGGTTAGGGT	244
169	169long	GGACATTCTGGCCAACGCCGTAAAGCACTAAAATCCTGTT	245
170	170shortTelo	CTAAAGGGCGACCAGTAATAAAAGTTTTTAGGGTTAGGGT	246
173	173long	AGATTCACCAGTCACAAGCCCCCGATTTAGAGCGGTCCAC	247
174	174shortTelo	GGAAAGCCGATTATTTACATTGGCTTTTTAGGGTTAGGGT	248
177	177long	TCAATCGTCTGAAATGGGCGAACGTGGCGAGATCACCGCC	249
178	178shortTelo	GAAGAAAGACCTACATTTTGACGCTTTTTAGGGTTAGGGT	250

[0109] Staple DNA with a staple number labeled “Telo” contains a telomere element sequence (including two G triplet repeats). The telomere element sequence forms a protruding structure consisting of single-stranded DNA without complementary binding to M13mp18. Among the three structures based on the conformation of the Holliday junction (FIG. **9**: provided that Cross is an intermediate structure that mediates structural changes between Antiparallel and Parallel), when taking Parallel, the protruding structure in the partial structure 1 and the protruding structure in the partial structure 2 are arranged so that these protruding structures face each other (i.e., staple DNA containing a telomere element sequence is arranged). The telomere element sequence in the partial structure 1 and the telomere element sequence in the partial structure 2 can be linked in the presence of a monovalent cation to form a G-quartet structure (nucleic acid enzyme: heme stacks can exhibit peroxidase (HRP) activity).

[0110] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} solution) was added, or not added, to a solution containing the prepared nucleic acid structure, followed by still standing for 5 minutes. AFM observation was then performed, and the presence of K^+ was visualized as a structural difference of the nucleic acid structure. The solution containing the nucleic acid structure contained 3.2 nM of the nucleic acid structure and 0 mM of KCl or 200 mM of KCl. Further, each structure of the nucleic acid structure was counted from the acquired images, and its proportion was examined.

[0111] FIG. **10** shows AFM observation images. Further, FIG. **11** shows the proportion of each structure. A structural change to the parallel type due to the addition of K^+ was confirmed.

Reference Example 3: Detection 1 of Enzymatic Activity

[0112] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} solution) was added, or not added, to a solution containing the nucleic acid structure (Reference Example 2), followed by still standing for 5 minutes. Then, hemin (dissolved in DMSO at 100 μM) was added, followed by still standing for 1 hour. The solution containing the nucleic acid structure contained 3.2 nM of the nucleic acid structure, 4.0 μM of hemin, and 0 mM of KCl or 200 mM of KCl. Then, an equivalent amount of 1-Step Ultra TMB-ELISA (catalog number: 34028, Thermo Scientific) was added, and the presence of K^+ was visualized as a color.

[0113] The results are shown in FIG. **12**. A strong blue color was developed when a large amount of parallel nucleic acid structure was contained (K^+ addition). This revealed that enzymatic activity can be exhibited by forming a G-quartet structure in the parallel nucleic acid structure.

Example 1: Preparation 3 of Nucleic Acid Structure

[0114] A nucleic acid structure was prepared in the same manner as in Reference Example 1, except that some staple DNAs were changed to other staple DNAs. Tables 8 and 9 show the corresponding staple numbers of Reference Example 1, and the staple numbers, base sequences, and SEQ ID Nos of the staple DNAs used. The configuration of the obtained nucleic acid structure was the same as that of the nucleic acid structure of Reference Example 1, except that the partial structure 1 and the partial structure 2 each contained a protruding structure consisting of single-stranded DNA of a telomere element sequence, the partial nucleic acid structure 1 contained a protruding structure containing a nucleic acid aptamer sequence (complementary sequence of miR-20 (Mature sequence hsa-miR-20a-5p: SEQ ID NO: 274)), and the partial nucleic acid structure 2 contained a sequence complementary to part of the nucleic acid aptamer sequence.

TABLE 8

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
65	65long	TCTTAAACAGCTTGATGTGCCGTCGAGAGGGTGAGCCGCC	251
66	66shortTelo	GTATAGCCGCTTTCGAGGTGAATTTTTTAGGGTTAGGGT	252
69	69long	ACAATGACAACAACCATAGCGGGGTTTTGCTCCAGGTCAG	253
70	70shortTelo	GCGGATAAACCGATAGTTGCGCCGTTTTTAGGGTTAGGGT	254
73	73LongCompM20	GTGCAGGTAGTTTTGATATATTCGGTCGCTAGGCTGAGAC TCCTCACAAATAAA	255
74	74shortTelo	ATTAGGATTCGCCCACGCATAACCTTTTTAGGGTTAGGGT	256
78s	78s-telo	GTATTAAGGAGGCTTGCAGGGAGTTTTTTAGGGTTAGGGT	257
166s	166s-telo	GTCGAGGTAGAGATAGAACCCTTCTTTTTAGGGTTAGGGT	258
169	169LongCompM20	GTGCAGGTAGTTTTGGAACATTTCTGGCCAACGCCGTAAA GCACTAAAATCCTGTT	259
170	170shortTelo	CTAAAGGGCGACCAGTAATAAAAGTTTTTAGGGTTAGGGT	260
173	173long	AGATTCACCAGTCACAAGCCCCCGATTTAGAGCGGTCCAC	261
174	174shortTelo	GGAAAGCCGATTATTTACATTGGCTTTTTAGGGTTAGGGT	262
177	177long	TCAATCGTCTGAAATGGGCGAACGTGGCGAGATCACCGCC	263
178	178shortTelo	GAAGAAAGACCTACATTTTGACGCTTTTTAGGGTTAGGGT	264

TABLE 9

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
81c	81cLong	CAATATTTTTGAATGGTATAAACAGTTAATGCGTCATACA	265
82s	82sBothLongM20	GTGCAGGTAGTTTTTAAAGGCCGCTTTTGCGAATACGTGG CACAGATTTTCTACCTGCACTATAAGCACTTTA	266
85c	85cLong	CGCGAACTGATAGCCCAACGGGGTCAGTGCCTACTGGTAA	267
86c	86cShortMainM20	AGTGCCCGCTATTAGTCTTTAATGTTTTCTACCTGCACTAT AAGCACTTTA	268
157c	157cLong	AAACGGCTACAGAGGCTACGTGGACTCCAACGTAGTCCACT	269
158c	158cShortMainM20	GAAAAACCTCGGAACGAGGGTAGCTTTTCTACCTGCACTAT AAGCACTTTA	270
161c	161cLong	GCAGCGAAAGACAGCAGTCTATCAGGGCGATGTAGGGTTG	271
162s	162sBothLongM20	GTGCAGGTAGTTTTTGACCTGAAAGCGTAAGGGATCGTCA CCCTCATTTTCTACCTGCACTATAAGCACTTTA	272

[0115] Staple DNA with a staple number labeled "Telo" contains a telomere element sequence (including two G triplet repeats). The structure formed by the telomere element sequence, its position, etc. are the same as those in Reference Example 2. The telomere element sequence in the partial structure 1 and the telomere element sequence in the partial structure 2 can be linked in the presence of miR-20 and a monovalent cation to form a G-quartet structure (nucleic acid enzyme: heme stacks can exhibit peroxidase (HRP) activity).

[0116] Staple DNA with a staple number labeled "M20" contains the complementary sequence of miR-20 or a sequence complementary to part of the sequence. These sequences form a protruding structure consisting of single-stranded DNA without complementary binding to M13mp18. Among the three structures based on the conformation of the Holliday junction (FIG. **9**: provided that Cross is an intermediate structure that mediates structural changes between Antiparallel and Parallel), when taking Antiparallel, the protruding structure in the partial structure 1 and the protruding structure in the partial structure 2 are arranged so that these protruding structures face each other (i.e., staple DNA containing the complementary sequence of miR-20 or a sequence complementary to part of the sequence is arranged). The complementary sequence of miR-20 in the partial structure 2 and the sequence complementary to part of the sequence in the partial structure 1 can be bonded based on complementary base pairing in the absence of miR-20.

[0117] That is, as shown in FIG. **13**, the nucleic acid structure of this Example is fixed at the antiparallel type based on complementary binding in the absence of miR-20, whereas in the presence of miR-20, the complementary bond dissociates to induce a structural change to the cross type, and the presence of a monovalent cation induces a structure change to the parallel type based on the formation of a G-quartet structure, resulting in the exhibition of peroxidase activity.

[0118] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} solution) was added to a solution containing the prepared nucleic acid structure, followed by still standing for 5 minutes. Then, a test substance, miR20 (dissolved in RNase-free Water (catalog number: 9012, TakaRa) at 200 μM), was added, followed by still standing for 1 hour. AFM observation was then performed, and the presence of miR20 was visualized as a structural difference of the nucleic acid structure. The solution containing the nucleic acid structure contained 3.2 nM of the nucleic acid structure, 1 μM of miR20, and 200 mM of KCl. Further, each structure of the nucleic acid structure was counted from the acquired images, and its proportion was examined.

[0119] FIG. **14** shows AFM observation images. Further, FIG. **15** shows the proportion of each structure. The antiparallel type was the majority before the addition of miR-20, indicating that the addition of miR-20 induced a structural change from the antiparallel type to the parallel type.

Example 2: Detection 2 of Enzymatic Activity

[0120] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} solution) was added to a solution containing the nucleic acid structure (Example 1), followed by still standing for 5 minutes. Then, a test substance, miR20 (dissolved in RNase-free Water (catalog number: 9012, TakaRa) at 200 μM), was added, followed by still standing for 1 hour. Further, hemin (dissolved in DMSO at 100 μM) was added, followed by still standing for 1 hour. The solution containing the nucleic acid structure contained 3.2 nM of the nucleic acid structure, 10 μM of miR20, 4.0 μM of hemin, and 200 mM of KCl. Then, an equivalent amount of 1-Step™ Ultra TMB-ELISA (catalog number: 34028, Thermo Scientific) was added, and the presence of miR20 was visualized as a color.

[0121] The results are shown in FIG. **16**. A strong blue color was developed when miR-20 was contained. This revealed that miR-20 can be detected as a color change using the nucleic acid structure.

[0122] In conventional DNA nano-structures, split G-quartet sequences are highly stable, and it is difficult to maintain the split state; however, in the present invention, the pre-fixation on the opposite side (antiparallel type) successfully prevented the leakage of enzymatic activity. By using the technique of the present invention, the enzymatic activity can be controlled, for example, by varying the number of DNA strands attached to biomolecules. Unlike immunochromatography, the technique of the present invention is superior in that it does not require the fixation of antibodies or aptamers to underlying membranes or the like. In addition, the technique of the present invention, which can be composed only of nucleic acids, allows mass synthesis in a short time, cheaply and easily, with less lot differences compared to antibodies.

Example 3: Preparation 4 of Nucleic Acid Structure

[0123] A nucleic acid structure was prepared in the same manner as in Reference Example 1, except that some staple DNAs were changed to other staple DNAs. Tables 10 and 11 show the corresponding staple numbers of Reference Example 1, and the staple numbers, base sequences, and SEQ ID Nos of the staple DNAs used. The configuration of the obtained nucleic acid structure was the same as that of the nucleic acid structure of Reference Example 1, except that the partial structure 1 and the partial structure 2 each contained a protruding structure consisting of single-stranded DNA of a telomere element sequence, the partial nucleic acid structure 1 contained a protruding structure containing a nucleic acid aptamer sequence (complementary sequence of R509 (SEQ ID NO: 297)), and the partial nucleic acid structure 2 contained a sequence complementary to part of the nucleic acid aptamer sequence. R509 is the complementary sequence of NIID_WH-1_F509 (National Institute of Infectious Diseases, Manual for the Detection of Pathogen 2019-nCOV Ver. 2.9.1).

TABLE 10

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
65	65long	TCTTAAACAGCTTGATGTGCCGTCGAGAGGGTGAGCCGCC	275
66	66shortTelo	GTATAGCCGCTTTCGAGGTGAATTTTTTTAGGGTTAGGGT	276
69	69long	ACAATGACAACAACCATAGCGGGGTTTTGCTCCAGGTCAG	277
70	70shortTelo	GCGGATAAACCGATAGTTGCGCCGTTTTTAGGGTTAGGGT	278
73	73LongCompCF509	CAGTTCGAGTTTTGATATATTCGGTCGCTAGGCTGAGACTCCT CACAAATAAA	279
74	74shortTelo	ATTAGGATTCGCCCACGCATAACCTTTTTAGGGTTAGGGT	280
78s	78s-telo	GTATTAAGGAGGCTTGCAGGGAGTTTTTTAGGGTTAGGGT	281
166s	166s-telo	GTCGAGGTAGAGATAGAACCCTTCTTTTTAGGGTTAGGGT	282
169	169LongCompCF509	CAGTTCGAGTTTTGGAACATTTCTGGCCAACGCCGTAAAGCAC TAAAATCCTGTT	283
170	170shortTelo	CTAAAGGGCGACCAGTAATAAAAGTTTTTAGGGTTAGGGT	284
173	173long	AGATTCACCAGTCACAAGCCCCCGATTTAGAGCGGTCCAC	285
174	174shortTelo	GGAAAGCCGATTATTTACATTGGCTTTTTAGGGTTAGGGT	286
177	177long	TCAATCGTCTGAAATGGGCGAACGTGGCGAGATCACCGCC	287
178	178shortTelo	GAAGAAAGACCTACATTTTGACGCTTTTTAGGGTTAGGGT	288

TABLE 11

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
81c	81cLong	CAATATTTTTGAATGGTATAAACAGTTAATGCGTCATACA	289
82s	82sBothLongCF509	CAGTTCGAGTTTTTAAAGGCCGCTTTTGCGAATACGTGGCACA GATTTTCTCGAACTGCACCTCATGG	290
85c	85cLong	CGCGAACTGATAGCCCAACGGGGTCAGTGCCTACTGGTAA	291
86c	86cShortCF509	AGTGCCCGCTATTAGTCTTTAATGTTTTCTCGAACTGCACCTCA TGG	292
157c	157cLong	AAACGGCTACAGAGGCTACGTGGACTCCAACGTAGTCCACT	293
158c	158cShortCF509	GAAAAACCTCGGAACGAGGGTAGCTTTTCTCGAACTGCACCTC ATGG	294

TABLE 11-continued

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
161c	161cLong	GCAGCGAAAGACAGCAGTCTATCAGGGCGATGTAGGGTTG	295
162s	162sBothLongCF509	CAGTTCGAGTTTTTGACCTGAAAGCGTAAGGGATCGTCACCCT CATTTTCTCGAACTGCACCTCATGG	296

[0124] Staple DNA with a staple number labeled "Telo" contains a telomere element sequence (including two G triplet repeats). The structure formed by the telomere element sequence, its position, etc. are the same as those in Reference Example 2. The telomere element sequence in the partial structure 1 and the telomere element sequence in the partial structure 2 can be linked in the presence of R509 and a monovalent cation to form a G-quartet structure (nucleic acid enzyme: heme stacks can exhibit peroxidase (HRP) activity).

[0125] Staple DNA with a staple number labeled "CF509" contains the complementary sequence of R509 or a sequence complementary to part of the sequence. These sequences form a protruding structure consisting of single-stranded DNA without complementary binding to M13mp18. Among the three structures based on the conformation of the Holliday junction (FIG. **9**: provided that Cross is an intermediate structure that mediates structural changes between Antiparallel and Parallel), when taking Antiparallel, the protruding structure in the partial structure 1 and the protruding structure in the partial structure 2 are arranged so that these protruding structures face each other (i.e., staple DNA containing the complementary sequence of R509 or a sequence complementary to part of the sequence is arranged). The complementary sequence of R509 in the partial structure 2 and the sequence complementary to part of the sequence in the partial structure 1 can be bonded based on complementary base pairing in the absence of R509.

[0126] That is, as shown in FIG. **17**, the nucleic acid structure of this Example is fixed at the antiparallel type based on complementary binding in the absence of R509, whereas in the presence of R509, the complementary bond dissociates to induce a structural change to the cross type, and the presence of a monovalent cation induces a structure change to the parallel type based on the formation of a G-quartet structure, resulting in the exhibition of peroxidase activity.

[0127] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} solution) was added to a solution containing the prepared nucleic acid structure, followed by still standing for 5 minutes. Then, a test substance, R509 (dissolved in RNase-free Water (catalog number: 9012, TakaRa) at 200 μM), or R29142 (SEQ ID NO: 344), which was not a test substance, was added, followed by still standing for 1 hour. AFM observation was then performed, and the presence of R509 was visualized as a structural difference of the nucleic acid structure. The solution containing the nucleic acid structure contained 3.2 nM of the nucleic acid structure, 1 μM of R509, and 200 mM of KCl. Further, each structure of the nucleic acid structure was counted from the acquired images, and its proportion was examined. R29142 is the complementary sequence of NIID_2019-nCOV_N_F2 (National Institute of Infectious Diseases, Manual for the Detection of Pathogen 2019-nCOV Ver. 2.9.1).

[0128] FIG. **18** shows AFM observation images. Further, FIG. **19** shows the proportion of each structure. The antiparallel type was the majority before the addition of R509, indicating that the addition of R509 induced a structural change from the antiparallel type to the parallel type. It was also indicated that the addition of R29142 did not induce a structural change from the antiparallel type to the parallel type.

Example 4: Detection 3 of Enzymatic Activity

[0129] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} solution) was added to a solution containing the nucleic acid structure (Example 3), followed by still standing for 5 minutes. Then, a test substance, R509 (dissolved in RNase-free Water (catalog number: 9012, TaKaRa) at 200 μM), was added, followed by still standing for 1 hour. Further, hemin (dissolved in DMSO at 100 μM) was added, followed by still standing for 1 hour. The solution containing the nucleic acid structure contained 3.2 nM of the nucleic acid structure, 10 μM of R509, 4.0 μM of hemin, and 200 mM of KCl. Then, an equivalent amount of 1-Step™ Ultra TMB-ELISA (catalog number: 34028, Thermo Scientific) was added, and the presence of R509 was visualized as a color.

[0130] The results are shown in FIG. **20**. A strong blue color was developed when R509 was contained. This revealed that R509 can be detected as a color change using the nucleic acid structure.

[0131] In conventional DNA nano-structures, split G-quartet sequences are highly stable, and it is difficult to maintain the split state; however, in the present invention, the pre-fixation on the opposite side (antiparallel type) successfully prevented the leakage of enzymatic activity. By using the technique of the present invention, the enzymatic activity can be controlled, for example, by varying the number of DNA strands attached to biomolecules. Unlike immunochromatography, the technique of the present invention is superior in that it does not require the fixation of antibodies or aptamers to underlying membranes or the like. In addition, the technique of the present invention, which can be composed only of nucleic acids, allows mass synthesis in a short time, cheaply and easily, with less lot differences compared to antibodies.

Example 5: Preparation 5 of Nucleic Acid Structure

[0132] A nucleic acid structure was prepared in the same manner as in Reference Example 1, except that some staple DNAs were changed to other staple DNAs. Tables 12 and 13 show the corresponding staple numbers of Reference Example 1, and the staple numbers, base sequences, and SEQ ID Nos of the staple DNAs used. The configuration of the obtained nucleic acid structure was the same as that of the nucleic acid structure of Reference Example 1, except that the partial structure 1 and the partial structure 2 each contained a protruding structure consisting of single-stranded DNA of a telomere element sequence, the partial nucleic acid structure 1 contained a protruding structure containing a nucleic acid aptamer sequence (complementary sequence of N1 (SEQ ID NO: 320)), and the partial nucleic acid structure 2 contained a sequence complementary to part of the nucleic acid aptamer sequence. The complementary sequence of N1 is a sequence complementary to the nucleic acid sequence of N_Sarbeco_P1 (National Institute of Infectious Diseases, Manual for the Detection of Pathogen 2019-nCOV Ver. 2.9.1).

TABLE 12

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
65	65long	TCTTAAACAGCTTGATGTGCCGTCGAGAGGGTGAGCCGCC	298
66	66shortTelo	GTATAGCCGCTTTCGAGGTGAATTTTTTAGGGTTAGGGT	299
69	69long	ACAATGACAACAACCATAGCGGGGTTTTGCTCCAGGTCAG	300
70	70shortTelo	GCGGATAAACCGATAGTTGCGCCGTTTTTAGGGTTAGGGT	301
73	73LongCompN1	AACATTGCCATTTTGATATATTCGGTCGCTAGGCTGAGACTCCT CACAAATAAA	302
74	74shortTelo	ATTAGGATTCGCCCACGCATAACCTTTTTAGGGTTAGGGT	303
78s	78s-telo	GTATTAAGGAGGCTTGCAGGGAGTTTTTTAGGGTTAGGGT	304
166s	166s-telo	GTCGAGGTAGAGATAGAACCCTTCTTTTTAGGGTTAGGGT	305
169	169LongCompN1	AACATTGCCATTTTGGACATTCTGGCCAACGCCGTAAAGCACTA AAATCCTGTT	306
170	170shortTelo	CTAAAGGGCGACCAGTAATAAAAGTTTTTAGGGTTAGGGT	307
173	173long	AGATTCACCAGTCACAAGCCCCCGATTTAGAGCGGTCCAC	308
174	174shortTelo	GGAAAGCCGATTATTTACATTGGCTTTTTAGGGTTAGGGT	309
177	177long	TCAATCGTCTGAAATGGGCGAACGTGGCGAGATCACCGCC	310
178	178shortTelo	GAAGAAAGACCTACATTTTGACGCTTTTTAGGGTTAGGGT	311

TABLE 13

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
81c	81cLong	CAATATTTTTGAATGGTATAAACAGTTAATGCGTCATACA	312
82s	82sBothLongN1	AACATTGCCATTTTTAAAGGCCGCTTTTGCGAATACGTGGCACA GATTTTTGGCAATGTTGTTCCTTGAGGAAGT	313
85c	85cLong	CGCGAACTGATAGCCCAACGGGGTCAGTGCCTACTGGTAA	314
86c	86cShortN1	AGTGCCCGCTATTAGTCTTTAATGTTTTTGGCAATGTTGTTCCT TGAGGAAGT	315
157c	157cLong	AAACGGCTACAGAGGCTACGTGGACTCCAACGTAGTCCACT	316
158c	158cShortN1	GAAAAACCTCGGAACGAGGGTAGCTTTTTGGCAATGTTGTTCC TTGAGGAAGT	317
161c	161cLong	GCAGCGAAAGACAGCAGTCTATCAGGGCGATGTAGGGTTG	318
162s	162sBothLongN1	AACATTGCCATTTTTGACCTGAAAGCGTAAGGGATCGTCACCC TCATTTTTGGCAATGTTGTTCCTTGAGGAAGT	319

[0133] Staple DNA with a staple number labeled "Telo" contains a telomere element sequence (including two G triple repeats). The structure formed by the telomere element sequence, its position, etc. are the same as those in Reference Example 2. The telomere element sequence in the partial structure 1 and the telomere element sequence in the partial structure 2 can be linked in the presence of N1 and a monovalent cation to form a G-quartet structure (nucleic acid enzyme: heme stacks can exhibit peroxidase (HRP) activity).

[0134] Staple DNA with a staple number labeled "N1" contains the complementary sequence of N1 or a sequence complementary to part of the sequence. These sequences form a protruding structure consisting of single-stranded DNA without complementary binding to M13mp18. Among the three structures based on the conformation of the Holliday junction (FIG. **9**: provided that Cross is an intermediate structure that mediates structural changes between Antiparallel and Parallel), when taking Antiparallel, the protruding structure in the partial structure 1 and the protruding structure in the partial structure 2 are arranged so that these protruding structures face each other (i.e., staple DNA containing the complementary sequence of N1 or a sequence complementary to part of the sequence is arranged). The complementary sequence of N1 in the partial structure 2 and the sequence complementary to part of the sequence in the partial structure 1 can be bonded based on complementary base pairing in the absence of N1.

[0135] That is, as shown in FIG. **21**, the nucleic acid structure of this Example is fixed at the antiparallel type based on complementary binding in the absence of N1, whereas in the presence of N1, the complementary bond dissociates to induce a structural change to the cross type, and the presence of a monovalent cation induces a structure change to the parallel type based on the formation of a G-quartet structure, resulting in the exhibition of peroxidase activity.

[0136] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} test substance, N1 (dissolved in RNase-free Water (catalog number: 9012, TakaRa) at 200 μM), was added, followed by still standing for 1 hour. AFM observation was then performed, and the presence of N1 was visualized as a structural difference of the nucleic acid structure. The solution containing the nucleic acid structure contained 3.2 nM of the nucleic acid structure, 1.0 μM of N1, and 200 mM of KCl. Further, each structure of the nucleic acid structure was counted from the acquired images, and its proportion was examined.

[0137] FIG. **22** shows AFM observation images. Further, FIG. **23** shows the proportion of each structure. The antiparallel type was the majority before the addition of N1, indicating that the addition of N1 induced a structural change from the antiparallel type to the parallel type.

Example 6: Detection 4 of Enzymatic Activity

[0138] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} solution) was added to a solution containing the nucleic acid structure (Example 5), followed by still standing for 5 minutes. Then, a test substance, N1 (dissolved in RNase-free Water (catalog number: 9012, TakaRa) at 200 μM), was added, followed by still standing for 15 minutes. Further, hemin (dissolved in DMSO at 100 μM) was added, followed by still standing for 30 minutes. The solution containing the nucleic acid structure contained 7.5 nM of the nucleic acid structure, 2.0 μM of N1, 5.0 μM of hemin, 0.0004% (w/v) of Triton X-100, and 200 mM of KCl. Then, an equivalent amount of 1-Step™ Ultra TMB-ELISA (catalog number: 34028, Thermo Scientific, pH: 4.9) mixed with 1×TAE/Mg^{2+} and 200 mM of KCl was added (pH of the solution after mixing: 5.5), and 10 minutes later, the presence of N1 was visualized as a color.

[0139] The results are shown in FIG. **24**. A strong blue color was developed when N1 was contained. This revealed that N1 can be detected as a color change using the nucleic acid structure.

[0140] In conventional DNA nano-structures, split G-quartet sequences are highly stable, and it is difficult to maintain the split state; however, in the present invention, the pre-fixation on the opposite side (antiparallel type) successfully prevented the leakage of enzymatic activity. By using the technique of the present invention, the enzymatic activity can be controlled, for example, by varying the number of DNA strands attached to biomolecules. Unlike immunochromatography, the technique of the present invention is superior in that it does not require the fixation of antibodies or aptamers to underlying membranes or the like. In addition, the technique of the present invention, which can be composed only of nucleic acids, allows mass synthesis in a short time, cheaply and easily, with less lot differences compared to antibodies.

Example 7: Preparation 6 of Nucleic Acid Structure

[0141] A nucleic acid structure was prepared in the same manner as in Reference Example 1, except that some staple DNAs were changed to other staple DNAs. Tables 14 and 15 show the corresponding staple numbers of Reference Example 1, and the staple numbers, base sequences, and SEQ ID Nos of the staple DNAs used. The configuration of the obtained nucleic acid structure was the same as that of the nucleic acid structure of Reference Example 1, except that the partial structure 1 and the partial structure 2 each contained a protruding structure consisting of single-stranded DNA of a telomere element sequence, the partial nucleic acid structure 1 contained a protruding structure containing a nucleic acid aptamer sequence (HD1 (thrombin aptamer, Bock, L. et al., (1992) Nature 355, 564-566, Munzar J. D. et al., (2018) Nat Commun. doi: 10.1038/s41467-017-02556-3): SEQ ID NO: 343), and the partial nucleic acid structure 2 contained a sequence complementary to part of the nucleic acid aptamer sequence.

TABLE 14

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
65	65long	TCTTAAACAGCTTGATGTGCCGTCGAGAGGGTGAGCCGCC	321
66	66shortTelo	GTATAGCCGCTTTCGAGGTGAATTTTTTAGGGTTAGGGT	322
69	69long	ACAATGACAACAACCATAGCGGGGTTTTGCTCCAGGTCAG	323
70	70shortTelo	GCGGATAAACCGATAGTTGCGCCGTTTTTAGGGTTAGGGT	324
73	73LongCompHD1T1	ACACCAACCTTTTTGATATATTCGGTCGCTAGGCTGAGACTCCT CACAAATAAA	325
74	74shortTelo	ATTAGGATTCGCCCACGCATAACCTTTTTAGGGTTAGGGT	326
78s	78s-telo	GTATTAAGGAGGCTTGCAGGGAGTTTTTTAGGGTTAGGGT	327
166s	166s-telo	GTCGAGGTAGAGATAGAACCCTTCTTTTTAGGGTTAGGGT	328
169	169LongCompHD1T1	ACACCAACCTTTTTGGACATTCTGGCCAACGCCGTAAAGCACTA AAATCCTGTT	329
170	170shortTelo	CTAAAGGGCGACCAGTAATAAAAGTTTTTAGGGTTAGGGT	330
173	173long	AGATTCACCAGTCACAAGCCCCCGATTTAGAGCGGTCCAC	331
174	174shortTelo	GGAAAGCCGATTATTTACATTGGCTTTTTAGGGTTAGGGT	332
177	177long	TCAATCGTCTGAAATGGGCGAACGTGGCGAGATCACCGCC	333
178	178shortTelo	GAAGAAAGACCTACATTTTGACGCTTTTTAGGGTTAGGGT	334

TABLE 15

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
81c	81cLong	CAATATTTTTGAATGGTATAAACAGTTAATGCGTCATACA	335
82s	82sBothLongHD1T1	ACACCAACCTTTTTTAAAGGCCGCTTTTGCGAATACGTGGCACA GATTTGTGGTTGGTGTGGTTGGTCATA	336
85c	85cLong	CGCGAACTGATAGCCCAACGGGGTCAGTGCCTACTGGTAA	337
86c	86cShortHD1T1	AGTGCCCGCTATTAGTCTTTAATGTTTTAGTCCGTGGTAGGGC AGGTTGGGGTGACT	338
157c	157cLong	AAACGGCTACAGAGGCTACGTGGACTCCAACGTAGTCCACT	339
158c	158cShortHD1T1	GAAAAACCTCGGAACGAGGGTAGCTTTTAGTCCGTGGTAGGGC AGGTTGGGGTGACT	340
161c	161cLong	GCAGCGAAAGACAGCAGTCTATCAGGGCGATGTAGGGTTG	341
162s	162sBothLongHD1T1	ACACCAACCTTTTTTGACCTGAAAGCGTAAGGGATCGTCACCC TCATTTGTGGTTGGTGTGGTTGGTCATA	342

[0142] Staple DNA with a staple number labeled "Telo" contains a telomere element sequence (including two G triple repeats). The structure formed by the telomere element sequence, its position, etc. are the same as those in Reference Example 2. The telomere element sequence in the partial structure 1 and the telomere element sequence in the partial structure 2 can be linked in the presence of thrombin and a monovalent cation to form a G-quartet structure (nucleic acid enzyme: heme stacks can exhibit peroxidase (HRP) activity).

[0143] Staple DNA with a staple number labeled "HD1T1" contains the complementary sequence of HD1 or a sequence complementary to part of the sequence. These sequences form a protruding structure consisting of single-stranded DNA without complementary binding to M13mp18. Among the three structures based on the conformation of the Holliday junction (FIG. **9**: provided that Cross is an intermediate structure that mediates structural changes between Antiparallel and Parallel), when taking Antiparallel, the protruding structure in the partial structure 1 and the protruding structure in the partial structure 2 are arranged so that these protruding structures face each other (i.e., staple DNA containing the complementary sequence of HD1T1 or a sequence complementary to part of the sequence is arranged). The complementary sequence of HD1T1 in the partial structure 2 and the sequence complementary to part of the sequence in the partial structure 1 can be bonded based on complementary base pairing in the absence of thrombin.

[0144] That is, as shown in FIG. **25**, the nucleic acid structure of this Example is fixed at the antiparallel type based on complementary binding in the absence of thrombin, whereas in the presence of thrombin, the complementary bond dissociates to induce a structural change to the cross type, and the presence of a monovalent cation induces a structure change to the parallel type based on the formation of a G-quartet structure, resulting in the exhibition of peroxidase activity.

[0145] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} test substance, thrombin, was added, followed by still standing for 1 hour. AFM observation was then performed, and the presence of thrombin was visualized as a structural difference of the nucleic acid structure. The solution containing the nucleic acid structure contained 3.2 nM of the nucleic acid structure, 2.0 μM of thrombin, and 200 mM of KCl. Further, each structure of the nucleic acid structure was counted from the acquired images, and its proportion was examined.

[0146] FIG. **26** shows AFM observation images. Further, FIG. **27** shows the proportion of each structure. The antiparallel type was the majority before the addition of thrombin, indicating that the addition of thrombin induced a structural change from the antiparallel type to the parallel type.

Example 8: Detection 5 of Enzymatic Activity

[0147] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} solution) was added to a solution containing the nucleic acid structure (Example 7), followed by still standing for 5 minutes. Then, a test substance, thrombin, was added, followed by still standing for 15 minutes. Further, hemin (dissolved in DMSO at 100 μM) was added, followed by still standing for 30 minutes. The solution containing the nucleic acid structure contained 7.5 nM of the nucleic acid structure, 2.0 μM of thrombin, 5.0 μM of hemin, 0.0004% (w/v) of Triton X-100, and 200 mM of KCl. Then, an equivalent amount of 1-Step™ Ultra TMB-ELISA (catalog number: 34028, Thermo Scientific) was added, and the presence of thrombin was visualized as a color.

[0148] The results are shown in FIG. **28**. A strong blue color was developed when thrombin was contained. This revealed that thrombin can be detected as a color change using the nucleic acid structure.

[0149] In conventional DNA nano-structures, split G-quartet sequences are highly stable, and it is difficult to maintain the split state; however, in the present invention, the pre-fixation on the opposite side (antiparallel type) successfully prevented the leakage of enzymatic activity. By using the technique of the present invention, the enzymatic activity can be controlled, for example, by varying the number of DNA strands attached to biomolecules. Unlike immunochromatography, the technique of the present invention is superior in that it does not require the fixation of antibodies or aptamers to underlying membranes or the like. In addition, the technique of the present invention, which can be composed only of nucleic acids, allows mass synthesis in a short time, cheaply and easily, with less lot differences compared to antibodies.

Example 9: Preparation 7 of Nucleic Acid Structure

[0150] NanoLuc (Promega) is a luciferase with furimazine as a substrate, and its luminescence intensity is very strong, about 100 times the luminescence intensity of firefly luciferase. LgBiT (Promega) and SmBIT (Promega) are subunits of NanoLuc, and they are reconstituted as a luciferase by interacting with each other. LgBiT is a large subunit of approximately 18 kDa, and SmBiT is a small subunit of 11 amino acid residues. LgBiT and SmBiT have a dissociation constant of 190 μM and extremely low affinity, and they do not express luciferase activity when simply mixed (Andrew, D., (2016) ACS Chem. Biol. 11, 400-408). In this Example, only when the nucleic acid structure is structurally changed to the parallel type, SmBiT and LgBiT are brought close enough to interact, so that SmBiT and LgBiT interact only in the presence of a test substance, resulting in the exhibition of luciferase activity. Specifically, a nucleic acid structure was prepared in the following manner.

[0151] SmBiT and 95w (SEQ ID NO: 95) were bonded together. SmBIT (VTGYRLFEEIL-orn (N3)) modified with an azide group on the C-terminal side was chemically synthesized (Sigma-Aldrich). On the other hand, 95w modified with an amine on the 5-terminal side was chemically synthesized (IDT), and 1/30 equivalent (molar ratio) thereof was mixed with DBCO-NHS Ester (Click Chemistry Tools). Azide group-modified SmBiT and DBCO-modified 95w were mixed at a molar ratio of 1:1, and the mixture was allowed to stand for 16 hours and then purified by HPLC, thereby obtaining SmBiT-95w.

[0152] LgBiT and 138w (SEQ ID NO: 138) were bonded together. Lgbit was inserted into pH6HTN His6HaloTag T7 Vector (Promega) using the In-Fusion HD Cloning Kit (Clontech), and transformed into DH5α (SMOBIO Technology, Inc.) to obtain an LgBiT expression vector. The vector was transformed into BL21 (SMOBIO Technology, Inc.) and cultured in LB medium, and protein expression was induced with 0.4 mM of IPTG (Nacalai). The resultant was purified using the His-Spin Protein Miniprep Kit (Zymo Research), thereby obtaining HaloTag-fused LgBiT. On the other hand, 138w modified with an amine on the 5-terminal side was chemically synthesized (IDT), and 1/30 equivalent (molar ratio) thereof was mixed with HaloTag (registered trademark) Succinimidyl Ester (04) Ligand (Promega). The mixture was allowed to stand for 16 hours, and then purified by HPLC, thereby obtaining HaloTag ligand-binding 138w staple. HaloTag-fused LgBiT and HaloTag ligand-binding 138w staple were mixed at a molar ratio of 1:1, and the mixture was allowed to stand for 16 hours. Then, unreacted 138w staple was removed using a centrifugal ultrafiltration filter (Amicon Ultra 0.5 mL-50 K, Millipore), thereby obtaining LgBiT-138w.

[0153] The obtained Lgbit-138w was confirmed by SDS polyacrylamide electrophoresis. FIG. **29** shows an electrophoretic image. From left, a marker (PageRuler Plus Prestain Protein Ladder, Thermo Scientific), HaloTag-fused LgBiT, and LgBiT-138w are shown. HaloTag-fused LgBiT migrated at a position similar to its expected molecular weight (approximately 54 kDa). In addition, the increase in molecular weight was confirmed based on the binding procedure of 138w shown above, suggesting that 138w bound to LgBiT.

[0154] A nucleic acid structure was prepared in the same manner as in Reference Example 1, except that some staple DNAs were changed to other staple DNAs. Tables 16 and 17 show the corresponding staple numbers of Reference Example 1, and the staple numbers, base sequences, and SEQ ID Nos of the staple DNAs used. LgBiT-138w was added at a molar ratio of 1:1 after the nucleic acid structure was prepared with other staple DNAs except for this. The configuration of the obtained nucleic acid structure was the same as that of the nucleic acid structure of Reference Example 1, except that the partial structure 1 and the partial structure 2 each contained a protruding structure consisting of single-stranded DNA of a telomere element sequence, the partial nucleic acid structure 1 contained a protruding structure containing a nucleic acid aptamer sequence (complementary sequence of miR-20 (Mature sequence hsa-miR-20a-5p: SEQ ID NO: 274)), and the partial nucleic acid structure 2 contained a sequence complementary to part of the nucleic acid aptamer sequence, and contained SmbiT and LgBiT.

TABLE 16

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
65	65long	TCTTAAACAGCTTGATGTGCCGTCGAGAGGGTGAGCCGCC	345
66	66shortTelo	GTATAGCCGCTTTCGAGGTGAATTTTTTAGGGTTAGGGT	346
69	69long	ACAATGACAACAACCATAGCGGGGTTTTGCTCCAGGTCAG	347
70	70shortTelo	GCGGATAAACCGATAGTTGCGCCGTTTTTAGGGTTAGGGT	348
73	73LongCompM20	GTGCAGGTAGTTTTGATATATTCGGTCGCTAGGCTGAGACTCC TCACAAATAAA	349
74	74shortTelo	ATTAGGATTCGCCCACGCATAACCTTTTTAGGGTTAGGGT	350
78s	78s-telo	GTATTAAGGAGGCTTGCAGGGAGTTTTTTAGGGTTAGGGT	351
166s	166s-telo	GTCGAGGTAGAGATAGAACCCTTCTTTTTAGGGTTAGGGT	352
169	169LongCompM20	GTGCAGGTAGTTTTGGAACATTTCTGGCCAACGCCGTAAAGCA CTAAAATCCTGTT	353
170	170shortTelo	CTAAAGGGCGACCAGTAATAAAAGTTTTTAGGGTTAGGGT	354
173	173long	AGATTCACCAGTCACAAGCCCCCGATTTAGAGCGGTCCAC	355
174	174shortTelo	GGAAAGCCGATTATTTACATTGGCTTTTTAGGGTTAGGGT	356
177	177long	TCAATCGTCTGAAATGGGCGAACGTGGCGAGATCACCGCC	357
178	178shortTelo	GAAGAAAGACCTACATTTTGACGCTTTTTAGGGTTAGGGT	358

TABLE 17

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
81c	81cLong	CAATATTTTTGAATGGTATAAACAGTTAATGCGTCATACA	359
82s	82sBothLongM20	GTGCAGGTAGTTTTTAAAGGCCGCTTTTGCGAATACGTGGCAC AGATTTTCTACCTGCACTATAAGCACTTTA	360
85c	85cLong	CGCGAACTGATAGCCCAACGGGGTCAGTGCCTACTGGTAA	361
86c	86cShortMainM20	AGTGCCCGCTATTAGTCTTTAATGTTTTCTACCTGCACTATAAG CACTTTA	362
157c	157cLong	AAACGGCTACAGAGGCTACGTGGACTCCAACGTAGTCCACT	363
158c	158cShortMainM20	GAAAAACCTCGGAACGAGGGTAGCTTTTCTACCTGCACTATAA GCACTTTA	364
161c	161cLong	GCAGCGAAAGACAGCAGTCTATCAGGGCGATGTAGGGTTG	365
162s	162sBothLongM20	GTGCAGGTAGTTTTGACCTGAAAGCGTAAGGGATCGTCACCC TCATTTTCTACCTGCACTATAAGCACTTTA	366

TABLE 17-continued

Corresponding staple No. of Ref. Ex. 1	staple No.	Sequence	SEQ ID No
95w	SmBIT-95w	SmBIT-AAGGAATTGAGGAAGG	367
138w	LgBiT-138w	LgBiT-ATCCGCGACCTGCTCC	368

[0155] Staple DNA with a staple number labeled "Telo" contains a telomere element sequence (including two G triple repeats). The structure formed by the telomere element sequence, its position, etc. are the same as those in Reference Example 2. The telomere element sequence in the partial structure 1 and the telomere element sequence in the partial structure 2 can be linked in the presence of miR-20 and a monovalent cation to form a G-quartet structure.

[0156] Staple DNA with a staple number labeled "M20" contains the complementary sequence of miR-20 or a sequence complementary to part of the sequence. These sequences form a protruding structure consisting of single-stranded DNA without complementary binding to M13mp18. Among the three structures based on the conformation of the Holliday junction (FIG. **9**: provided that Cross is an intermediate structure that mediates structural changes between Antiparallel and Parallel), when taking Antiparallel, the protruding structure in the partial structure 1 and the protruding structure in the partial structure 2 are arranged so that these protruding structures face each other (i.e., staple DNA containing the complementary sequence of miR-20 or a sequence complementary to part of the sequence is arranged). The complementary sequence of miR-20 in the partial structure 2 and the sequence complementary to part of the sequence in the partial structure 1 can be bonded based on complementary base pairing in the absence of miR-20.

[0157] That is, as shown in FIG. **30**, the nucleic acid structure of this Example is fixed at the antiparallel type based on complementary binding in the absence of miR-20, whereas in the presence of miR-20, the complementary bond dissociates to induce a structural change to the cross type, and the presence of a monovalent cation induces a structure change to the parallel type based on the formation of a G-quartet structure. As a result, SmBiT and LgBiT are in close proximity and interaction to be reconstituted as a luciferase.

[0158] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} test substance, miR20 (dissolved in RNase-free Water (catalog number: 9012, TakaRa) at 200 μM), was added, followed by still standing for 1 hour. AFM observation was then performed, and the presence of miR20 was visualized as a structural difference of the nucleic acid structure. The solution containing the nucleic acid structure contained 3.2 nM of the nucleic acid structure, 5 μM of miR20, and 200 mM of KCl. Further, each structure of the nucleic acid structure was counted from the acquired images, and its proportion was examined.

[0159] FIG. **31** shows AFM observation images. Further, FIG. **32** shows the proportion of each structure. The antiparallel type was the majority before the addition of miR-20, indicating that the addition of miR-20 induced a structural change from the antiparallel type to the parallel type.

Example 10: Detection 6 of Enzymatic Activity

[0160] A KCl solution (1 M KCl was dissolved in 1×TAE/Mg^{2+} solution) was added to a solution containing the nucleic acid structure (Example 9), followed by still standing for 5 minutes. Then, a test substance, miR20 (dissolved in RNase-free Water (catalog number: 9012, TakaRa) at 200 μM), was added, followed by still standing for 15 minutes. The solution containing the nucleic acid structure contained 3.2 nM of the nucleic acid structure, 1 μM or 5 μM of miR20, and 200 mM of KCl. Then, furimazine (Nano-Glo Substrate) attached to the Nano-Glo HiBiT Blotting System (Promega) was added, and the luminescence intensity was quantified by a plate reader (SpectraMax iD5, Molecular Devices).

[0161] The results are shown in FIG. **33**. The luminescence intensity increased when miR-20 was contained. This revealed that miR-20 can be detected using the nucleic acid structure.

[0162] SmBiT and LgBiT have a dissociation constant of about 190 μM and extremely low affinity, and they do not express luciferase activity when simply mixed (Andrew, D., (2016) ACS Chem. Biol. 11, 400-408). In the present invention, only when the nucleic acid structure was structurally changed to the parallel type, SmBiT and LgBiT were in close proximity and interaction to succeed in significantly increasing luciferase activity in the presence of a test substance. Further, the luciferase NanoLuc, which is reconstituted by the interaction of SmBiT and LgBiT, is known to have a luminescence intensity about 100 times that of firefly luciferase, and can be photographed with a smartphone camera (Arts, R. et al., (2016) Anal. Chem. 88, 4525-4532). Therefore, visualization is possible by increasing the concentration of the nucleic acid structure.

SEQUENCE LISTING

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 368

<210> SEQ ID NO 1
<211> LENGTH: 38
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 1s

<400> SEQUENCE: 1

cctaatttt ttatcctgaa tcttaccaac gctatttt 38

<210> SEQ ID NO 2
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 2s

<400> SEQUENCE: 2

ttttacgagc gtctgcagca ccgttttt 28

<210> SEQ ID NO 3
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 3s

<400> SEQUENCE: 3

caagtttgaa tgaaaccatc gatattccag ag 32

<210> SEQ ID NO 4
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 4s

<400> SEQUENCE: 4

agctacaagc cagttacaaa ataaggccgg aa 32

<210> SEQ ID NO 5
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 5s

<400> SEQUENCE: 5

acgtcacccc tttagcgtca gactacaacg cc 32

<210> SEQ ID NO 6
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 6s

<400> SEQUENCE: 6

attatttaaa gattagttgc tattttgcac cc 32

<210> SEQ ID NO 7
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 7s

<400> SEQUENCE: 7

-continued

ttttcatccc attaccatta gcaaacagcc at 32

<210> SEQ ID NO 8
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 8s

<400> SEQUENCE: 8

ttgaagcctt aaatctccca at 22

<210> SEQ ID NO 9
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 9s

<400> SEQUENCE: 9

ccaaataaga aacgatttag agccagca 28

<210> SEQ ID NO 10
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 10

<400> SEQUENCE: 10

ttttaatcag tagcatagtt agcgtttt 28

<210> SEQ ID NO 11
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 11

<400> SEQUENCE: 11

tcgtcttttc cacagacagc cctcgacaga at 32

<210> SEQ ID NO 12
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 12

<400> SEQUENCE: 12

tttttaacga tctaaagttt tg 22

<210> SEQ ID NO 13
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 13s

<400> SEQUENCE: 13

aaatcaccag tagcaggcat tt 22

-continued

<210> SEQ ID NO 14
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 14s

<400> SEQUENCE: 14

tgccatctga gccatttggg aatttttgtt ta 32

<210> SEQ ID NO 15
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 15s

<400> SEQUENCE: 15

gcgttttaaa tgaaaatagc agcctcaccg tc 32

<210> SEQ ID NO 16
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 16s

<400> SEQUENCE: 16

accgactttt tcataatcaa aatcctcatt tt 32

<210> SEQ ID NO 17
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 17s

<400> SEQUENCE: 17

taaaaacaat agaaggctta tccggtattc taagaacgcg ag 42

<210> SEQ ID NO 18
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 18s

<400> SEQUENCE: 18

aggtgaatta tttacagaga gaataaca 28

<210> SEQ ID NO 19
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 19s

<400> SEQUENCE: 19

accggaacgg aaattattca ttaa 24

<210> SEQ ID NO 20
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 20s

<400> SEQUENCE: 20

aatcagatgg gaagcgcatt agacgaaggt aa 32

<210> SEQ ID NO 21
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 21s

<400> SEQUENCE: 21

atattgaccg cctccctcag agccagaacc gc 32

<210> SEQ ID NO 22
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 22s

<400> SEQUENCE: 22

taactgaaca ttaccgcgcc caatagcaag ca 32

<210> SEQ ID NO 23
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 23

<400> SEQUENCE: 23

tgtatggggt caccagtaca aactgtagcg cg 32

<210> SEQ ID NO 24
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 24

<400> SEQUENCE: 24

tgtagcatcc agacgttagt aaatgaattt tc 32

<210> SEQ ID NO 25
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 25s

<400> SEQUENCE: 25

cagaaccgaa ccgattgagg gagggggaga at 32

<210> SEQ ID NO 26
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 26s

<400> SEQUENCE: 26

-continued

```
gtaggaatca ccctgaacaa agtccaaaag gg                              32


<210> SEQ ID NO 27
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 27s

<400> SEQUENCE: 27

cgacattccc accctcagag ccacccgtac tc                              32


<210> SEQ ID NO 28
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 28s

<400> SEQUENCE: 28

attgagcgaa gcaagccgtt tttattttca tc                              32


<210> SEQ ID NO 29
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 29s

<400> SEQUENCE: 29

gagccgcctt accagcgcca aagaagaggg ta                              32


<210> SEQ ID NO 30
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 30s

<400> SEQUENCE: 30

tcgagaacct aatatcagag agatagaaaa tt                              32


<210> SEQ ID NO 31
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 31s

<400> SEQUENCE: 31

catatggtac cagaaccacc accatgatat aa                              32


<210> SEQ ID NO 32
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 32s

<400> SEQUENCE: 32

agaattgagt attaaaccaa gtaccgcact ca                              32
```

-continued

<210> SEQ ID NO 33
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 33

<400> SEQUENCE: 33

tcggtcatgt aacactgagt ttcattttgc 30

<210> SEQ ID NO 34
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 34

<400> SEQUENCE: 34

cccatgtacc agcccctta ttagcgtt 28

<210> SEQ ID NO 35
<211> LENGTH: 26
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 35

<400> SEQUENCE: 35

taaacaactt tcaacagttt cagcgg 26

<210> SEQ ID NO 36
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 36s

<400> SEQUENCE: 36

gccagcattt ttgtcacaat caataaccca ca 32

<210> SEQ ID NO 37
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 37s

<400> SEQUENCE: 37

aagaacgggt taagcccaat aataccacgg aa 32

<210> SEQ ID NO 38
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 38s

<400> SEQUENCE: 38

taagtttatg acaggaggtt gaggagtacc ag 32

<210> SEQ ID NO 39
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 39s

<400> SEQUENCE: 39

aaacaatgtc ggctgtcttt ccttatcatt cc 32

<210> SEQ ID NO 40
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 40s

<400> SEQUENCE: 40

acgattggaa gaaacgcaaa gacaagagca ag 32

<210> SEQ ID NO 41
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 41

<400> SEQUENCE: 41

agtgagaagc aagcccaata ggaa 24

<210> SEQ ID NO 42
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 42s

<400> SEQUENCE: 42

atcaataaaa atagcaatag ctataggtgg ca 32

<210> SEQ ID NO 43
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 43s

<400> SEQUENCE: 43

acatataacc ttgatattca caaaagagaa gg 32

<210> SEQ ID NO 44
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 44s

<400> SEQUENCE: 44

agccctttta atttacgagc atgtagaaac ca 32

<210> SEQ ID NO 45
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 45s

<400> SEQUENCE: 45

-continued

tcctcattga aaatacatac ataacttacc ga 32

<210> SEQ ID NO 46
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 46s

<400> SEQUENCE: 46

aaataatatc ccatccttaa gaaaagtaag catatgttag 40

<210> SEQ ID NO 47
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 47s

<400> SEQUENCE: 47

caaacgtaaa agccagaatg gaaaacatga aa 32

<210> SEQ ID NO 48
<211> LENGTH: 26
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 48s

<400> SEQUENCE: 48

gaatcataat aaggcgttaa ataaga 26

<210> SEQ ID NO 49
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 49s

<400> SEQUENCE: 49

aacaaagtaa gtcctgaaca agaaaaacac cg 32

<210> SEQ ID NO 50
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 50s

<400> SEQUENCE: 50

tctgaattac tccttattac gcaggatagc cg 32

<210> SEQ ID NO 51
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 51s

<400> SEQUENCE: 51

tgtgataaat tactagaaaa agcctttatc aa 32

-continued

<210> SEQ ID NO 52
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 52

<400> SEQUENCE: 52

agccaccacc accggaacca gagccacc 28

<210> SEQ ID NO 53
<211> LENGTH: 34
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 53

<400> SEQUENCE: 53

cagggatata gaaaggaaca actaaaggaa ttgc 34

<210> SEQ ID NO 54
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 54

<400> SEQUENCE: 54

gaataataat tttttcgaac cgccaccctc ag 32

<210> SEQ ID NO 55
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 55s

<400> SEQUENCE: 55

caatagatta ccagaaggaa accgaactgg ca 32

<210> SEQ ID NO 56
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 56s

<400> SEQUENCE: 56

tgattaagta ccgttccagt aagccccctg cc 32

<210> SEQ ID NO 57
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 57

<400> SEQUENCE: 57

gctccaaata gtaccgccac cctcgccacc ct 32

<210> SEQ ID NO 58
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 58

<400> SEQUENCE: 58

caccctcaac gttgaaaatc tccaaaaaaa ag 32

<210> SEQ ID NO 59
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 59s

<400> SEQUENCE: 59

atcatatgtt taatggtttg aaataccgac cg 32

<210> SEQ ID NO 60
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 60s

<400> SEQUENCE: 60

caataataat gcagaacgcg cctgtgttta gt 32

<210> SEQ ID NO 61
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 61

<400> SEQUENCE: 61

atcagcttcg gaataggtgt atcacaccct ca 32

<210> SEQ ID NO 62
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 62

<400> SEQUENCE: 62

aggaggttag gagcctttaa ttgtatcggt tt 32

<210> SEQ ID NO 63
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 63s

<400> SEQUENCE: 63

tggcttttac ggaataccca aaagaggaaa cg 32

<210> SEQ ID NO 64
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 64s

<400> SEQUENCE: 64

-continued

gacctaaacg ttatacaaat tcttacaaca tg   32

<210> SEQ ID NO 65
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 65

<400> SEQUENCE: 65

agcttgatgt gccgtcgaga gggtgagccg cc   32

<210> SEQ ID NO 66
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 66

<400> SEQUENCE: 66

gtatagccgc tttcgaggtg aatttcttaa ac   32

<210> SEQ ID NO 67
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 67s

<400> SEQUENCE: 67

ttcagctaag gtttaacgtc agataattgc gt   32

<210> SEQ ID NO 68
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 68s

<400> SEQUENCE: 68

aaagccaaat attttagtta atttcatctt ct   32

<210> SEQ ID NO 69
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 69

<400> SEQUENCE: 69

aacaaccata gcggggtttt gctccaggtc ag   32

<210> SEQ ID NO 70
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 70

<400> SEQUENCE: 70

gcggataaac cgatagttgc gccgacaatg ac   32

-continued

<210> SEQ ID NO 71
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 71s

<400> SEQUENCE: 71

agtaacagca gacgacgaca ataaaccagt at 32

<210> SEQ ID NO 72
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 72s

<400> SEQUENCE: 72

tttcaaatcg ctcaacagta gggcgtaaag ta 32

<210> SEQ ID NO 73
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 73

<400> SEQUENCE: 73

cggtcgctag gctgagactc ctcacaaata aa 32

<210> SEQ ID NO 74
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 74

<400> SEQUENCE: 74

attaggattc gcccacgcat aaccgatata tt 32

<210> SEQ ID NO 75
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 75s

<400> SEQUENCE: 75

attctgtcta ccttttacat cgggaaaatt at 32

<210> SEQ ID NO 76
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 76s

<400> SEQUENCE: 76

gaatcgccaa gacaaagaac gcgagaaaac tt 32

<210> SEQ ID NO 77
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 77s

<400> SEQUENCE: 77

tatttcggaa cctattattc tgaagcgcag tc 32

<210> SEQ ID NO 78
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 78s

<400> SEQUENCE: 78

gtattaagga ggcttgcagg gagt 24

<210> SEQ ID NO 79
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 79s

<400> SEQUENCE: 79

taacggataa agtaccgaca aaagttaatt ga 32

<210> SEQ ID NO 80
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 80s

<400> SEQUENCE: 80

ccaatcgcat atttaacaac gccaagtaat aa 32

<210> SEQ ID NO 81
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 81c

<400> SEQUENCE: 81

ttgaatggta taaacagtta atgcgtcata ca 32

<210> SEQ ID NO 82
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 82s

<400> SEQUENCE: 82

taaaggccgc ttttgcgaat acgtggcaca gacaatattt 40

<210> SEQ ID NO 83
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 83w

<400> SEQUENCE: 83

-continued

agattttcga tgatacagga gtgttgagta ac 32

<210> SEQ ID NO 84
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 84w

<400> SEQUENCE: 84

taagttttaa aacagaaata aagagaatat ac 32

<210> SEQ ID NO 85
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 85c

<400> SEQUENCE: 85

gatagcccaa cggggtcagt gcctactggt aa 32

<210> SEQ ID NO 86
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 86c

<400> SEQUENCE: 86

agtgcccgct attagtcttt aatgcgcgaa ct 32

<210> SEQ ID NO 87
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 87w

<400> SEQUENCE: 87

ttgcacgtgg tgaggcggtc agtacagaag at 32

<210> SEQ ID NO 88
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 88w

<400> SEQUENCE: 88

gcctgcaata gaacctacca tatcagaaac aa 32

<210> SEQ ID NO 89
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 89w

<400> SEQUENCE: 89

cgaaccacca gttaacacc 19

-continued

<210> SEQ ID NO 90
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 90w

<400> SEQUENCE: 90

aaaacagata aaacatcgc 19

<210> SEQ ID NO 91
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 91w

<400> SEQUENCE: 91

cattaaaaat accgaa 16

<210> SEQ ID NO 92
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 92w

<400> SEQUENCE: 92

ggaagggtca gtgccacgct gaga 24

<210> SEQ ID NO 93
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 93w

<400> SEQUENCE: 93

caaatgaagt ttggattata cttcaatacc aa 32

<210> SEQ ID NO 94
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 94w

<400> SEQUENCE: 94

gcaaatcaac agttgagcca gcag 24

<210> SEQ ID NO 95
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 95w

<400> SEQUENCE: 95

aaggaattga ggaagg 16

<210> SEQ ID NO 96
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 96w

<400> SEQUENCE: 96

tcctgattaa atctaaagca tcacaatatc tg 32

<210> SEQ ID NO 97
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 97w

<400> SEQUENCE: 97

acctcaaaga tgatggcaat tcatttattc at 32

<210> SEQ ID NO 98
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 98w

<400> SEQUENCE: 98

ctaacaacta tcaaaccctc aatccttgct ga 32

<210> SEQ ID NO 99
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 99w

<400> SEQUENCE: 99

gtcagttgtt atctaaaata tctttaggag ca 32

<210> SEQ ID NO 100
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 100w

<400> SEQUENCE: 100

gattatcaac gttattaatt ttaataaatc ct 32

<210> SEQ ID NO 101
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 101w

<400> SEQUENCE: 101

gtaacattga gcggaattat catctgatga aa 32

<210> SEQ ID NO 102
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 102w

<400> SEQUENCE: 102

-continued

taatacatat tcgacaactc gtataagttt ga 32

<210> SEQ ID NO 103
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 103w

<400> SEQUENCE: 103

ttgcccgata atagattaga gccgtcaata ga 32

<210> SEQ ID NO 104
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 104w

<400> SEQUENCE: 104

accagaagat cattttgcgg aacatttttt agactt 36

<210> SEQ ID NO 105
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 105s

<400> SEQUENCE: 105

aacaatttaa gaaaacaaaa ttaattttaa gaaacc 36

<210> SEQ ID NO 106
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 106w

<400> SEQUENCE: 106

tacaaacatt gaggatttag aagttttt 28

<210> SEQ ID NO 107
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 107s

<400> SEQUENCE: 107

caaacatcca tttgaattac ctttaacata gc 32

<210> SEQ ID NO 108
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 108w

<400> SEQUENCE: 108

aaacagtacc tgagcaaaag aagaatattc ct 32

-continued

<210> SEQ ID NO 109
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 109s

<400> SEQUENCE: 109

ttcaattaca taaatcaata tatgctatta at 32

<210> SEQ ID NO 110
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 110w

<400> SEQUENCE: 110

taaccttgat cgcgcagagg cgaacaatat aa 32

<210> SEQ ID NO 111
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 111s

<400> SEQUENCE: 111

gagaatattc gcctgattgc tttgtgaata at 32

<210> SEQ ID NO 112
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 112s

<400> SEQUENCE: 112

tttaggcact atatgtaaat gctgatgcaa at 32

<210> SEQ ID NO 113
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 113s

<400> SEQUENCE: 113

gttacaaaga ggcattttcg agccacatgt aa 32

<210> SEQ ID NO 114
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 114s

<400> SEQUENCE: 114

ttatataact tctgtaaatc gtcgtgagtg aa 32

<210> SEQ ID NO 115
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 115s

<400> SEQUENCE: 115

taattttctt ttaacctccg gcttaggttg gg 32

<210> SEQ ID NO 116
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 116s

<400> SEQUENCE: 116

gactacctcc ttagaatcct tgaatttaat gg 32

<210> SEQ ID NO 117
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 117s

<400> SEQUENCE: 117

gatagctttt tatcaaaatc ataggtctga ga 32

<210> SEQ ID NO 118
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 118s

<400> SEQUENCE: 118

gagtcaatag tgaaagatta agacgctgag ttttttacat tt 42

<210> SEQ ID NO 119
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 119s

<400> SEQUENCE: 119

accataaatc aaaagttcag aaaacgagaa ttttaaaatg tt 42

<210> SEQ ID NO 120
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 120s

<400> SEQUENCE: 120

tttaaacaat caggtcttta ccctgactat ta 32

<210> SEQ ID NO 121
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 121s

<400> SEQUENCE: 121

-continued

tagtcagaat tcattgaatc cccccggaat cg 32

<210> SEQ ID NO 122
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 122s

<400> SEQUENCE: 122

gagtagatag caaagcggat tgcatcaaaa ag 32

<210> SEQ ID NO 123
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 123w

<400> SEQUENCE: 123

agaccaggaa agaggacaga tgaatttt 28

<210> SEQ ID NO 124
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 124w

<400> SEQUENCE: 124

agtgaatatc aacgtaacaa agctttttcg gtgtac 36

<210> SEQ ID NO 125
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 125s

<400> SEQUENCE: 125

aaaagaagtt ttgcatagcg agaggctttt ttttgctcat tc 42

<210> SEQ ID NO 126
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 126w

<400> SEQUENCE: 126

caactttgcg cataggctgg ctgaatattc at 32

<210> SEQ ID NO 127
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 127w

<400> SEQUENCE: 127

tacccaaaag gcttgccctg acgagacgat aa 32

-continued

<210> SEQ ID NO 128
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 128s

<400> SEQUENCE: 128

aaaccaaaca gagggggtaa tagt 24

<210> SEQ ID NO 129
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 129w

<400> SEQUENCE: 129

aagagtaaca atcataaggg aaccgaactg ac 32

<210> SEQ ID NO 130
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 130w

<400> SEQUENCE: 130

agaacgagtc ttgacaagaa ccggccttca tc 32

<210> SEQ ID NO 131
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 131w

<400> SEQUENCE: 131

tagactggcc ctcgtttacc agacgaaaca cc 32

<210> SEQ ID NO 132
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 132w

<400> SEQUENCE: 132

cagacggtac aaagtacaac ggagttatac ca 32

<210> SEQ ID NO 133
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 133w

<400> SEQUENCE: 133

agcgcgaata gtaaattggg cttgagcaac ac 32

<210> SEQ ID NO 134
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 134s

<400> SEQUENCE: 134

tatcataaat agcgtccaat actgtcaaat gc 32

<210> SEQ ID NO 135
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 135w

<400> SEQUENCE: 135

catcgcctat gttacttagc cggaacgagg cg 32

<210> SEQ ID NO 136
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 136w

<400> SEQUENCE: 136

taatttcact ttgaccccca gcgaatttgt at 32

<210> SEQ ID NO 137
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 137w

<400> SEQUENCE: 137

tcataaatta cgaggcatag taagagatgg tt 32

<210> SEQ ID NO 138
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 138w

<400> SEQUENCE: 138

atccgcgacc tgctcc 16

<210> SEQ ID NO 139
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 139w

<400> SEQUENCE: 139

gataaattgt gtcgaaacac taaa 24

<210> SEQ ID NO 140
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 140w

<400> SEQUENCE: 140

-continued

acactcatac tttaatcatt gtgataacgc ca 32

<210> SEQ ID NO 141
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 141s

<400> SEQUENCE: 141

aaaggaattt agtttgacca ttagctgcga ac 32

<210> SEQ ID NO 142
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 142w

<400> SEQUENCE: 142

atgcgattaa agaggcaaaa gaat 24

<210> SEQ ID NO 143
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 143s

<400> SEQUENCE: 143

cgcaaatgaa ctaatgcaga tacaattacc tt 32

<210> SEQ ID NO 144
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 144w

<400> SEQUENCE: 144

agactttttc atgagg 16

<210> SEQ ID NO 145
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 145w

<400> SEQUENCE: 145

aagtttccat tgcaccaac 19

<210> SEQ ID NO 146
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 146w

<400> SEQUENCE: 146

ctaaaacgtt aagaactggc tcattaggaa ta 32

-continued

<210> SEQ ID NO 147
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 147s

<400> SEQUENCE: 147

ccacattcgt caataacctg tttagtttca tt 32

<210> SEQ ID NO 148
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 148w

<400> SEQUENCE: 148

aaaatacgtt gaggactaa 19

<210> SEQ ID NO 149
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 149w

<400> SEQUENCE: 149

tcaggacgta atgccactac gaagaaacgg gt 32

<210> SEQ ID NO 150
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 150s

<400> SEQUENCE: 150

ttcatttgtt catcagttga gatttatacc ag 32

<210> SEQ ID NO 151
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 151w

<400> SEQUENCE: 151

attaaagatt gggaagaaaa atctattaca gg 32

<210> SEQ ID NO 152
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 152s

<400> SEQUENCE: 152

tagaaagagg gcgcgagctg aaaattaaat at 32

<210> SEQ ID NO 153
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 153w

<400> SEQUENCE: 153

aaaacgaatc cagtttggaa caagcaaagg gc 32

<210> SEQ ID NO 154
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 154s

<400> SEQUENCE: 154

caattctact aacggaacaa cattacgtta at 32

<210> SEQ ID NO 155
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 155s

<400> SEQUENCE: 155

agtgttgttg aggggacgac gacacgtgca tc 32

<210> SEQ ID NO 156
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 156s

<400> SEQUENCE: 156

tgccagttct aatagtagta gcatttgctg aa 32

<210> SEQ ID NO 157
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 157c

<400> SEQUENCE: 157

cagaggctac gtggactcca acgtagtcca ct 32

<210> SEQ ID NO 158
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 158c

<400> SEQUENCE: 158

gaaaaacctc ggaacgaggg tagcaacggc ta 32

<210> SEQ ID NO 159
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 159s

<400> SEQUENCE: 159

-continued

aataaatcat gggcgcatcg taacgtatcg gc 32

<210> SEQ ID NO 160
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 160s

<400> SEQUENCE: 160

ctcaggaaaa agaatagccc gagagcccac ta 32

<210> SEQ ID NO 161
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 161c

<400> SEQUENCE: 161

agacagcagt ctatcagggc gatgtagggt tg 32

<210> SEQ ID NO 162
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 162s

<400> SEQUENCE: 162

tgacctgaaa gcgtaaggga tcgtcaccct cagcagcgaa 40

<210> SEQ ID NO 163
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 163s

<400> SEQUENCE: 163

gaattagcaa aattaatgac cgtaatggga tagctttccg 40

<210> SEQ ID NO 164
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 164s

<400> SEQUENCE: 164

ataaatcaga tcgcactcca gccaggtcac gt 32

<210> SEQ ID NO 165
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 165s

<400> SEQUENCE: 165

cgtgaaccat cacccaaatc aagtaatccc tt 32

-continued

<210> SEQ ID NO 166
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 166s

<400> SEQUENCE: 166

gtcgaggtag agatagaacc cttc 24

<210> SEQ ID NO 167
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 167s

<400> SEQUENCE: 167

gcaccgctgt tccgaaatcg gcaatttttg gg 32

<210> SEQ ID NO 168
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 168s

<400> SEQUENCE: 168

ctaaatcgcg tcggattctc cgtgaggcaa ag 32

<210> SEQ ID NO 169
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 169

<400> SEQUENCE: 169

tggccaacgc cgtaaagcac taaaatcctg tt 32

<210> SEQ ID NO 170
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 170

<400> SEQUENCE: 170

ctaaagggcg accagtaata aaagggacat tc 32

<210> SEQ ID NO 171
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 171s

<400> SEQUENCE: 171

cgccattcgc cccagcaggc gaaatcggaa cc 32

<210> SEQ ID NO 172
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 172s

<400> SEQUENCE: 172

tgatggtgtc tggtgccgga aaccggaaca aa 32

<210> SEQ ID NO 173
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 173

<400> SEQUENCE: 173

cagtcacaag cccccgattt agagcggtcc ac 32

<210> SEQ ID NO 174
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 174

<400> SEQUENCE: 174

ggaaagccga ttatttacat tggcagattc ac 32

<210> SEQ ID NO 175
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 175s

<400> SEQUENCE: 175

ggaagggcag agagttgcag caagcttgac gg 32

<210> SEQ ID NO 176
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 176s

<400> SEQUENCE: 176

gctggtttgc cattcaggct gcgctgagcg ag 32

<210> SEQ ID NO 177
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 177

<400> SEQUENCE: 177

ctgaaatggg cgaacgtggc gagatcaccg cc 32

<210> SEQ ID NO 178
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 178

<400> SEQUENCE: 178

-continued

gaagaaagac ctacattttg acgctcaatc gt 32

<210> SEQ ID NO 179
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 179s

<400> SEQUENCE: 179

tacgccagaa cagctgattg ccctaaggaa gg 32

<210> SEQ ID NO 180
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 180s

<400> SEQUENCE: 180

tggccctgga tcggtgcggg cctcttcctg ta 32

<210> SEQ ID NO 181
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 181

<400> SEQUENCE: 181

atggaaatcg aaaggagcgg gcgccaccag tg 32

<210> SEQ ID NO 182
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 182

<400> SEQUENCE: 182

tggcaagtgc cattgcaaca ggaaaaacgc tc 32

<210> SEQ ID NO 183
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 183s

<400> SEQUENCE: 183

aggcgattgg gtggtttttc tttttagggc gc 32

<210> SEQ ID NO 184
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 184s

<400> SEQUENCE: 184

agacgggcct ggcgaaaggg ggatggaacg cc 32

-continued

<210> SEQ ID NO 185
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 185

<400> SEQUENCE: 185

taccgccagt agcggtcacg ctgctgcgta tt 32

<210> SEQ ID NO 186
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 186

<400> SEQUENCE: 186

accacacctg ctggtaatat ccagaacaat at 32

<210> SEQ ID NO 187
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 187s

<400> SEQUENCE: 187

ccagtcacgc ggggagaggc ggttgcgtaa cc 32

<210> SEQ ID NO 188
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 188s

<400> SEQUENCE: 188

gggcgccaaa gttgggtaac gccattttgt ta 32

<210> SEQ ID NO 189
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 189

<400> SEQUENCE: 189

ctcaaactat cggcctcgcc gcgcttaatg cg 32

<210> SEQ ID NO 190
<211> LENGTH: 34
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 190

<400> SEQUENCE: 190

tatggttgta acatcacttg cctgagtaga agaa 34

<210> SEQ ID NO 191
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

```
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 191

<400> SEQUENCE: 191

ccgctacagg agctgcatta atgaatcg                                    28


<210> SEQ ID NO 192
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 192s

<400> SEQUENCE: 192

atgcctgcgg aaacctgtcg tgccgcgcgt ac                               32


<210> SEQ ID NO 193
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 193s

<400> SEQUENCE: 193

gccaacgcga cgttgtaaaa cgac                                        24


<210> SEQ ID NO 194
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 194s

<400> SEQUENCE: 194

ggccagtgcc taaattgtaa acgttaat                                    28


<210> SEQ ID NO 195
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 195s

<400> SEQUENCE: 195

attttgtttt caaaagggtg agaaaggccg gagacagtca aa                    42


<210> SEQ ID NO 196
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 196s

<400> SEQUENCE: 196

ggtaaagaaa aattcgcatt aaatgggttt tc                               32


<210> SEQ ID NO 197
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 197s

<400> SEQUENCE: 197
```

-continued

aatcagctaa tgcaatgcct gagtaatgtg ta 32

<210> SEQ ID NO 198
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 198s

<400> SEQUENCE: 198

atattttaca ttttttaacc aatagtgctg ca 32

<210> SEQ ID NO 199
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 199s

<400> SEQUENCE: 199

atcaaaaaga taaaaatttt tagaaccctc at 32

<210> SEQ ID NO 200
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 200s

<400> SEQUENCE: 200

aacgcaagta attcgcgtct ggccttcgct at 32

<210> SEQ ID NO 201
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 201s

<400> SEQUENCE: 201

gccagctttt ttgcgggaga agcctttatt tc 32

<210> SEQ ID NO 202
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 202s

<400> SEQUENCE: 202

tgtaatactc atcaacatta aatgaactgt tg 32

<210> SEQ ID NO 203
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 203s

<400> SEQUENCE: 203

taacaaccgt tgtaccaaaa acattatgac cc 32

-continued

<210> SEQ ID NO 204
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 204

<400> SEQUENCE: 204

attagtaact ttgacgagca cgta 24

<210> SEQ ID NO 205
<211> LENGTH: 26
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 205

<400> SEQUENCE: 205

ttaaccgttg tagcaatact tctttg 26

<210> SEQ ID NO 206
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 206

<400> SEQUENCE: 206

aattgcgttc ctcgttagaa tcaacgcaaa 30

<210> SEQ ID NO 207
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 207

<400> SEQUENCE: 207

taacgtgctt tgcgctcact gcccgctt 28

<210> SEQ ID NO 208
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 208s

<400> SEQUENCE: 208

tccagtcgag gtcgactcta gaggccaaaa ac 32

<210> SEQ ID NO 209
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 209s

<400> SEQUENCE: 209

tcaccatctg tataagcaaa tattaagctt gc 32

<210> SEQ ID NO 210
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 210s

<400> SEQUENCE: 210

accgagctcg aattctcaca tt 22

<210> SEQ ID NO 211
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 211s

<400> SEQUENCE: 211

ttgataatca gaaaagccat ccccgggt 28

<210> SEQ ID NO 212
<211> LENGTH: 34
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 212s

<400> SEQUENCE: 212

aggaagataa tatgatattc aaccgttcta gctg 34

<210> SEQ ID NO 213
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 213s

<400> SEQUENCE: 213

gagctaacgt aatcatggtc atagcatgtc aa 32

<210> SEQ ID NO 214
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 214s

<400> SEQUENCE: 214

ataaattaat gccggacccc gg 22

<210> SEQ ID NO 215
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 215s

<400> SEQUENCE: 215

tcatatgtag agggtagcta tttttgagag at 32

<210> SEQ ID NO 216
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 216s

<400> SEQUENCE: 216

-continued

tgtgtgaata aagcctgggg tgccggaggc cg 32

<210> SEQ ID NO 217
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 217s

<400> SEQUENCE: 217

ctacaaaggt aatcgtaaaa ctagctgttt cc 32

<210> SEQ ID NO 218
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 218s

<400> SEQUENCE: 218

ataaagtgat tgttatccgc tcacagagaa tc 32

<210> SEQ ID NO 219
<211> LENGTH: 38
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 219s

<400> SEQUENCE: 219

gatgaacggc tatcaggtca ttgcctgaga gtcttttt 38

<210> SEQ ID NO 220
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 220s

<400> SEQUENCE: 220

ttttggagca aacaaattcc acactttt 28

<210> SEQ ID NO 221
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 221

<400> SEQUENCE: 221

tgtccatcga gcgggagcta aacataatga gt 32

<210> SEQ ID NO 222
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 222

<400> SEQUENCE: 222

attaaagggt gaggccaccg agtaaaagag tc 32

-continued

<210> SEQ ID NO 223
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 223s

<400> SEQUENCE: 223

cggcggatgc aataaagcct cagagcataa ag 32

<210> SEQ ID NO 224
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 224s

<400> SEQUENCE: 224

tggtgtagat acaggcaagg caaagaggtc at 32

<210> SEQ ID NO 225
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 225s

<400> SEQUENCE: 225

ttttgcggtt aattgctcct tttg 24

<210> SEQ ID NO 226
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 226s

<400> SEQUENCE: 226

gagtacctat ggcttagagc ttaataacat cc 32

<210> SEQ ID NO 227
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 227s

<400> SEQUENCE: 227

tataatgcaa actccaacag gtcaggatta ga 32

<210> SEQ ID NO 228
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 228s

<400> SEQUENCE: 228

ccggaagctg tagctcaaca tgttggtggc at 32

<210> SEQ ID NO 229
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 229s

<400> SEQUENCE: 229

gcaactaaat tcgagcttca aagcgaacca ga 32

<210> SEQ ID NO 230
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 230s

<400> SEQUENCE: 230

gcgttttaag tacggtgtct ggaagctata tt 32

<210> SEQ ID NO 231
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 231s

<400> SEQUENCE: 231

ccatataaga agcccgaaag acttcaaata tc 32

<210> SEQ ID NO 232
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 232s

<400> SEQUENCE: 232

attaagagca gttgattccc aattatacat tt 32

<210> SEQ ID NO 233
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 233

<400> SEQUENCE: 233

tataatcaga ttttagacag gaacccggaa gc 32

<210> SEQ ID NO 234
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 234

<400> SEQUENCE: 234

ttttaatcct gagaagtgtt tt 22

<210> SEQ ID NO 235
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 235

<400> SEQUENCE: 235

-continued

ttttaacata cgagggtacg ccagtttt 28

<210> SEQ ID NO 236
<211> LENGTH: 34
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 8s-2

<400> SEQUENCE: 236

acgtcaaagc gaacctcccg acttgcggga ggtt 34

<210> SEQ ID NO 237
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 65long

<400> SEQUENCE: 237

tcttaaacag cttgatgtgc cgtcgagagg gtgagccgcc 40

<210> SEQ ID NO 238
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 66shortTelo

<400> SEQUENCE: 238

gtatagccgc tttcgaggtg aattttttta gggttagggt 40

<210> SEQ ID NO 239
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 69long

<400> SEQUENCE: 239

acaatgacaa caaccatagc ggggttttgc tccaggtcag 40

<210> SEQ ID NO 240
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 70shortTelo

<400> SEQUENCE: 240

gcggataaac cgatagttgc gccgttttta gggttagggt 40

<210> SEQ ID NO 241
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 73long

<400> SEQUENCE: 241

gatatattcg gtcgctaggc tgagactcct cacaaataaa 40

-continued

<210> SEQ ID NO 242
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 74shortTelo

<400> SEQUENCE: 242

attaggattc gcccacgcat aacctttta gggttagggt 40

<210> SEQ ID NO 243
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 78s-telo

<400> SEQUENCE: 243

gtattaagga ggcttgcagg gagtttttta gggttagggt 40

<210> SEQ ID NO 244
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 166s-telo

<400> SEQUENCE: 244

gtcgaggtag agatagaacc cttcttttta gggttagggt 40

<210> SEQ ID NO 245
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 169long

<400> SEQUENCE: 245

ggacattctg gccaacgccg taaagcacta aaatcctgtt 40

<210> SEQ ID NO 246
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 170shortTelo

<400> SEQUENCE: 246

ctaaagggcg accagtaata aaagttttta gggttagggt 40

<210> SEQ ID NO 247
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 173long

<400> SEQUENCE: 247

agattcacca gtcacaagcc cccgatttag agcggtccac 40

<210> SEQ ID NO 248
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 174shortTelo

<400> SEQUENCE: 248

ggaaagccga ttatttacat tggcttttta gggttagggt 40

<210> SEQ ID NO 249
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 177long

<400> SEQUENCE: 249

tcaatcgtct gaaatgggcg aacgtggcga gatcaccgcc 40

<210> SEQ ID NO 250
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 178shortTelo

<400> SEQUENCE: 250

gaagaaagac ctacattttg acgcttttta gggttagggt 40

<210> SEQ ID NO 251
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 65long

<400> SEQUENCE: 251

tcttaaacag cttgatgtgc cgtcgagagg gtgagccgcc 40

<210> SEQ ID NO 252
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 66shortTelo

<400> SEQUENCE: 252

gtatagccgc tttcgaggtg aatttttta gggttagggt 40

<210> SEQ ID NO 253
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 69long

<400> SEQUENCE: 253

acaatgacaa caaccatagc ggggttttgc tccaggtcag 40

<210> SEQ ID NO 254
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 70shortTelo

<400> SEQUENCE: 254

-continued

gcggataaac cgatagttgc gccgttttta gggttagggt 40

<210> SEQ ID NO 255
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 73LongCompM20

<400> SEQUENCE: 255

gtgcaggtag ttttgatata ttcggtcgct aggctgagac tcctcacaaa taaa 54

<210> SEQ ID NO 256
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 74shortTelo

<400> SEQUENCE: 256

attaggattc gcccacgcat aaccttttta gggttagggt 40

<210> SEQ ID NO 257
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 78s-telo

<400> SEQUENCE: 257

gtattaagga ggcttgcagg gagtttttta gggttagggt 40

<210> SEQ ID NO 258
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 166s-telo

<400> SEQUENCE: 258

gtcgaggtag agatagaacc cttcttttta gggttagggt 40

<210> SEQ ID NO 259
<211> LENGTH: 56
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 169LongCompM20

<400> SEQUENCE: 259

gtgcaggtag ttttggaaca tttctggcca acgccgtaaa gcactaaaat cctgtt 56

<210> SEQ ID NO 260
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 170shortTelo

<400> SEQUENCE: 260

ctaaagggcg accagtaata aaagttttta gggttagggt 40

-continued

<210> SEQ ID NO 261
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 173long

<400> SEQUENCE: 261

agattcacca gtcacaagcc cccgatttag agcggtccac 40

<210> SEQ ID NO 262
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 174shortTelo

<400> SEQUENCE: 262

ggaaagccga ttatttacat tggcttttta gggttagggt 40

<210> SEQ ID NO 263
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 177long

<400> SEQUENCE: 263

tcaatcgtct gaaatgggcg aacgtggcga gatcaccgcc 40

<210> SEQ ID NO 264
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 178shortTelo

<400> SEQUENCE: 264

gaagaaagac ctacattttg acgcttttta gggttagggt 40

<210> SEQ ID NO 265
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 81cLong

<400> SEQUENCE: 265

caatattttt gaatggtata aacagttaat gcgtcataca 40

<210> SEQ ID NO 266
<211> LENGTH: 73
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 82sBothLongM20

<400> SEQUENCE: 266

gtgcaggtag tttttaaagg ccgcttttgc gaatacgtgg cacagatttt ctacctgcac 60

tataagcact tta 73

<210> SEQ ID NO 267
<211> LENGTH: 40

-continued

<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 85cLong

<400> SEQUENCE: 267

cgcgaactga tagcccaacg gggtcagtgc ctactggtaa 40

<210> SEQ ID NO 268
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 86cShortMainM20

<400> SEQUENCE: 268

agtgcccgct attagtcttt aatgttttct acctgcacta taagcacttt a 51

<210> SEQ ID NO 269
<211> LENGTH: 41
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 157cLong

<400> SEQUENCE: 269

aaacggctac agaggctacg tggactccaa cgtagtccac t 41

<210> SEQ ID NO 270
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 158cShortMainM20

<400> SEQUENCE: 270

gaaaaacctc ggaacgaggg tagcttttct acctgcacta taagcacttt a 51

<210> SEQ ID NO 271
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 161cLong

<400> SEQUENCE: 271

gcagcgaaag acagcagtct atcagggcga tgtagggttg 40

<210> SEQ ID NO 272
<211> LENGTH: 73
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 162sBothLongM20

<400> SEQUENCE: 272

gtgcaggtag tttttgacct gaaagcgtaa gggatcgtca ccctcatttt ctacctgcac 60

tataagcact tta 73

<210> SEQ ID NO 273
<211> LENGTH: 7249
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, M13mp18

<400> SEQUENCE: 273

aatgctacta ctattagtag aattgatgcc accttttcag ctcgcgcccc aaatgaaaat 60

atagctaaac aggttattga ccatttgcga aatgtatcta atggtcaaac taaatctact 120

cgttcgcaga attgggaatc aactgttata tggaatgaaa cttccagaca ccgtacttta 180

gttgcatatt taaaacatgt tgagctacag cattatattc agcaattaag ctctaagcca 240

tccgcaaaaa tgacctctta tcaaaaggag caattaaagg tactctctaa tcctgacctg 300

ttggagtttg cttccggtct ggttcgcttt gaagctcgaa ttaaaacgcg atatttgaag 360

tctttcgggc ttcctcttaa tctttttgat gcaatccgct ttgcttctga ctataatagt 420

cagggtaaag acctgatttt tgatttatgg tcattctcgt tttctgaact gtttaaagca 480

tttgaggggg attcaatgaa tatttatgac gattccgcag tattggacgc tatccagtct 540

aaacatttta ctattacccc ctctggcaaa acttcttttg caaaagcctc tcgctatttt 600

ggtttttatc gtcgtctggt aaacgagggt tatgatagtg ttgctcttac tatgcctcgt 660

aattcctttt ggcgttatgt atctgcatta gttgaatgtg gtattcctaa atctcaactg 720

atgaatcttt ctacctgtaa taatgttgtt ccgttagttc gttttattaa cgtagatttt 780

tcttcccaac gtcctgactg gtataatgag ccagttctta aaatcgcata aggtaattca 840

caatgattaa agttgaaatt aaaccatctc aagcccaatt tactactcgt tctggtgttt 900

ctcgtcaggg caagccttat tcactgaatg agcagctttg ttacgttgat ttgggtaatg 960

aatatccggt tcttgtcaag attactcttg atgaaggtca gccagcctat gcgcctggtc 1020

tgtacaccgt tcatctgtcc tctttcaaag ttggtcagtt cggttccctt atgattgacc 1080

gtctgcgcct cgttccggct aagtaacatg gagcaggtcg cggatttcga cacaatttat 1140

caggcgatga tacaaatctc cgttgtactt tgtttcgcgc ttggtataat cgctgggggt 1200

caaagatgag tgttttagtg tattcttttg cctctttcgt tttaggttgg tgccttcgta 1260

gtggcattac gtattttacc cgtttaatgg aaacttcctc atgaaaaagt ctttagtcct 1320

caaagcctct gtagccgttg ctaccctcgt tccgatgctg tctttcgctg ctgagggtga 1380

cgatcccgca aaagcggcct ttaactccct gcaagcctca gcgaccgaat atatcggtta 1440

tgcgtgggcg atggttgttg tcattgtcgg cgcaactatc ggtatcaagc tgtttaagaa 1500

attcacctcg aaagcaagct gataaaccga tacaattaaa ggctcctttt ggagcctttt 1560

ttttggagat tttcaacgtg aaaaaattat tattcgcaat tcctttagtt gttcctttct 1620

attctcactc cgctgaaact gttgaaagtt gtttagcaaa atcccataca gaaaattcat 1680

ttactaacgt ctggaaagac gacaaaactt tagatcgtta cgctaactat gagggctgtc 1740

tgtggaatgc tacaggcgtt gtagtttgta ctggtgacga aactcagtgt tacggtacat 1800

gggttcctat tgggcttgct atccctgaaa atgagggtgg tggctctgag ggtggcggtt 1860

ctgagggtgg cggttctgag ggtggcggta ctaaacctcc tgagtacggt gatacaccta 1920

ttccgggcta tacttatatc aaccctctcg acggcactta tccgcctggt actgagcaaa 1980

accccgctaa tcctaatcct tctcttgagg agtctcagcc tcttaatact ttcatgtttc 2040

agaataatag gttccgaaat aggcaggggg cattaactgt ttatacgggc actgttactc 2100

aaggcactga ccccgttaaa acttattacc agtacactcc tgtatcatca aaagccatgt 2160

-continued

```
atgacgctta ctggaacggt aaattcagag actgcgcttt ccattctggc tttaatgagg   2220
atttatttgt ttgtgaatat caaggccaat cgtctgacct gcctcaacct cctgtcaatg   2280
ctggcggcgg ctctggtggt ggttctggtg gcggctctga gggtggtggc tctgagggtg   2340
gcggttctga gggtggcggc tctgagggag gcggttccgg tggtggctct ggttccggtg   2400
attttgatta tgaaaagatg gcaaacgcta ataagggggc tatgaccgaa aatgccgatg   2460
aaaacgcgct acagtctgac gctaaaggca aacttgattc tgtcgctact gattacggtg   2520
ctgctatcga tggtttcatt ggtgacgttt ccggccttgc taatggtaat ggtgctactg   2580
gtgattttgc tggctctaat tcccaaatgg ctcaagtcgg tgacggtgat aattcacctt   2640
taatgaataa tttccgtcaa tatttacctt ccctccctca atcggttgaa tgtcgccctt   2700
ttgtctttgg cgctggtaaa ccatatgaat tttctattga ttgtgacaaa ataaacttat   2760
tccgtggtgt ctttgcgttt cttttatatg ttgccacctt tatgtatgta ttttctacgt   2820
ttgctaacat actgcgtaat aaggagtctt aatcatgcca gttcttttgg gtattccgtt   2880
attattgcgt ttcctcggtt tccttctggt aactttgttc ggctatctgc ttacttttct   2940
taaaaagggc ttcggtaaga tagctattgc tatttcattg tttcttgctc ttattattgg   3000
gcttaactca attcttgtgg gttatctctc tgatattagc gctcaattac cctctgactt   3060
tgttcagggt gttcagttaa ttctcccgtc taatgcgctt ccctgttttt atgttattct   3120
ctctgtaaag gctgctattt tcatttttga cgttaaacaa aaaatcgttt cttatttgga   3180
ttgggataaa taatatggct gtttattttg taactggcaa attaggctct ggaaagacgc   3240
tcgttagcgt tggtaagatt caggataaaa ttgtagctgg gtgcaaaata gcaactaatc   3300
ttgatttaag gcttcaaaac ctcccgcaag tcgggaggtt cgctaaaacg cctcgcgttc   3360
ttagaatacc ggataagcct tctatatctg atttgcttgc tattgggcgc ggtaatgatt   3420
cctacgatga aaataaaaac ggcttgcttg ttctcgatga gtgcggtact tggtttaata   3480
cccgttcttg gaatgataag gaaagacagc cgattattga ttggtttcta catgctcgta   3540
aattaggatg ggatattatt tttcttgttc aggacttatc tattgttgat aaacaggcgc   3600
gttctgcatt agctgaacat gttgtttatt gtcgtcgtct ggacagaatt actttacctt   3660
ttgtcggtac tttatattct cttattactg gctcgaaaat gcctctgcct aaattacatg   3720
ttggcgttgt taaatatggc gattctcaat taagccctac tgttgagcgt tggctttata   3780
ctggtaagaa tttgtataac gcatatgata ctaaacaggc tttttctagt aattatgatt   3840
ccggtgttta ttcttattta acgccttatt tatcacacgg tcggtatttc aaaccattaa   3900
atttaggtca gaagatgaaa ttaactaaaa tatatttgaa aaagttttct cgcgttcttt   3960
gtcttgcgat tggatttgca tcagcattta catatagtta tataacccaa cctaagccgg   4020
aggttaaaaa ggtagtctct cagacctatg attttgataa attcactatt gactcttctc   4080
agcgtcttaa tctaagctat cgctatgttt tcaaggattc taagggaaaa ttaattaata   4140
gcgacgattt acagaagcaa ggttattcac tcacatatat tgatttatgt actgtttcca   4200
ttaaaaaagg taattcaaat gaaattgtta aatgtaatta attttgtttt cttgatgttt   4260
gtttcatcat cttcttttgc tcaggtaatt gaaatgaata attcgcctct gcgcgatttt   4320
gtaacttggt attcaaagca atcaggcgaa tccgttattg tttctcccga tgtaaaaggt   4380
actgttactg tatattcatc tgacgttaaa cctgaaaatc tacgcaattt ctttatttct   4440
```

-continued

```
gttttacgtg caaataattt tgatatggta ggttctaacc cttccattat tcagaagtat   4500
aatccaaaca atcaggatta tattgatgaa ttgccatcat ctgataatca ggaatatgat   4560
gataattccg ctccttctgg tggtttcttt gttccgcaaa atgataatgt tactcaaact   4620
tttaaaatta ataacgttcg ggcaaaggat ttaatacgag ttgtcgaatt gtttgtaaag   4680
tctaatactt ctaaatcctc aaatgtatta tctattgacg gctctaatct attagttgtt   4740
agtgctccta aagatatttt agataacctt cctcaattcc tttcaactgt tgatttgcca   4800
actgaccaga tattgattga gggtttgata tttgaggttc agcaaggtga tgctttagat   4860
ttttcatttg ctgctggctc tcagcgtggc actgttgcag gcggtgttaa tactgaccgc   4920
ctcacctctg ttttatcttc tgctggtggt tcgttcggta tttttaatgg cgatgtttta   4980
gggctatcag ttcgcgcatt aaagactaat agccattcaa aaatattgtc tgtgccacgt   5040
attcttacgc tttcaggtca gaagggttct atctctgttg gccagaatgt cccttttatt   5100
actggtcgtg tgactggtga atctgccaat gtaaataatc catttcagac gattgagcgt   5160
caaaatgtag gtatttccat gagcgttttt cctgttgcaa tggctggcgg taatattgtt   5220
ctggatatta ccagcaaggc cgatagtttg agttcttcta ctcaggcaag tgatgttatt   5280
actaatcaaa gaagtattgc tacaacggtt aatttgcgtg atggacagac tcttttactc   5340
ggtggcctca ctgattataa aaacacttct caggattctg gcgtaccgtt cctgtctaaa   5400
atccctttaa tcggcctcct gtttagctcc cgctctgatt ctaacgagga aagcacgtta   5460
tacgtgctcg tcaaagcaac catagtacgc gccctgtagc ggcgcattaa gcgcggcggg   5520
tgtggtggtt acgcgcagcg tgaccgctac acttgccagc gccctagcgc ccgctccttt   5580
cgctttcttc ccttcctttc tcgccacgtt cgccggcttt ccccgtcaag ctctaaatcg   5640
ggggctccct ttagggttcc gatttagtgc tttacggcac ctcgacccca aaaaacttga   5700
tttgggtgat ggttcacgta gtgggccatc gccctgatag acggtttttc gccctttgac   5760
gttggagtcc acgttcttta atagtggact cttgttccaa actggaacaa cactcaaccc   5820
tatctcgggc tattcttttg atttataagg gattttgccg atttcggaac caccatcaaa   5880
caggattttc gcctgctggg gcaaaccagc gtggaccgct tgctgcaact ctctcagggc   5940
caggcggtga agggcaatca gctgttgccc gtctcactgg tgaaaagaaa aaccaccctg   6000
gcgcccaata cgcaaaccgc ctctccccgc gcgttggccg attcattaat gcagctggca   6060
cgacaggttt cccgactgga aagcgggcag tgagcgcaac gcaattaatg tgagttagct   6120
cactcattag gcaccccagg ctttacactt tatgcttccg gctcgtatgt tgtgtggaat   6180
tgtgagcgga taacaatttc acacaggaaa cagctatgac catgattacg aattcgagct   6240
cggtacccgg ggatcctcta gagtcgacct gcaggcatgc aagcttggca ctggccgtcg   6300
ttttacaacg tcgtgactgg gaaaaccctg gcgttaccca acttaatcgc cttgcagcac   6360
atcccccttt cgccagctgg cgtaatagcg aagaggcccg caccgatcgc ccttcccaac   6420
agttgcgcag cctgaatggc gaatggcgct ttgcctggtt tccggcacca gaagcggtgc   6480
cggaaagctg gctggagtgc gatcttcctg aggccgatac tgtcgtcgtc ccctcaaact   6540
ggcagatgca cggttacgat gcgcccatct acaccaacgt gacctatccc attacggtca   6600
atccgccgtt tgttcccacg gagaatccga cgggttgtta ctcgctcaca tttaatgttg   6660
atgaaagctg gctacaggaa ggccagacgc gaattatttt tgatggcgtt cctattggtt   6720
```

-continued

aaaaaatgag ctgatttaac aaaaatttaa tgcgaatttt aacaaaatat taacgtttac 6780

aatttaaata tttgcttata caatcttcct gtttttgggg cttttctgat tatcaaccgg 6840

ggtacatatg attgacatgc tagttttacg attaccgttc atcgattctc ttgtttgctc 6900

cagactctca ggcaatgacc tgatagcctt tgtagatctc tcaaaaatag ctaccctctc 6960

cggcattaat ttatcagcta gaacggttga atatcatatt gatggtgatt tgactgtctc 7020

cggcctttct caccctttg aatctttacc tacacattac tcaggcattg catttaaaat 7080

atatgagggt tctaaaaatt tttatccttg cgttgaaata aaggcttctc ccgcaaaagt 7140

attacagggt cataatgttt ttggtacaac cgatttagct ttatgctctg aggctttatt 7200

gcttaatttt gctaattctt tgccttgcct gtatgattta ttggatgtt 7249

<210> SEQ ID NO 274
<211> LENGTH: 23
<212> TYPE: RNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 274

uaaagugcuu auagugcagg uag 23

<210> SEQ ID NO 275
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 65long

<400> SEQUENCE: 275

tcttaaacag cttgatgtgc cgtcgagagg gtgagccgcc 40

<210> SEQ ID NO 276
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 66shortTelo

<400> SEQUENCE: 276

gtatagccgc tttcgaggtg aattttttta gggttagggt 40

<210> SEQ ID NO 277
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 69long

<400> SEQUENCE: 277

acaatgacaa caaccatagc ggggttttgc tccaggtcag 40

<210> SEQ ID NO 278
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 70shortTelo

<400> SEQUENCE: 278

gcggataaac cgatagttgc gccgttttta gggttagggt 40

-continued

<210> SEQ ID NO 279
<211> LENGTH: 53
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 73LongCompCF509

<400> SEQUENCE: 279

cagttcgagt tttgatatat tcggtcgcta ggctgagact cctcacaaat aaa 53

<210> SEQ ID NO 280
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 74shortTelo

<400> SEQUENCE: 280

attaggattc gcccacgcat aacctttta gggttagggt 40

<210> SEQ ID NO 281
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 78s-telo

<400> SEQUENCE: 281

gtattaagga ggcttgcagg gagtttttta gggttagggt 40

<210> SEQ ID NO 282
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 166s-telo

<400> SEQUENCE: 282

gtcgaggtag agatagaacc cttcttttta gggttagggt 40

<210> SEQ ID NO 283
<211> LENGTH: 55
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 169LongCompCF509

<400> SEQUENCE: 283

cagttcgagt tttggaacat ttctggccaa cgccgtaaag cactaaaatc ctgtt 55

<210> SEQ ID NO 284
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 170shortTelo

<400> SEQUENCE: 284

ctaaagggcg accagtaata aaagttttta gggttagggt 40

<210> SEQ ID NO 285
<211> LENGTH: 40

-continued

<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 173long

<400> SEQUENCE: 285

agattcacca gtcacaagcc cccgatttag agcggtccac 40

<210> SEQ ID NO 286
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 174shortTelo

<400> SEQUENCE: 286

ggaaagccga ttatttacat tggcttttta gggttagggt 40

<210> SEQ ID NO 287
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 177long

<400> SEQUENCE: 287

tcaatcgtct gaaatgggcg aacgtggcga gatcaccgcc 40

<210> SEQ ID NO 288
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 178shortTelo

<400> SEQUENCE: 288

gaagaaagac ctacattttg acgcttttta gggttagggt 40

<210> SEQ ID NO 289
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 81cLong

<400> SEQUENCE: 289

caatattttt gaatggtata aacagttaat gcgtcataca 40

<210> SEQ ID NO 290
<211> LENGTH: 68
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 82sBothLongCF509

<400> SEQUENCE: 290

cagttcgagt ttttaaaggc cgcttttgcg aatacgtggc acagattttc tcgaactgca 60

cctcatgg 68

<210> SEQ ID NO 291
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 85cLong

<400> SEQUENCE: 291

cgcgaactga tagcccaacg gggtcagtgc ctactggtaa 40

<210> SEQ ID NO 292
<211> LENGTH: 47
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 86cShortCF509

<400> SEQUENCE: 292

agtgcccgct attagtcttt aatgttttct cgaactgcac ctcatgg 47

<210> SEQ ID NO 293
<211> LENGTH: 41
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 157cLong

<400> SEQUENCE: 293

aaacggctac agaggctacg tggactccaa cgtagtccac t 41

<210> SEQ ID NO 294
<211> LENGTH: 47
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 158cShortCF509

<400> SEQUENCE: 294

gaaaaacctc ggaacgaggg tagcttttct cgaactgcac ctcatgg 47

<210> SEQ ID NO 295
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 161cLong

<400> SEQUENCE: 295

gcagcgaaag acagcagtct atcagggcga tgtagggttg 40

<210> SEQ ID NO 296
<211> LENGTH: 68
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 162sBothLongCF509

<400> SEQUENCE: 296

cagttcgagt tttgacctg aaagcgtaag ggatcgtcac cctcattttc tcgaactgca 60

cctcatgg 68

<210> SEQ ID NO 297
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, R509

-continued

<400> SEQUENCE: 297

ccatgaggtg cagttcgag 19

<210> SEQ ID NO 298
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 65long

<400> SEQUENCE: 298

tcttaaacag cttgatgtgc cgtcgagagg gtgagccgcc 40

<210> SEQ ID NO 299
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 66shortTelo

<400> SEQUENCE: 299

gtatagccgc tttcgaggtg aattttttta gggttagggt 40

<210> SEQ ID NO 300
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 69long

<400> SEQUENCE: 300

acaatgacaa caaccatagc ggggttttgc tccaggtcag 40

<210> SEQ ID NO 301
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 70shortTelo

<400> SEQUENCE: 301

gcggataaac cgatagttgc gccgttttta gggttagggt 40

<210> SEQ ID NO 302
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 73LongCompN1

<400> SEQUENCE: 302

aacattgcca ttttgatata ttcggtcgct aggctgagac tcctcacaaa taaa 54

<210> SEQ ID NO 303
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 74shortTelo

<400> SEQUENCE: 303

attaggattc gcccacgcat aacctttttta gggttagggt 40

-continued

<210> SEQ ID NO 304
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 78s-telo

<400> SEQUENCE: 304

gtattaagga ggcttgcagg gagtttttta gggttagggt 40

<210> SEQ ID NO 305
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 166s-telo

<400> SEQUENCE: 305

gtcgaggtag agatagaacc cttcttttta gggttagggt 40

<210> SEQ ID NO 306
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 169LongCompN1

<400> SEQUENCE: 306

aacattgcca ttttggacat tctggccaac gccgtaaagc actaaaatcc tgtt 54

<210> SEQ ID NO 307
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 170shortTelo

<400> SEQUENCE: 307

ctaaagggcg accagtaata aaagttttta gggttagggt 40

<210> SEQ ID NO 308
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 173long

<400> SEQUENCE: 308

agattcacca gtcacaagcc cccgatttag agcggtccac 40

<210> SEQ ID NO 309
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 174shortTelo

<400> SEQUENCE: 309

ggaaagccga ttatttacat tggcttttta gggttagggt 40

<210> SEQ ID NO 310
<211> LENGTH: 40

-continued

<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 177long

<400> SEQUENCE: 310

tcaatcgtct gaaatgggcg aacgtggcga gatcaccgcc 40

<210> SEQ ID NO 311
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 178shortTelo

<400> SEQUENCE: 311

gaagaaagac ctacattttg acgcttttta gggttagggt 40

<210> SEQ ID NO 312
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 81cLong

<400> SEQUENCE: 312

caatattttt gaatggtata aacagttaat gcgtcataca 40

<210> SEQ ID NO 313
<211> LENGTH: 75
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 82sBothLongN1

<400> SEQUENCE: 313

aacattgcca tttttaaagg ccgcttttgc gaatacgtgg cacagatttt tggcaatgtt 60

gttccttgag gaagt 75

<210> SEQ ID NO 314
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 85cLong

<400> SEQUENCE: 314

cgcgaactga tagcccaacg gggtcagtgc ctactggtaa 40

<210> SEQ ID NO 315
<211> LENGTH: 53
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 86cShortN1

<400> SEQUENCE: 315

agtgcccgct attagtcttt aatgtttttg gcaatgttgt tccttgagga agt 53

<210> SEQ ID NO 316
<211> LENGTH: 41
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 157cLong

<400> SEQUENCE: 316

aaacggctac agaggctacg tggactccaa cgtagtccac t 41

<210> SEQ ID NO 317
<211> LENGTH: 53
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 158cShortN1

<400> SEQUENCE: 317

gaaaaacctc ggaacgaggg tagctttttg gcaatgttgt tccttgagga agt 53

<210> SEQ ID NO 318
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 161cLong

<400> SEQUENCE: 318

gcagcgaaag acagcagtct atcagggcga tgtagggttg 40

<210> SEQ ID NO 319
<211> LENGTH: 75
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 162sBothLongN1

<400> SEQUENCE: 319

aacattgcca tttttgacct gaaagcgtaa gggatcgtca ccctcatttt tggcaatgtt 60

gttccttgag gaagt 75

<210> SEQ ID NO 320
<211> LENGTH: 25
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, N1

<400> SEQUENCE: 320

acttcctcaa ggaacaacat tgcca 25

<210> SEQ ID NO 321
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 65long

<400> SEQUENCE: 321

tcttaaacag cttgatgtgc cgtcgagagg gtgagccgcc 40

<210> SEQ ID NO 322
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 66shortTelo

-continued

<400> SEQUENCE: 322

gtatagccgc tttcgaggtg aatttttta gggttagggt 40

<210> SEQ ID NO 323
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 69long

<400> SEQUENCE: 323

acaatgacaa caaccatagc ggggttttgc tccaggtcag 40

<210> SEQ ID NO 324
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 70shortTelo

<400> SEQUENCE: 324

gcggataaac cgatagttgc gccgttttta gggttagggt 40

<210> SEQ ID NO 325
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 73LongCompHD1T1

<400> SEQUENCE: 325

acaccaacct ttttgatata ttcggtcgct aggctgagac tcctcacaaa taaa 54

<210> SEQ ID NO 326
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 74shortTelo

<400> SEQUENCE: 326

attaggattc gcccacgcat aacctttta gggttagggt 40

<210> SEQ ID NO 327
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 78s-telo

<400> SEQUENCE: 327

gtattaagga ggcttgcagg gagtttttta gggttagggt 40

<210> SEQ ID NO 328
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 166s-telo

<400> SEQUENCE: 328

gtcgaggtag agatagaacc cttcttttta gggttagggt 40

-continued

<210> SEQ ID NO 329
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 169LongCompHD1T1

<400> SEQUENCE: 329

acaccaacct ttttggacat tctggccaac gccgtaaagc actaaaatcc tgtt 54

<210> SEQ ID NO 330
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 170shortTelo

<400> SEQUENCE: 330

ctaaagggcg accagtaata aaagttttta gggttagggt 40

<210> SEQ ID NO 331
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 173long

<400> SEQUENCE: 331

agattcacca gtcacaagcc cccgatttag agcggtccac 40

<210> SEQ ID NO 332
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 174shortTelo

<400> SEQUENCE: 332

ggaaagccga ttatttacat tggcttttta gggttagggt 40

<210> SEQ ID NO 333
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 177long

<400> SEQUENCE: 333

tcaatcgtct gaaatgggcg aacgtggcga gatcaccgcc 40

<210> SEQ ID NO 334
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 178shortTelo

<400> SEQUENCE: 334

gaagaaagac ctacattttg acgcttttta gggttagggt 40

<210> SEQ ID NO 335
<211> LENGTH: 40

-continued

<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 81cLong

<400> SEQUENCE: 335

caatattttt gaatggtata aacagttaat gcgtcataca 40

<210> SEQ ID NO 336
<211> LENGTH: 71
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 82sBothLongHD1T1

<400> SEQUENCE: 336

acaccaacct tttttaaagg ccgcttttgc gaatacgtgg cacagatttg tggttggtgt 60

ggttggtcat a 71

<210> SEQ ID NO 337
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 85cLong

<400> SEQUENCE: 337

cgcgaactga tagcccaacg gggtcagtgc ctactggtaa 40

<210> SEQ ID NO 338
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 86cShortHD1T1

<400> SEQUENCE: 338

agtgcccgct attagtcttt aatgttttag tccgtggtag ggcaggttgg ggtgact 57

<210> SEQ ID NO 339
<211> LENGTH: 41
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 157cLong

<400> SEQUENCE: 339

aaacggctac agaggctacg tggactccaa cgtagtccac t 41

<210> SEQ ID NO 340
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 158cShortHD1T1

<400> SEQUENCE: 340

gaaaaacctc ggaacgaggg tagcttttag tccgtggtag ggcaggttgg ggtgact 57

<210> SEQ ID NO 341
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 161cLong

<400> SEQUENCE: 341

gcagcgaaag acagcagtct atcagggcga tgtagggttg 40

<210> SEQ ID NO 342
<211> LENGTH: 71
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 162sBothLongHD1T1

<400> SEQUENCE: 342

acaccaacct tttttgacct gaaagcgtaa gggatcgtca ccctcatttg tggttggtgt 60

ggttggtcat a 71

<210> SEQ ID NO 343
<211> LENGTH: 25
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, HD1

<400> SEQUENCE: 343

tttgtggttg gtgtggttgg tcata 25

<210> SEQ ID NO 344
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, R29142

<400> SEQUENCE: 344

gttcctggtc cccaaaattt 20

<210> SEQ ID NO 345
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 65long

<400> SEQUENCE: 345

tcttaaacag cttgatgtgc cgtcgagagg gtgagccgcc 40

<210> SEQ ID NO 346
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 66shortTelo

<400> SEQUENCE: 346

gtatagccgc tttcgaggtg aattttttta gggttagggt 40

<210> SEQ ID NO 347
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 69long

-continued

<400> SEQUENCE: 347

acaatgacaa caaccatagc ggggttttgc tccaggtcag 40

<210> SEQ ID NO 348
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 70shortTelo

<400> SEQUENCE: 348

gcggataaac cgatagttgc gccgttttta gggttagggt 40

<210> SEQ ID NO 349
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 73LongCompM20

<400> SEQUENCE: 349

gtgcaggtag ttttgatata ttcggtcgct aggctgagac tcctcacaaa taaa 54

<210> SEQ ID NO 350
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 74shortTelo

<400> SEQUENCE: 350

attaggattc gcccacgcat aacctttta gggttagggt 40

<210> SEQ ID NO 351
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 78s-telo

<400> SEQUENCE: 351

gtattaagga ggcttgcagg gagtttttta gggttagggt 40

<210> SEQ ID NO 352
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 166s-telo

<400> SEQUENCE: 352

gtcgaggtag agatagaacc cttcttttta gggttagggt 40

<210> SEQ ID NO 353
<211> LENGTH: 56
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 169LongCompM20

<400> SEQUENCE: 353

gtgcaggtag ttttggaaca tttctggcca acgccgtaaa gcactaaaat cctgtt 56

-continued

<210> SEQ ID NO 354
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 170shortTelo

<400> SEQUENCE: 354

ctaaagggcg accagtaata aaagttttta gggttagggt 40

<210> SEQ ID NO 355
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 173long

<400> SEQUENCE: 355

agattcacca gtcacaagcc cccgatttag agcggtccac 40

<210> SEQ ID NO 356
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 174shortTelo

<400> SEQUENCE: 356

ggaaagccga ttatttacat tggcttttta gggttagggt 40

<210> SEQ ID NO 357
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 177long

<400> SEQUENCE: 357

tcaatcgtct gaaatgggcg aacgtggcga gatcaccgcc 40

<210> SEQ ID NO 358
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 178shortTelo

<400> SEQUENCE: 358

gaagaaagac ctacattttg acgcttttta gggttagggt 40

<210> SEQ ID NO 359
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 81cLong

<400> SEQUENCE: 359

caatattttt gaatggtata aacagttaat gcgtcataca 40

<210> SEQ ID NO 360
<211> LENGTH: 73

-continued

<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 82sBothLongM20

<400> SEQUENCE: 360

gtgcaggtag tttttaaagg ccgcttttgc gaatacgtgg cacagatttt ctacctgcac 60

tataagcact tta 73

<210> SEQ ID NO 361
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 85cLong

<400> SEQUENCE: 361

cgcgaactga tagcccaacg gggtcagtgc ctactggtaa 40

<210> SEQ ID NO 362
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 86cShortMainM20

<400> SEQUENCE: 362

agtgcccgct attagtcttt aatgttttct acctgcacta taagcacttt a 51

<210> SEQ ID NO 363
<211> LENGTH: 41
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 157cLong

<400> SEQUENCE: 363

aaacggctac agaggctacg tggactccaa cgtagtccac t 41

<210> SEQ ID NO 364
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 158cShortMainM20

<400> SEQUENCE: 364

gaaaaacctc ggaacgaggg tagcttttct acctgcacta taagcacttt a 51

<210> SEQ ID NO 365
<211> LENGTH: 40
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 161cLong

<400> SEQUENCE: 365

gcagcgaaag acagcagtct atcagggcga tgtagggttg 40

<210> SEQ ID NO 366
<211> LENGTH: 73
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

```
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, 162sBothLongM20

<400> SEQUENCE: 366

gtgcaggtag tttttgacct gaaagcgtaa gggatcgtca ccctcatttt ctacctgcac     60

tataagcact tta                                                        73


<210> SEQ ID NO 367
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, SmBiT-95w

<400> SEQUENCE: 367

aaggaattga ggaagg                                                     16


<210> SEQ ID NO 368
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthesized seqence, LgBiT-138w

<400> SEQUENCE: 368

atccgcgacc tgctcc                                                     16
```

1. A nucleic acid structure comprising a partial nucleic acid structure 1 and a partial nucleic acid structure 2,

the partial nucleic acid structure 1 and the partial nucleic acid structure 2 being linked by a Holliday junction,

the partial nucleic acid structure 1 comprising a protruding structure 1a containing a nucleic acid aptamer sequence, and a protruding structure 1b containing a split domain 1 of an enzyme, and

the partial nucleic acid structure 2 comprising a protruding structure 2a containing a sequence complementary to part or all of the nucleic acid aptamer sequence, and a protruding structure 2b containing a split domain 2 paired with the split domain 1.

2. The nucleic acid structure according to claim **1**, wherein the partial structure 1 and the partial structure 2 each have an elongated shape.

3. The nucleic acid structure according to claim **1**, which is made of a material containing a single-stranded circular nucleic acid and a staple nucleic acid containing a sequence complementary to the single-stranded circular nucleic acid.

4. The nucleic acid structure according to claim **1**, wherein the protruding structures are arranged so that, among three structures (structural type 1, structural type 2, and structural type 3: provided that the structural type 2 is an intermediate structure that mediates structural changes between the structural type 1 and the structural type 3) based on the conformation of the Holliday junction, when taking the structural type 1, the protruding structure 1a and the protruding structure 2a face each other, and when taking the structural type 3, the protruding structure 1b and the protruding structure 2b face each other.

5. The nucleic acid structure according to claim **4**, wherein the protruding structures are arranged so that the protruding structure 1a and the protruding structure 2a face each other and bind together based on complementary base pairing in the absence of a recognition substance for the nucleic acid aptamer sequence.

6. The nucleic acid structure according to claim **4**, wherein the protruding structures are arranged so that the protruding structure 1b and the protruding structure 2b face each other to form the enzyme in the presence of a recognition substance for the nucleic acid aptamer sequence and in the presence of a monovalent cation.

7. The nucleic acid structure according to claim **1**, wherein the enzyme is a nucleic acid enzyme containing two or more G-quartets.

8. The nucleic acid structure according to claim **1**, wherein the recognition substance for the nucleic acid aptamer sequence is a biological material.

9. The nucleic acid structure according to claim **1**, wherein the recognition substance for the nucleic acid aptamer sequence is at least one member selected from the group consisting of nucleic acids, proteins, peptides, low-molecular-weight compounds, physiologically active substances, and metal ions.

10. The nucleic acid structure according to claim **1**, which is a nano-structure.

11. A test substance detection composition comprising the nucleic acid structure according to claim **1**.

12. The test substance detection composition according to claim **11**, which is a reagent or a test agent.

13. A method for detecting a test substance, comprising bringing the nucleic acid structure according to claim **1** into contact with a sample that may contain the test substance.

* * * * *